



Sacramento River Bank Protection Project Long-Term Aquatic Monitoring Program Fiscal Years 2011 through 2015

Annual Report 2011

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Sacramento River Bank Protection Project

Long-Term Aquatic Monitoring Program

Fiscal Years 2011 through 2015

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Executive Summary

Major levee repairs and emergency levee repairs within the Sacramento River system have been traditionally conducted by the United States Army Corps of Engineers (USACE) through their Sacramento River Bank Protection Project (SRBPP), and the PL84-99 Rehabilitation Programs. ("PL 84-99" refers to federal Public Law 84-99, the Flood and Coastal Storm Emergencies Act). On February 24, 2006, Governor Arnold Schwarzenegger declared a State of Emergency for California's levee system, releasing up to \$500 million to repair and evaluate State and Federal project levees. These emergency levee repair funds are now being administered through the existing USACE's levee repair programs and a new program, the Sacramento-San Joaquin Erosion Repairs Project (SSJERP). The USACE is the lead agency for the SRBPP and the PL 84-99 Rehabilitation Program, while the California State Department of Water Resources (DWR) is the lead agency for the SSJERP.

When the levees were originally constructed, maintained, and repaired (starting in the mid to late 19th century), mitigation measures were not required, but since the early 1990s, winter-run and spring-run Chinook salmon (*Oncorhynchus tshawytscha*), steelhead (*O. mykiss*), and the delta smelt (*Hypomesus transpacificus*) have become listed species in the Sacramento River through the state and federal Endangered Species Acts. As a result, mitigation measures for channel and river bank modifications were developed for the post-2006 levee repairs and were designed based on the Standardized Assessment Methodology (SAM; USACE 2004, 2006; USACE and CVFPB 2009). The SAM is a predictive model used to quantify potential impacts to threatened and endangered fish species due to construction of bank revetment and other bank protection measures, and can also be utilized to assess the fish habitat value of experimental mitigation and compensation measures. The SAM quantifies habitat values in terms of a bank line or area-weighted species response index; habitat variables include bank slope, floodplain inundation ratio, bank substrate size, instream structure, aquatic vegetation, and overhanging shade.

Based on SAM results, four experimental levee repair designs were developed and constructed in the SAM planning area. For these post-2006 emergency levee repairs covered by the Governor's declaration, the National Marine Fisheries Service (NMFS) issued Biological Opinions (Opinion) pertaining to Chinook salmon and steelhead in June 2006 and August 2007; and the United States Fish and Wildlife Service (USFWS) issued Opinions pertaining to delta smelt in June 2006 and April 2007. The NMFS and USFWS Opinions support approval of critical levee repairs at 57 sites (along the Sacramento and Bear rivers; and Steamboat, Sutter, and Cache sloughs). DWR undertook repairs at 21 of the sites, and the USACE undertook repairs at the remaining 36 sites. These sites were identified as "critical," as defined by bank erosion that could threaten the structural integrity of the flood control system, and therefore required immediate work to prevent levee failure during the next flood (Ayres Associates 2007).

The NMFS and USFWS Opinions also required post-construction monitoring to evaluate whether "on-site compensations [at post-2006 emergency levee Repair Sites] are functioning in a manner that enhances habitat value and offset adverse bank protection effects" (NMFS 2001, 2004, 2006a, 2006b, 2007; USFWS 2004a, 2004b, 2006a, 2006b, 2006c, 2007). A fish and habitat monitoring program (Sacramento River Bank Protection Project Long-Term Aquatic Monitoring Program) was developed in 2008 (H.T. Harvey & Associates et al. 2008) and revised

in 2011 (FISHBIO 2011), which established monitoring goals and objectives, as well as monitoring protocols to meet the post-construction monitoring requirements of these Opinions. Annual monitoring has been conducted since 2007, but habitat data beginning in 2008 and fisheries data beginning in 2009 is considered the baseline dataset for long-term monitoring evaluations.

This report documents and evaluates the results of the annual fish and habitat monitoring conducted in 2011, as well as comparisons to fish and habitat monitoring conducted from 2008-2010 (H.T. Harvey & Associates and PRBO 2011). In 2011, the SAM habitat features were measured, and fish presence and abundance was evaluated at several types of sites by using electrofishing and telemetry. Site types include post-2006 emergency levee repair sites (SAM Repair Sites) that incorporate habitat mitigation features (n = 11 sites), Historic pre-2006 levee repair sites (Historic Repair Sites) that generally consist of riprap without additional habitat enhancement (n=5 sites), and natural bank sites (Natural Sites) that have not been engineered, are not rip-rapped, and are dominated by native, naturally established vegetation (n=6 sites). Through this monitoring effort, the following study objectives could be addressed:

- 1) identify habitat features and design types of levee repair sites that promote use by focus species (anadromous salmonids and delta smelt), so that future repair sites may maximize benefits for species of concern, and
- 2) determine if focus fish species preferentially use naturalized sites or levee repair sites.

Key results include:

- Fish communities were most similar between Natural Sites and SAM Repair Sites.
- Chinook salmon appear to prefer SAM Repair Sites containing mitigated habitat features compared to Historic Repair Sites consisting primarily of rip-rap.
- Based on electrofishing, Chinook salmon appear to prefer the 10:1 Slope/Bench Site Type over both Natural Sites and other SAM Repair Site Types.
- Regardless of SAM Repair Site Type, salmon preferentially inhabit shallow water areas, and juveniles are more commonly found in areas with faster current.
- Chinook salmon fry abundance decreased as bank coverage of terrestrial vegetation and terrestrial woody material increased.
- Predation risk to fry and juvenile salmon appears to be highest at Wetland Trench Sites, with some potential at No Bench Sites.
- Effect of river region and SAM Repair Site Type on Chinook salmon size.
- Based on telemetry, SAM Repair Sites appear to be preferred by Chinook salmon, steelhead, and striped bass over Natural Sites.

- Chinook salmon and steelhead residency is more likely in sites with tidal influence.
- Residency at sites generally short, but fish in tidal areas frequently visited individual sites more than once.

1.0 INTRODUCTION AND BACKGROUND

The Sacramento River Bank Protection Project (SRBPP) Long-Term Aquatic Monitoring Program (Program) is part of ongoing work associated with the SRBPP. Authorization & Appropriation for the Program are being provided through: Public Law 86-645 (Flood Control Act of 1960) in accordance with the recommendation of the Chief of Engineers in Senate Document No. 103, 86th Congress, Second Session, "Sacramento River Flood Control Project, California."

The Program has been designed to evaluate habitat features of levee repair sites that promote fish use, so that future repair sites can maximize values for fish, and to determine if fish use is similar between naturalized sites and levee repair sites. The Program was developed in 2008 (H.T. Harvey & Associates et al. 2008) and revised in 2011 (FISHBIO 2011), on behalf of DWR, USACE, NMFS, USFWS, and CFDG to identify monitoring goals and objectives, as well as monitoring protocols to meet the post-construction monitoring requirements of Biological Opinions for emergency levee repairs (see Background). Annual monitoring has been conducted since 2007, but habitat data beginning in 2008 and fisheries data beginning in 2009 is considered the baseline dataset for long-term monitoring evaluations.

This report presents the monitoring results from 2011, with some comparisons to results from monitoring conducted in 2009 and 2010 (H.T. Harvey & Associates and PRBO 2011). Electrofishing was used to document fish presence and estimated abundance at three different site types: 1) **SAM Repair Sites** (i.e., post-2006 emergency levee Repair Sites that incorporate habitat mitigation features; n = 11 sites), 2) **Historic Repair Sites** (i.e., Historic pre-2006 levee Repair Sites that generally consist of riprap without additional habitat enhancement; n=5 sites), and 3) **Natural Sites** (i.e., natural bank sites that have not been engineered, are not rip-rapped, and are dominated by native, naturally established vegetation; n=6 sites). Monitoring at these sites allowed us to compare the efficacy of the habitat improvements incorporated into SAM Repair Sites to that of non-mitigated repair designs associated with Historic Repair Sites, as well as to non-engineered conditions at Natural Sites.

1.1 Background

On February 24, 2006, Governor Arnold Schwarzenegger declared a State of Emergency for California's levee system, commissioning up to \$500 million of state funds to repair and evaluate State/Federal project levees. Repairs to both State and Federal project levees are being conducted under three main programs: the Sacramento River Bank Protection Project (SRBPP), the PL84-99 Rehabilitation Program, and the Sacramento-San Joaquin Erosion Repairs Project (SSJERP). The USACE is the lead agency for the SRBPP and the PL 84-99 Rehabilitation Program, while the California State Department of Water Resources (DWR) is the lead agency for the SSJERP.

When the levees were originally constructed, maintained, and repaired (starting in the mid to late 19th century), mitigation measures were not required, but since the early 1990s, winter-run and spring-run Chinook salmon (*Oncorhynchus tshawytscha*), steelhead (*O. mykiss*), and the delta smelt (*Hypomesus transpacificus*) have become listed species in the Sacramento River through the state and federal Endangered Species Acts. As a result, mitigation measures for channel and

river bank modifications were developed for the post-2006 levee repairs and were designed based on the Standardized Assessment Methodology (SAM; USACE 2004, 2006; USACE and CVFPB 2009). The SAM is a predictive model used to quantify potential impacts to threatened and endangered fish species due to construction of bank revetment and other bank protection measures, and it can also be utilized to assess the fish habitat value of experimental mitigation and compensation measures. The SAM quantifies habitat values in terms of a bank line or area-weighted species response index; habitat variables include bank slope, floodplain inundation ratio, bank substrate size, instream structure, aquatic vegetation, and overhanging shade.

Based on SAM results, four experimental levee repair designs were developed and constructed in the SAM planning area. Recently completed (post-2006) emergency levee repairs (SAM Repair Sites) include constructed habitat features to offset the impacts from loss of riparian habitat due to the levee repair action. Purposes of constructed habitat features include: 1) protecting the toe and upper slopes of the bank with riprap and/or riprap mixed with soil; 2) establishing a berm around the mean summer water level to provide aquatic habitat during lower and higher river stages in winter and spring; 3) placing instream woody material (IWM) to enhance fish habitat value; and 4) planting live pole cuttings, container plantings and grasses to stabilize the bank and provide riparian and shaded riverine aquatic habitat features. Riprap provides bank and levee protection, while the other habitat features provides benefits to fish and other aquatic and terrestrial organisms.

As part of the authorization for these bank protection actions, the USACE and DWR participated in Endangered Species Act (ESA) Section 7 consultations with the National Marine Fisheries Service (NMFS) and U.S. Fish and Wildlife Service (USFWS), which culminated in issuance of Biological Opinions (Opinions) by NMFS and the USFWS (NMFS 2006a, 2006b, 2007, 2008, 2009; USFWS 2006a, 2006b, 2006c, 2007). The NMFS and USFWS Opinions support approval of critical levee repairs at 57 sites (along the Sacramento and Bear rivers; and Steamboat, Sutter, and Cache sloughs). DWR undertook repairs at 21 of the sites, and the USACE undertook repairs at the remaining 36 sites. These sites were identified as “critical,” as defined by bank erosion that could threaten the structural integrity of the flood control system, and therefore required immediate work to prevent levee failure during the next flood (Ayres Associates 2007).

The terms and conditions of these Opinions required long-term monitoring of habitat features and fish at SRBPP sites for the purpose of characterizing the success of the habitat features in replacing the natural features at the site. In accordance with these Opinions, the USACE used the SAM, in coordination with USFWS, NMFS, cooperating state agencies, and consultants, to quantify potential impacts to focus species (i.e., threatened, endangered fish, and candidate species under the federal ESA and fish species with Essential Fish Habitat designations under the Magnuson-Stevens Fisheries Act) from “bank revetment and other bank protection projects as well as experimental mitigation and compensation measures that may be implemented within the SRBPP planning area” (USACE 2004, 2006; USACE and CVFPB 2009). Focus species and potential predators of the focus fish species addressed in the SAM include:

- Sacramento River winter-run Chinook salmon (Focus species; federally Endangered)
- Central Valley spring-run Chinook salmon (Focus species; federally Threatened)
- Central Valley fall/late fall-run Chinook salmon (Focus species; federally Species of

Concern)

- Central Valley steelhead (Focus species; federally Threatened)
- Delta smelt (Focus species; federally Threatened)
- Sacramento pikeminnow (native Predator)
- Inland silverside (introduced predator of delta smelt eggs/larvae only)
- Striped bass (introduced Predator)
- Black bass- Largemouth and smallmouth bass (introduced Predators)

In June 2008, H.T. Harvey & Associates (2008) prepared the first fish and habitat monitoring plan, on behalf of DWR, USACE, NMFS, USFWS, and CDFG, to meet the monitoring goals and objectives determined by the Technical Task Group (8 February 2008). The monitoring goals and objectives were formulated to meet NMFS and USFWS Opinion requirements that the data evaluate whether “the various experimental ... on-site compensations are functioning in a manner that enhances habitat value and offsets adverse bank protection effects” (NMFS 2001, 2004, 2006a, 2006b, 2007; USFWS 2004a, 2004b, 2006a, 2006b, 2006c, 2007). A revised plan was developed and implemented by FISHBIO in 2011.

2.0 STUDY AREA DESCRIPTION

The Sacramento River originates on the slopes of Mount Eddy in the Trinity Mountains within the Klamath Mountain Range (Thompson 1961). It joins with the McCloud and Pit rivers at Lake Shasta to form the largest river in California, with a drainage basin approximately 335 mi (540 km) long and 168 mi (270 km) wide (Larsen and Greco 2002). The Sacramento River basin climate is Mediterranean with mean annual rainfall of 36 in (914 mm) and average temperatures ranging from 54°F (12°C) in January to 88°F (31°C) in July (Domagalski et al. 2000). The Sacramento River system flows are regulated by dams built for water supply and flood control, which have stabilized and altered flow regimes, changing the river’s erosion and sediment transport characteristics (Mount 1995).

The SAM planning area within the Sacramento River mainstem extends from the lower Sacramento River near Collinsville at RM 0 to Chico Landing at RM 194. Based on differences in reach geomorphology and vegetation as described below (and in further detail in Stillwater Sciences and Dean Ryan Consultants 2004), four reach designations of the Sacramento River are identified in the SAM:

- Reach 1a: from Collinsville to (near) Isleton (RM 0-20)
- Reach 1b: from (near) Isleton to Verona (RM 20–80)
- Reach 2: from Verona to Colusa (RM 80–143)
- Reach 3: from Colusa to Chico Landing (RM 143-194)

The lower reaches of the Sacramento River (1a and 1b) along RM 0-80 run past the City of Sacramento to the Delta (Stillwater Sciences and Dean Ryan Consultants 2004). Here, the majority of the river is narrowly confined by large levees. The riparian vegetation tends to consist of discontinuous narrow bands of riparian vegetation along the levees. The soils along these reaches are primarily composed of Colombia and Sycamore series (SRAC 2003). A portion

of the reach (RM 0 to approx. RM 30) is tidally influenced.

The middle reach of the Sacramento River extends from RM 80, near the Feather River confluence, north towards RM 143 near Colusa (Reach 2; (Stillwater Sciences and Dean Ryan Consultants 2004). The channel is generally wider than in the lower reaches; however the river is confined within levees. A few relict stands of riparian vegetation occur on sands deposited over the riprapped banks.

The levees along the uppermost reach from RM 143-194 (Reach 3) are generally setback and spaced wide enough to allow for river migration (Stillwater Sciences and Dean Ryan Consultants 2004). In areas not fortified with riprap or other erosion control features, the meandering river forms cut banks, gravel bars, and associated floodplains. Sparse remnant stands of riparian forest occur along the upper reach and early successional species are established on alluvial deposits.

Numerous critical erosion repairs have been conducted in the SAM planning area since the State of Emergency for California's levee system was declared in 2006. The majority of levee repairs have occurred in Reaches 1a and 1b, with fewer in Reaches 2 and 3. The monitoring sites for this study included levee Repair Sites (both pre-2006 Historic Repair Sites and post-2006 SAM repair Sites) and Natural Sites between RM 24 and 91.9 on the Sacramento River (that is, within SAM Reaches 1b and 2).

Critical erosion repairs include several different habitat features that vary in construction cost, are constrained by site-specific conditions (e.g., hydraulics), and are hypothesized to provide varying values to fish. SRBPP monitoring sites include representative conditions at four (4) different erosion repair design types (Figures 1-4). They are:

- 1) **10:1 Slope design.** Sites with the 10:1 Slope design tend to have more gradual slopes (10:1) than the other design types;
- 2) **Bench design.** The Bench sites have a wetland bench designed to flood during periods of increased flows or high tide events, depending on location;
- 3) **Wetland Trench design.** Wetland Trench (a.k.a., Dietl ditch) sites have an off-channel ditch, parallel to the main channel, designed for aquatic habitat. These trenches are connected to the main channel and flood on a regular basis;
- 4) **No Bench design.** No Bench sites typically have steeper slopes (2:1 or 3:1) and do not include a wetland bench.

SAM Repair Sites were selected to exemplify conditions at the four design types and include a number of variables including SAM reach designations, geomorphic setting, percent vegetation, and bank slope. Habitat feature diversity among the available SAM Repair Sites is less than the overall diversity of habitat features throughout the study area. Therefore, Historic Repair and Natural Sites were selected with the intent of maximizing the range of habitat variables represented. Additional details regarding site selection strategies are provided in H.T. Harvey & Associates and PRBO (2011).

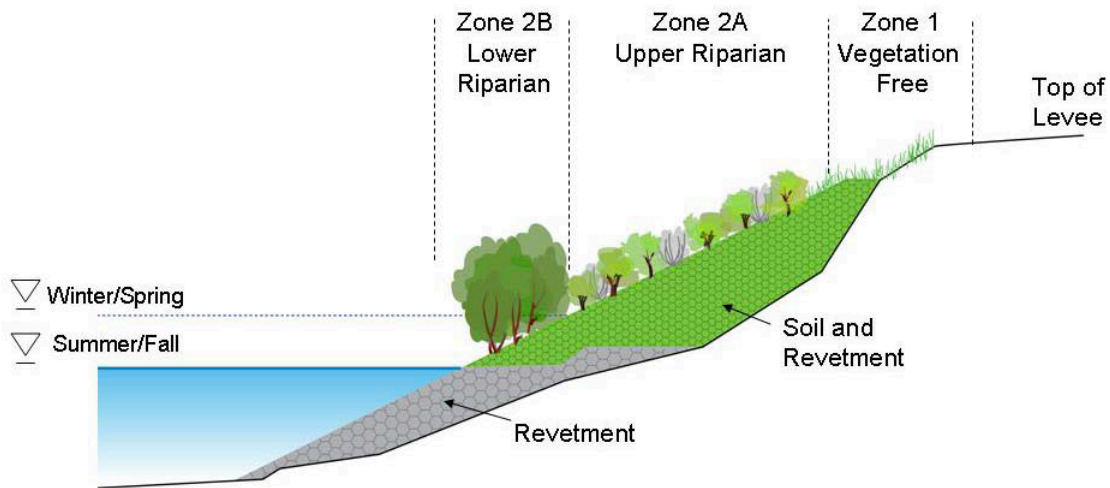


Figure 1. Bank fill rock slope with revegetation (Source: NMFS 2008).

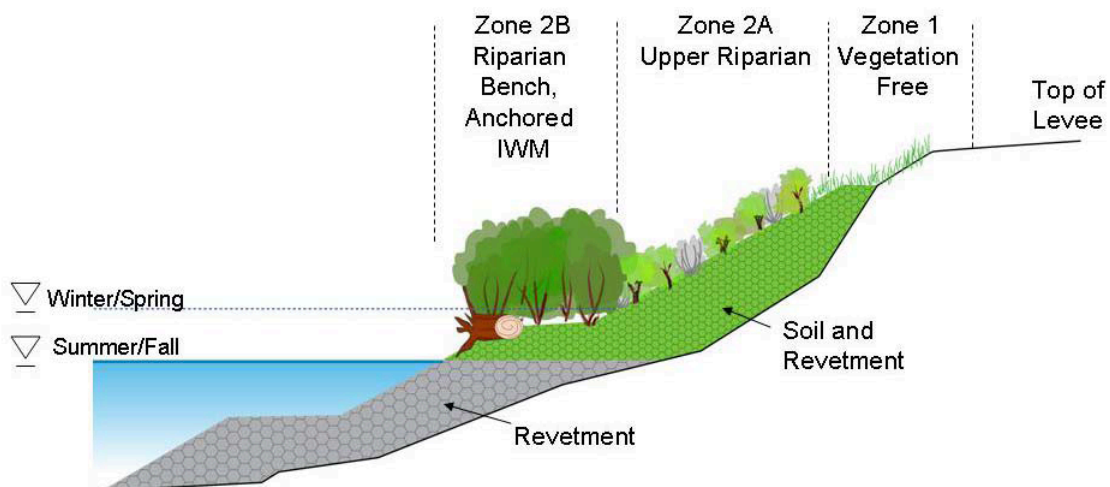


Figure 2. Low riparian bench with revegetation and anchored in-stream woody material enhancements *above* the summer/fall waterline (Source: NMFS 2008).

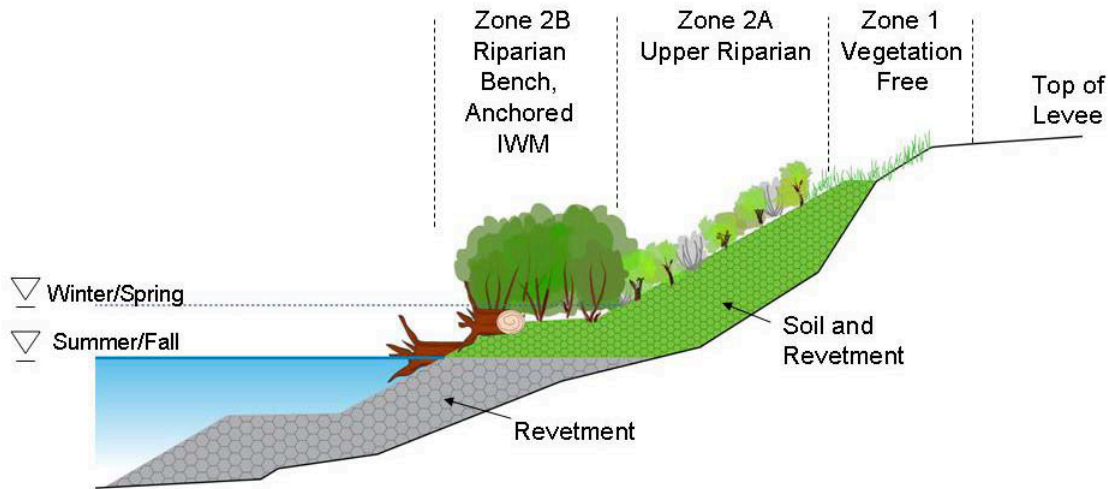


Figure 3. Low riparian bench with revegetation and anchored in-stream woody material enhancements *above and below* the summer/fall waterline (Source: NMFS 2008).

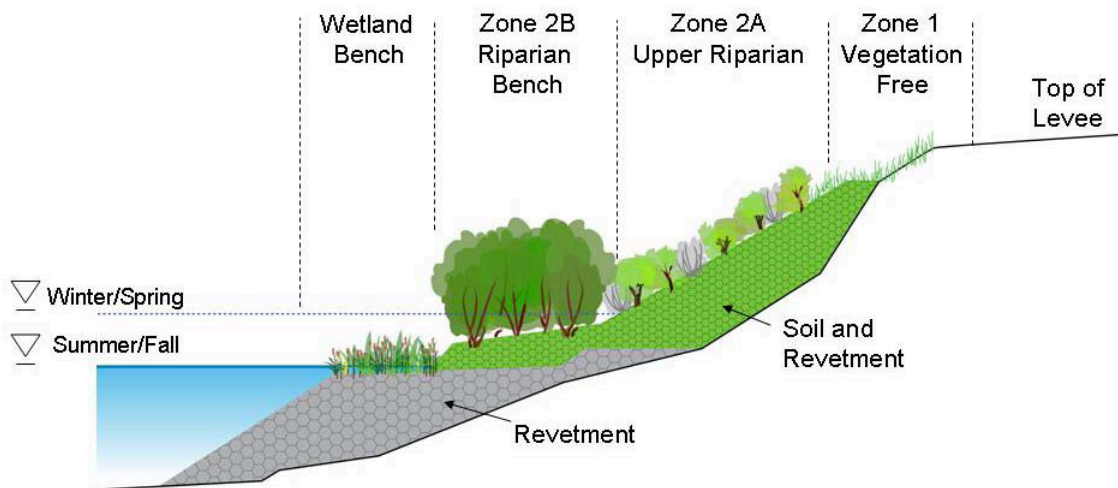


Figure 4. Low riparian and wetland benches with revegetation in tidally influenced areas (Source: NMFS 2008).

3.0 MONITORING OBJECTIVES

Specific goals and objectives have differed slightly between years. During initial monitoring, Stillwater Sciences (2009) identified two objectives and stated that results of monitoring could be “used to improve designs of future bank protection projects, to better inform offsite habitat compensation needs, and to update the SAM model (USACE 2004).” Objectives included:

Objective 1. Compare fish abundance and species composition at bank protection sites and natural bank areas, and

Objective 2. Evaluate the relationship between fish abundance and the different constructed habitat features.

In 2008, objectives were further refined and specific goals were identified (H.T. Harvey & Associates et al. 2008). These goals and several of the refined objectives were implemented in 2011, as follows:

Goal 1. Determine if on-site mitigation at SAM Repair Sites is approximating ‘Natural banks’¹, and

Goal 2. Identify which habitat features of levee Repair Sites maximize fish use and are of greatest value to focus fish species.

Fish Objective 1. Determine the fish species present at the constructed habitat features and levee benches at various seasonal flow levels at the bank protection sites after site construction and at other reference sites identified for monitoring (i.e., Natural and Historic Repair).

Fish Objective 2. Evaluate fish community abundance and composition using electrofishing at selected Natural and Repaired Sites. Information from these surveys will be used to determine salmonid presence and abundance (use) at sites and microhabitats within sites, as well as the relative risk of predation to juvenile salmonids at these sites.

Fish Objective 3. Determine the value of on-site mitigation features to juvenile salmonids based on migration timing in the SAM reaches and comparison of residency times at Repair versus Natural Sites using juvenile Chinook salmon and steelhead tagged with acoustic pingers.

Habitat Objective 1. Characterize the habitat features located at various seasonal water elevations at selected bank protection and reference sites. These data will be collected using a repeatable survey design and in a format that can be used to model focus fish species response per the SAM model requirements. Each selected site will be monitored with photography using two to six permanent photo points established at each site. In addition, a panoramic method for photographic recording of each site will be implemented.

4.0 METHODS

Fish abundance, telemetry and habitat monitoring were conducted according to the objectives of the Program. Survey methodologies have evolved and are continuously improved to adapt to

¹ “Natural banks” are those river banks that have not been engineered and are not riprapped; they are dominated by native, naturally established vegetation.

unforeseen challenges and incorporate insights gained in this ongoing monitoring and evaluation effort.

4.1 Fisheries Monitoring (Fish Objectives 1 and 2)

Boat electrofishing is one of the few effective sampling methods for large rivers and was used for sampling areas upstream of river mile (RM) 20 to meet Fish Objectives 1 and 2. SRBPP project sites downstream of RM 20 were restricted from using any type of sampling method (i.e., electrofishing or trapping) to avoid mortalities of delta smelt. Sites were electrofished using an SR-18 electrofishing boat (Smith-Root, Inc., Vancouver, Washington). Sampling occurred on two occasions in 2011 (March 2-5 and April 18-22) to encompass outmigration timing of fall-run, late fall-run, spring-run, and winter-run Chinook salmon and steelhead (Snider and Titus 2000, Williams 2006). The first sampling event was initiated in response to an observed increase of catch per unit effort at the California Department of Fish and Game's (CDFG) rotary screw trap (RST) at Knights Landing in the Sacramento River at RM 87.7. The second sampling event was initiated in response to increases in turbidity at Knight's Landing RST and flows at Wilkins's Slough (WLK, RM 117.6, CDEC)

The number and location of electrofishing monitoring sites has varied since 2007 (Table 1). Seventeen of the 22 sites included in the 2011 sampling effort were located within SAM Reach 1b; the remaining five were located in SAM Reach 2. Similar to previous years (but with slight modifications to accommodate newly added sites) the study sites were categorized into Regions based on clusters of study sites or proximity by river mile, and were identified as: Knights Landing (RM 71 to 95), Garcia Bend (RM 40 to 70), Ko-ke (RM 21 to 39), and Rio Vista (RM 10 to 20).

Twenty sites were sampled in March and 21 sites in April (Table 1 and Figure 5). Fifteen of the sites had been included in prior assessments (2008-2010; H.T. Harvey & Associates and PRBO 2009, 2010, 2011), while the remaining seven sites had not been sampled previously. Sites added to the monitoring program in 2011 included one Undulating Bench Site (RM 72.2R), one Future SAM Repair Site (RM 57.2) and five Historic Repair Sites (i.e., pre-2006 historic levee repair sites; RM 24.0R, RM 38.7R, RM 56.0R, RM 85.6L, and RM 90.8L, Table 1). Most sites were sampled during both events; exceptions were sites 57.2R (sampled only in March), and 24.0L and 90.8L (sampled only in April). Sites ranged from approximately 352 to 709 feet in length.

The sampling approach for 2011 was designed to sample a greater proportion of each site than conducted in previous years. In lieu of the previous grid point sampling method, each sampling

Table 1. Location and characteristics of electrofishing sampling sites including SAM reach, region, site type, sample unit length (feet), sampling years, and number of points sampled at each site (ratio of site length to number of points in each sampling event) and repair design status (activated or not during sampling) for 2011 monitoring. Sites are identified by river mile (RM) and right (R) or left (L) bank.

Site (RM)	SAM Reach	Region	Site Type	Sample Unit Length (ft)	2007 and/ or 2008	2009 and 2010	2011			
							Number of points sampled (feet/point)		Repair Design Activated (Y/N/P)	
							March	April	March	April
91.9R	2	Knights Landing	Natural	586		X	38 (15.4)	16 (36.6)	-	-
90.8R			Natural	553		X	34 (16.7)	NV	-	-
90.8L			Historic	637			NS	NV	-	-
85.6R			No Bench	536		X	40 (1.4)	34 (15.8)	P	Y
85.6L			Historic	518			27 (13.6)	23 (22.5)	-	-
72.2R	1B		Bench	657	X		36 (18.3)	NV	P	Y
69.9R	1B	Garcia Bend	No Bench	644		X	31 (20.8)	NV	P	Y
69.1L			Natural	610		X	35 (17.4)	31 (19.7)	-	-
57.2R			Setback	528			26 (20.3)	NS	-	-
56.0R			Historic	581			20 (29.1)	21 (27.7)	-	-
50.3R			Natural	352		X	21 (16.8)	16 (22.0)	-	-
50.2L			No Bench	652	X	X	34 (19.2)	28 (23.3)	P	Y
47.9R			10:1 slope	592	X	X	37(16.0)	21 (28.2)	P	Y
47.0L			10:1 slope	669	X	X	31 (21.6)	25 (26.8)	P	Y
44.7R			10:1 slope	709	X	X	55 (12.9)	38 (18.7)	P	P
43.7R			10:1 slope	552	X	X	37 (14.9)	29 (19.0)	P	P
38.7R	1B	Ko-ket	Historic	579			21 (27.6)	18 (32.2)	-	-
34.5R			Wetland Tr.	585	X	X	45 (13.0)	24 (24.4)	N	P
26.9L			Wetland Tr.	463	X	X	35 (13.2)	32 (14.5)	P	P
25.0L			Natural	652	X	X	48 (13.6)	40 (16.3)	-	-
24.0R			Historic	677		X	46 (14.7)	15 (45.1)	-	-
24.0L			Natural	693		X	NS	31 (22.4)	-	-

Codes: Historic= pre-2006 levee repair; NS =not sampled; NV=no video; Y=yes, N=no, P=partial.

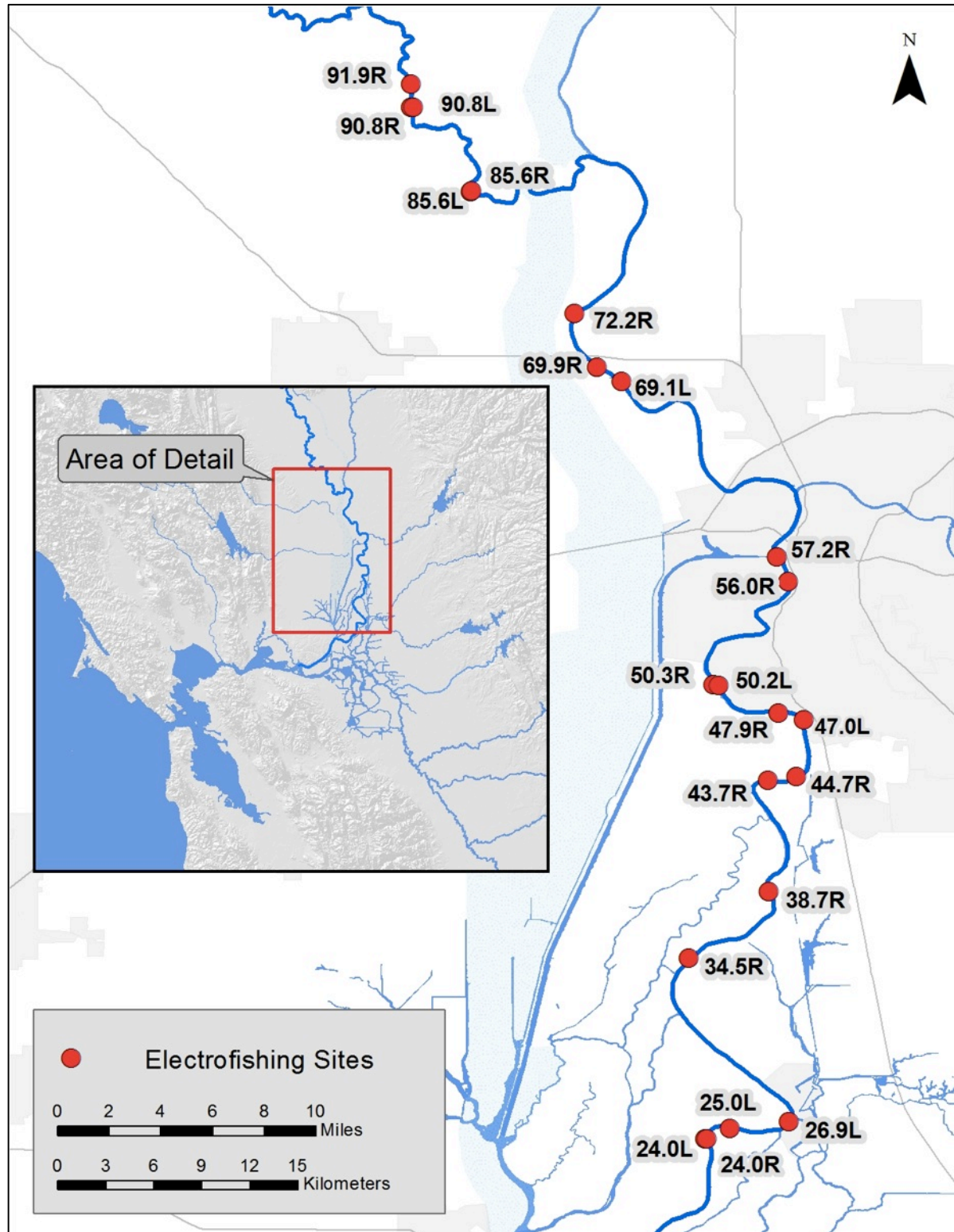


Figure 5. Location of 2011 electrofishing sites. Note that site 57.2R was only sampled in March and sites 24.0R and 90.8L were only sampled in April; all other sites were sampled during both events.

site was considered a unit and was electrofished in its entirety. At each site, sampling was initiated by approaching the shoreline with the electrodes activated for about 10-15 seconds, then turning off the electrodes while maneuvering away from the shoreline. This process was repeated multiple times, beginning at the downstream boundary of the sampling site and continuing with repeated approaches to the shoreline until the upstream boundary of the unit was reached.

Sampling in each unit was conducted as close as possible to the shoreline (1 – 5 meters; 3-16 feet) and each approach to the shoreline was designated as a sampling point. The number of points sampled in each sampling site varied between 15 and 48, depending on site and sampling event (Table 1).

While at some sites the design features intended to benefit fish were either partially or not activated (i.e., inundated) at the time of sampling, sampling occurred regardless of inundation level (Table 1). A partially activated site indicates that some, but not all, design features were inundated. For example, during the first sampling event, the bench at 72.2R was not inundated; however, as with most partially activated sites, the rip-rap bank along the periphery of the site contained designed habitat features (e.g., IWM) that were inundated. Design activation levels were determined through review and comparison of electrofishing GPS tracks and site photos taken during each electrofishing event with photos taken during SAM summer habitat surveys. For Wetland Trench Sites, the entire margin (outside of trench) was always sampled and sampling within the trench was conducted whenever possible; however, dense vegetation and shallow water sometimes prevented sampling within the entire trench and was limited to the inlets and outlets.

At each sampling site, voltage, pulse rate, and amperage ranged from low (50-500) to high voltage (100-1000), 30-60 pulses/s, and duty cycle of 20%-60%, respectively; starting voltages, pulse rates, and amperages were determined by final settings at the previous sampling site and adjustments were made based on conductivity or fish response. During electrofishing, a continuous track of the boat movement was recorded via GPS (Trimble; Geo XH 2005), and a video camera (with audio) was used to document activities. Sampling start and stop times were recorded.

Personnel, stationed at the front of the boat, captured fish via dip nets. Missed fish (i.e., fish that were observed but escaped the net) were noted, with fish species and approximate size recorded whenever possible. Fish collected were placed in an aerated livebox until fish processing (i.e., measuring and recording) was complete. A partition in the livebox allowed separation of potential predators from prey-sized fish to eliminate harmful interactions. Sampling was conducted continuously, and all captured fish were processed simultaneously following completion of sampling at each unit.

Fish Processing

During processing, all fish captured were identified to species when possible and measured to the nearest millimeter (total length, or fork length for species with forked tails, such as salmon and steelhead). Whenever large numbers of an individual species were captured during a single sampling event in a particular sample site, all fish were counted and identified, but measurements

were only collected from a random sub-sample of 50 individuals (for salmon and *O. mykiss*) and 20 individuals per species for non-salmonids. Sub-samples of salmonids were stratified by lifestage, with 50 measurements collected within each lifestage (i.e., fry <50 mm, parr=50-69 mm, smolts $\geq$ 70 mm for Chinook salmon; and fry <100 mm, juveniles 100-299 mm, and adults >300 mm for steelhead). Condition of all fish was rated as alive, injured, or mortality, and smolt index (seven category scale, where 1 = yolk-sac fry, 2 = fry, 3 = parr, 4 = silvery parr, 5 = smolt, 6 = mature adult, and IAD = immature adult; Interagency Ecological Program, unpublished) was recorded for all salmonids. Sensitive species/lifestages were anesthetized prior to processing using a solution of Tricaine methanesulfonate (Western Chemical, Inc.; Tricaine-S) at a 26.4 mg/L concentration, and were allowed to recover prior to release at the upstream boundary of the sampling unit.

Physical Characteristics of Each Site

In 2011, various environmental and physical parameters were recorded within each site to document the general physical characteristics present at each site during the electrofishing survey. Prior to electrofishing each site, environmental conditions were recorded immediately adjacent to the site in the mid-channel. After electrofishing each site, physical parameters were measured along several evenly spaced transects oriented perpendicular to the shoreline. A minimum of three transects were surveyed during the March sampling event but the minimum number of transects was increased to five for the April sampling event to ensure the variability of each site was adequately captured. At each transect, depth and mean column velocity were measured, and substrate type (sand/silt, gravel, or revetment) was recorded at 3 ft, 7 ft, and 12 ft (March) or 5 ft., 10 ft., and 15 ft. (April) from the bank for a total of 9-18 data points for each parameter at each site. The distance from shore of the transect points was extended during the April survey to more closely reflect the distance from shore where electrofishing was initiated during each approach. Depth and velocity measurements were averaged to estimate mean depth and velocity for each sampled unit. Substrate percent composition was calculated based on the 9-18 data points for each sampling unit. Measurement collections included the following:

- Water temperature and dissolved oxygen: measured in Celsius and mg/L, respectively, with an optical dissolved oxygen meter (YSI; ProODO).
- Conductivity and salinity: measured in μ S/cm and ppmS, respectively, with an Extech EC400 multi-range sensor.
- Water clarity: measured with a Secchi (depth in feet; LaMotte Company); and by water sample collected and analyzed immediately using a turbidimeter (Oakton; model T-100). Water samples were collected prior to electrofishing adjacent to site towards mid-channel to ensure samples were not biased due to substrate disturbance caused by boat wakes or other wave activity.
- Water depth: measured in feet with a marked polyvinyl chloride (PVC) rod.
- Mean column water velocity: measured in meters per second (m/s) with a digital flow probe (Marsh-McBirney or Global Water Instrumentation) at 0.6 of total depth in locations less than 2.5 ft; and average of readings at 0.2 and 0.8 of depth in locations 2.5 ft deep or more.
- Substrate type: characterized as sand/silt, gravel, or revetment by probing the substrate with a PVC rod; a ponar grab utilized during the first sampling event proved unreliable at

characterizing substrate.

- The presence/absence of aquatic vegetation was documented based on visual observations at transect points during the April survey (a ponar grab utilized during the first sampling event proved unreliable at characterizing vegetation). All other fixed structures (dock, pump, IWM, riparian vegetation, etc.) were recorded during summer habitat surveys. Although we intended to conduct complete habitat mapping of each sampling unit during the SAM summer habitat survey, we were unable to accomplish this task due to high flow conditions.
- The location of shoreline (water's edge) was recorded using a sub-meter GPS.

Fish Presence/Absence and Habitat Features at Sampling Points

Video/audio recordings and written notes were reviewed to determine fish presence and abundance (actual catch plus missed fish identifiable to species), and habitat features at each sampling point. Technical issues with the video camera system prevented recording of sampling sites 69.9R, 72.2R, 90.8L and 90.8R; therefore, presence/absence data was unavailable for these sites.

Habitat features at each sampling point were categorized as:

- Presence/absence of emergent structure (emergent vegetation and/or emergent woody material²);
- density of emergent vegetation and emergent woody material (none, low, medium, high);
- substrate (riprap versus mud/silt/sand).

4.2 Telemetry (Fish Objective 3)

Acoustic receivers were deployed by the USACE at 15 sites to meet Fish Objective 3 by detecting fish released during other studies in the Sacramento River and tributaries by members of the California Fish Tracking Consortium (CFTC), an allied group of researchers using acoustic telemetry to study fish movement patterns in the Central Valley. Some CFTC releases had the potential to encounter the USACE's array of receivers, and, by accessing the CFTC database, we were able to identify tags detected by the USACE array. The CFTC database identified (at a minimum) the species originally implanted with the tag, size at release, release date, and release location. Methods of receiver deployment and retrieval, data processing, and analysis of residency rates are described in more detail in the following sections.

Receiver Deployment and Retrieval

VEMCO 69KHz receivers were deployed at nine repair and six natural sites between RM 69.9 and RM 10.7 during late December/early January, and provided for continuous detection of

² Emergent vegetation and woody material were categorized based on reflected prevalence of sclerenchyma or collenchyma in the plant's supporting cell-structure where emergent vegetation was defined as any elastic plant structures with green coloration or light yellow coloration and emergent woody material was defined as any inelastic plant structures with grey, brown, or dark yellow colors.

tagged fish at these locations until the receivers were retrieved in mid to late April (Table 2). Receivers were not deployed in any of the historical repair sites. Specific receiver retrieval dates are not known at this time; therefore, retrieval was estimated based on dates of the last detections at each site. The detection ranges, depths, and distance to shore of each receiver are also unknown at this time.

Four receivers could not be retrieved and therefore no data is available from the receivers operated at sites 24.0L, 47.0L, 50.2L, and 50.3R. Tag detection data from the successfully retrieved receivers allowed for calculation of residency times at seven repair sites and at four natural sites.

Table 2. Location, site type and deployment/retrieval dates of VEMCO acoustic receivers deployed by the USACE during 2011. Asterisk indicates that specific receiver retrieval dates are not known at this time; therefore, retrieval was estimated based on dates of the last detections at each site, ND= no data, and NR=not retrieved.

Region	Site	Site Type	Receiver Number	Receiver Deployed	Receiver Retrieved*
Garcia Bend	69.9R	No Bench	104317	12/28/2010	4/26/2011
	69.1L	Natural	104318	12/28/2010	4/26/2011
	50.3R	Natural	104325	12/28/2010	NR
	50.2L	No Bench	104320	12/28/2010	NR
	47.9R	10:1 slope	104319	12/28/2010	4/14/2011
	47.0L	10:1 slope	104309	12/28/2010	NR
	44.7R	10:1 slope	104311	12/30/2011	4/14/2011
	43.7R	10:1 slope	104313	12/30/2011	4/14/2011
Ko-ket	25.0L	Natural	104315	12/30/2011	4/19/2011
	24.0L	Natural	ND	ND	NR
Rio Vista	16.9R	Natural	104310	1/6/2011	4/19/2011
	15.0R	Natural	104304	1/6/2011	4/19/2011
	12.7L	Bench	104308	1/6/2011	4/19/2011
	12.5L	Bench	104305	1/6/2011	4/18/2011
	10.7L	Wetland Trench	104306	1/6/2011	4/19/2011

Acoustic Data Processing

Data downloaded from the VEMCO receivers provided the tag code, date, and time that each acoustic transmitter was detected by each receiver. A total of 307 different transmitters were detected by the SRBPP receivers during the 2011 monitoring season, with a total of 80,204 individual detections. Of the 307 transmitters detected, 229 (74.6%) were successfully matched to the CFTC database, providing information on species, size at release, release date, and release

location. Transmitters detected by the receivers were from releases of tagged Chinook salmon (n=154, size range 140 mm to 180 mm), steelhead (n=61, size range 181 mm to 263 mm), green sturgeon (n=5, size range 1,570 mm to 1,870 mm), and striped bass (n=9, size range 245 mm to 625 mm), and a summary of the number of individuals of each species detected at each site is provided in Table 3.

Metadata from the CFTC database for these transmitters show that Chinook salmon were released at the Coleman National Fish Hatchery Weir in Battle Creek, in the Sacramento River at Jelly Ferry (RM 266.8), and in the Mokelumne River; steelhead were released in the Sacramento River at Jelly Ferry (RM 266.8) and Elk Landing (RM 70.4); green sturgeon were released at the confluence of Antelope Creek with the Sacramento River (RM 234); and striped bass were released in the Sacramento River at Brannan Island (RM 20.9) or Miller Park (RM 57.5), at the Lake Washington Outboard Motor Club in the Port of Sacramento Deep Ship Channel, in the San Joaquin River at Antioch (RM 5.1), and in sloughs bordering King Island.

Table 3. Summary of acoustic tag detections by site and by species during 2011.

Region	Site	Type	Receiver Number	Number of Tags Detected			
				Chinook Salmon	Steelhead	Green Sturgeon	Striped Bass
Garcia Bend	69.9R	No bench	104317	5	1	4	1
	69.1L	Natural	104318	30	16	3	2
	47.9R	10:1 slope	104319	57	27	4	2
	44.7R	10:1 slope	104311	110	46	3	2
	43.7R	10:1 slope	104313	92	39	3	4
Ko-ket	25.0L	Natural	104315	63	34	1	1
Rio Vista	16.9R	Natural	104310	59	33	1	1
	15.0R	Natural	104304	66	32	2	2
	12.7L	Bench	104308	75	28	4	3
	12.5L	Bench	104305	98	37	3	3
	10.7L	Wetland trench	104306	83	29	4	5

4.3 Habitat Monitoring (Habitat Objective 1)

Characterization of habitat features at seasonal water elevations and photo documentation of SRBPP long term monitoring sites was conducted to meet Habitat Objective 1. The principal focus of habitat monitoring was to typify habitat conditions at each site for comparison to previous surveys in order to evaluate the evolving conditions at each site over time and to provide input data for the SAM model to facilitate evaluation of the influence of repair design features on fish use.

Habitat conditions (bank substrate, aquatic vegetation, shade, instream woody material, and large

woody debris) were characterized in September 2011 according to protocols detailed in Normandeau Associates 2011, which primarily used SAM survey methods described in previous study plans and survey reports (SAM Habitat Survey) USACE 2004, 2006, 2008; USACE and CVFPB 2009; H.T. Harvey & Associates et al. 2008; H.T. Harvey & Associates and PRBO 2010).

Prior to 2011, SAM habitat measurements were collected from 19 to 21 SAM Repair and Natural Sites between RM 10.7 and RM 91.9. A list of twenty-five levee sites to be surveyed in 2011 was issued in March 2011 by the Interagency Work Group (IWG), which oversees this project. During the summer of 2011 and prior to the habitat surveys, ESA issues involving associated fish sampling on listed delta smelt (*Hypomesus transpacificus*) prohibited further biological sampling downstream of RM 20 and resulted in the elimination of several levee habitat monitoring sites in the lower river downstream of Isleton. Several additional sites were variously added and eliminated from the IWG list by the USACE (Brain Mulvey, personal communication). Based on these considerations, 19 Sacramento River levee sites, located between RM 24 and RM 92 were surveyed in the early fall 2011. These 19 sites correspond to all but four of the 2011 electrofishing sites identified in Table 1; non-surveyed electrofishing sites include three Historic Repair Sites (RM 24.0R, RM 38.7R, RM 56.0R) and one Setback Levee Site (RM 57.2R). One 10:1 Slope (RM 87.0R) that was not part of the electrofishing surveys was also added the habitat survey.

All sites were located using a GPS unit with sub-meter accuracy; latitude and longitude for each site are provided in Appendix A. During the first five days of habitat surveys, GPS data was collected using a GeoXH 2008-3000 sub-meter GPS with a Zephyr external antenna (Trimble Navigation Limited, Sunnyvale, CA). On the final day of the survey, a GeoXH 2005 sub-meter GPS with a Zephyr external antenna (Trimble Navigation Limited, Sunnyvale, CA) was used.

At each survey site, information on the bank substrate, and amount of instream woody material (IWM), aquatic vegetation, and overhanging shade was collected along two transects running parallel to the bank. These bank-line surveys were conducted along two established transects that represented the bank at the mean low water surface elevations and the mean high water surface elevations at that river location. Most locations had the transect endpoints marked with ½" inch rebar reference pins identified with aluminum tags, though at several of the sites only the high water transect was monumented and the low water transect was located using known distances from the high water reference points from previous survey notes. GPS coordinates were also available for each of the transect endpoints.

Occasional inconsistencies between site locations provided in previous reports and data received from USACE necessitated using several sources and some judgment to relocate site boundaries and habitat transects while in the field. The GPS locations provided in Appendix A reflect a compilation of data provided by USACE and points recorded in the field during the spring 2011 electrofishing surveys.

Photo Documentation

Photo documentation occurred during electrofishing and SAM Habitat Surveys in the spring and summer, respectively, as well as during establishment of permanent photo point monitoring locations in winter 2012.

During electrofishing surveys in March and April, several overlapping photographs were taken of each sampling site from the boat while drifting downstream positioned approximately mid-channel.

During the SAM Habitat Survey in September, each sampling site was photographed from four to six different points. Most photos were taken along the low water bank-line from the upstream, downstream, and middle portions of transects. Photos from 2011 were compared with those taken in 2008 (H.T. Harvey & Associates and PRBO 2011; Appendix B). However, previously established photo points (H.T. Harvey & Associates and PRBO 2011) were not recorded with sub-meter accuracy GPS units and were difficult or impossible to relocate; therefore, photo orientations are not exactly identical for many sites. For those that could be located, views were often obstructed in 2011 due to the dense riparian willow/cottonwood forest that has become established at most sites. Although photos may not be directly comparable, they nonetheless provide an indication of habitat changes over time.

Permanent photo point monitoring locations were established in February 2012 using a sub-meter GPS (Trimble; Geo-XH 2005 or Geo-XH 6000) and a laser rangefinder (Nikon Archers Choice) to document distance to shore. All GPS data was differentially corrected to further reduce positional error, converted to shapefile format and imported to GIS where additional relevant information was amended to the attributes of each position. A 12.1 megapixel/6X zoom digital camera (Cannon PowerShot G9) was used to take several overlapping images that encompassed the entirety of each site. Whenever possible, images were stitched together to produce a semi-seamless panoramic of individual sites. Steep banks and dense vegetation encountered at many SRBPP areas made traditional photo point monitoring impractical at many sites. Furthermore, channel widths encountered were typically too wide to permit establishment of photo points on opposing banks due to equipment focal limitations. It was determined that photos taken from the boat, mid-channel, would better document site conditions under most circumstances. Clear and repeatable photo perspectives at all but five sites were only obtainable when taken from the boat, mid-channel. At one of the five sites (90.8L) not taken from the boat, photos were taken from within the project site looking upstream and downstream from SAM mean high/low water transects end points. At the remaining four sites, photos were taken from the opposing bank. For land-based photo sites, a tripod was used to steady the camera while several consecutive, overlapping, images were taken of the target site. At project sites where land-based photo points were not practical, the boat was positioned approximately mid-channel, at a point near the longitudinal center of the site that afforded clear visibility of the sites upstream and downstream bounds. At some sites, the orientation of the sun created glare that necessitated boat positioning in a manner different than previously described, generally closer towards either the upstream or downstream end of the project site.

5.0 DATA REPORTING AND ANALYSIS

5.1 Fisheries Monitoring (Fish Objectives 1 and 2)

Steelhead were not captured in sufficient numbers ($n=7$) during 2011 to warrant their inclusion during analysis; therefore, only Chinook salmon were evaluated for investigations pertaining to salmonids. Also, due to limited data, the repair type “Setback Levee” was excluded from all analyses.

5.1.1 Species Captured (Fish Objective 1) and Fish Community Abundance and Composition (Fish Objective 2)

In order to determine the fish species present at selected Natural and Repaired Sites, the total number of different fish species recorded (captured and missed) are identified. Size (length) distributions of captured focus fish and predatory species were examined via histograms.

In order to evaluate fish community abundance and composition at selected natural and repaired sites, community composition at four different types of SAM Repair Sites (10:1 Slope, Bench, No Bench, and Wetland Trench), Natural Sites, and Historic Repair Sites (i.e., six site types) were recorded and quantified. Similarities between community composition at these six different sites were examined for all pair-wise comparisons with Morisita’s Index (Morisita 1959), as revised by Horn (1966) calculated using the *vegdist* function in the R package “vegan” version 2.0-2 (RDCT, 2008).

$$C_{\lambda} = \frac{2\Sigma(X_{ij} * X_{ik})}{(\lambda_j + \lambda_k)(\Sigma X_{ij} * \Sigma X_{ik})}$$

Where C_{λ} is the index of similarity between site type j and k , X is the number of individuals of species i at site type j and k , and

$$\lambda_j = \frac{\Sigma X_{ij}^2}{(\Sigma X_{ij})^2} \quad \lambda_k = \frac{\Sigma X_{ik}^2}{(\Sigma X_{ik})^2}$$

Total capture for each species was converted to proportions of the total combined capture for all species at each site type before calculating the indices. For the first assessment, the catch frequency data were combined for both sampling events (i.e., March & April) and for all sites within a site type. Indices were calculated to examine similarities between combined catch data from all four repair types (i.e., All Repair = No Bench, Bench, Wetland Trench, 10:1 Slope) and either Natural or Historic. Then, to compare species composition between sampling events, Morisita-Horn indices were calculated between months for each of the six site types. An index value of 0 indicates completely distinct samples (with no species in common) and an index of approximately 1 indicates that the species composition of the samples is identical (Horn 1966). The community composition indices are parameters to measure similarity, and should not be interpreted as statistical comparisons (Horn 1966).

5.1.2 Salmon Presence and Abundance (Use) at Sites and Microhabitats within Sites, and Relative Risk of Predation to Juvenile Salmon at Sites (Fish Objective 2)

In order to determine Chinook salmon abundance at Repair Sites and microhabitats within sites, we examined the association between abundance and a suite of habitat and environmental variables through Generalized Linear Model (GLM) analyses. Since Bench and 10:1 design types are similar (all are bench-type designs) and there was only one Bench Site, we combined these types and used the term “10:1 Slope/Bench” in our analysis.

GLMs were used to analyze the catch data from the two electrofishing events in 2011, with the intention to

- (1) determine the difference in Chinook salmon abundance among the different repair types (with repair type as the categorical predictor variable).
- (2) assess whether specific microhabitat features (such as flow, depth, density of emergent vegetation, shade.) are associated with Chinook salmon abundance.
- (3) investigate the relative risk of predation to juvenile salmonids at various Repair Sites and its relationship to various habitat characteristics.
- (4) determine if size distributions of Chinook salmon (FL) found at different site types are related to repair type and microhabitat

The models used in this analysis often included an interaction term between two or more of the explanatory variables, generally denoted by “x” in model descriptions (e.g. flow x depth). In this analysis, both numerical variables (e.g. counts, depths, percent shade...) and categorical variables are used in model fitting. Categorical variables are represented in the model by a dummy variable.

The process of model fitting and selection can be best described as a three step process:

- During first step of data analysis, models were developed that incorporate the effects of various spatial and temporal components (e.g. river region, month) as well as the different design types (described earlier in this report) on the abundance of focal fish species. Habitat Features (such as vegetation, IWD or shade) are not included in this first step to avoid confounding their effects with those of design type, which complicates the interpretation of the model output.
- During the second step of data analysis, a different set of models was developed, incorporating the effects of different Habitat Features in addition to temporal and spatial variables. Repair Types were not included in these models to independently interpret habitat effects on fish abundance.

- Effects of both, different Repair Types and Habitat Features were included in the third set of models (in addition to the spatial and temporal variables).

Through this process significant differences in fish distribution can be discerned, and relationships between fish abundance and habitat types (or habitat features) can be detected, while controlling for the confounding effects of variables not related thereto (such as sampling events or river region). When investigating the effects of habitat features, a suite of models is fit to the data, including a broad variety of predictor categories and their interactions.

Note that for the purpose of this report, catch data for the March and April sampling events were combined in all models, but month was retained as a categorical predictor variable in all model fits to account for abundance variation related to the timing of the sampling events. Furthermore, while a variety of piscivorous species were encountered during these surveys, black bass (*Micropterus spp.*), striped bass (*Morone saxatilis*), and Sacramento pikeminnow (*Ptychocheilus grandis*) were combined under a single category in one of the models relating to predator abundance.

Model Summary

Site Type

A GLM model of fry and juvenile abundance (fry < 50mm FL < juvenile, residuals follow a Poisson distribution) was first used to assess whether abundance is affected by Site Type. Site Types used in this model include SAM Repair Sites (categories: 10:1 Slope/Bench, No Bench, Wetland Trench [WLT]), Historic Repair Sites, and Natural Sites. To determine if Site Type has a significant effect on fry abundance after accounting for variability introduced by other spatial and temporal variable, this model also included

- Region (Ko-Ket, Garcia Bend, and Knights Landing)
- Month (sampling was conducted in both, March and April)
- Region x Month interaction
- Site Type
- Site Type x Month interaction

SAM Repair Type

A GLM model of was then used to assess whether abundance of Chinook salmon fry and/or juveniles, and predator abundance is affected by SAM Repair Site Type. SAM Repair Site Types used in this model include SAM Repair Sites (categories: 10:1 slope, Bench, No Bench, Wetland Trench [WLT]), and Natural Sites. To determine if SAM Repair Site Type has a significant effect on fry abundance after accounting for variability introduced by other spatial and temporal variable, this model also included

- Region (Ko-Ket, Garcia Bend, and Knights Landing)
- Month (sampling was conducted in both, March and April)
- Region x Month interaction

Microhabitat

The GLM models used to evaluate the use of different microhabitats by Chinook salmon and fish predators accounted for the following habitat variables:

- Flow (mean flow in cfs at the sampling site)
- Depth (mean depth at the sampling site in feet)
- Shade (present/absent)
- Emergent Vegetation (absent, low/medium/high density)
- Emergent Woody Material
- Terrestrial Vegetation (percent coverage)
- Terrestrial Woody Material (percent coverage)

To determine whether microhabitat has a significant effect on fry abundance after accounting for variability introduced by other spatial and temporal variable, this model also included

- Region (Ko-Ket, Garcia Bend, and Knights Landing)
- Month (sampling was conducted in both, March and April)
- Region x Month interaction

The habitat variables analyzed in this report largely overlap with those included in analyses conducted in previous years (H.T. Harvey & Associates and PRBO 2011), yet differences in habitat measurements only allow for comparison and interpretation of results rather than a combination of data across years and subsequent meta-analysis. For example, analyses of fish abundance in this report include shade as a categorical variable, yet in past years the percentage of shade cover was used in analyses. Furthermore, some habitat features included in past analyses (such as submerged vegetation or substrate composition) were not measured in during the 2011 survey. Also, even though point measurements of physical habitat (e.g. IWM diversity and dominant IWM size) are available from past surveys, these measurements do not directly correspond to points sampled during 2011. Therefore, inclusion of these habitat measurements in this analysis would render the validity of subsequent inferences questionable.

However, this analysis does incorporate effects of some variables not taken into account previously (e.g., flow and density of emergent woody material). Ultimately, while some differences exist between data collection and analysis in 2011 compared to years past, this report offers valuable insights on the overall effectiveness of various bank repair types and a suite of biologically relevant habitat features, and allows for the interpretation of relevant results in a broader temporal framework.

5.2 Telemetry (Fish Objectives 3)

Residence time was expressed in terms of hours, and was estimated as the difference between the first and last times of detection of a transmitter at a given site. An interval of one hour between the first and last detections was selected as the minimum criteria for determining residency, which is consistent with the approach used in analyzing the 2009 and 2010 acoustic telemetry data (H.T. Harvey & Associates and PRBO 2011). When the elapsed time between detections was less than one hour, an individual was considered to be passing through the site.

Several assumptions underlie the interpretation of 2011 detection data. First, the detection patterns for a given transmitter were assumed to be representative of the species tagged. However, predation is known to be a confounding factor in interpreting acoustic tag data in the Central Valley, and it is possible that tagged salmon and steelhead may have been consumed by predators. When this occurs, tag detections subsequent to predation are representative of the behavior of the predator that consumed the tagged individual rather than of the salmon or steelhead that was tagged.

Secondly, large time intervals (days or weeks) between detections of an individual tag at a given site introduce ambiguity to the interpretation of these detections, as it is unknown whether the individual left the site and subsequently returned, or if it remained at the site but out of detection range of the receiver. Since the detection ranges of the receivers are not known at this time, the potential for an individual to remain undetected while onsite could not be assessed. In some instances it is clear that the individual left and then returned to the site, as detections were made by receivers at other sites, which most commonly occurred when receivers were located at sites nearby. However, it remains unclear whether individuals remained at these sites during prolonged periods between detections, as no receivers were located nearby. For the purpose of this analysis it was assumed that an individual remained at the site during these gaps unless it was detected at another location. If this assumption is incorrect, residency durations for sites with a long distance to the next receiver (i.e., sites 47.9R and 25.0L) may be overestimated.

Furthermore, data indicates that timestamps of detection for one or more of the receivers may be inaccurate. On several occasions individuals were detected at different sites within very short time periods (seconds to minutes). The minimum distance between receivers with known coordinates was 0.4 miles (sites 12.5L and 12.7L, Table 4). The short distance between these two stations (assuming nearly overlapping detection ranges) may explain some of these unusual detection patterns, however, similar patterns were observed between site 10.7L and sites 12.5L (or 12.7L). In these instances, it is clear that fish could not have moved between the receivers (over distances greater than 2 miles) as rapidly as the detection data indicate. Based on this information, it appears that the timestamps for data collected by one or more of the receivers may have been inaccurate. Since this could neither be confirmed nor corrected, the subsequently reported residency durations are based on the detection data provided.

5.3 Habitat Monitoring (Habitat Objective 1)

In order to characterize the habitat features located at various seasonal water elevations at selected bank protection and reference sites, data was collected using a repeatable survey design and in a format that can be used to model focus fish species response per the SAM model requirements. SAM Modeling was not performed but may be conducted by others using the data collected. Also, permanent photopoints were established at select sites and panoramic photos were recorded.

5.3.1 Characterize Habitat Features at Seasonal Water Elevations

A number of comparisons were made for many of the habitat features (bank substrate, aquatic vegetation, shade, instream woody material, individual large woody debris pieces) between Site

Table 4. Distances in miles between acoustic receivers during 2011. Brackets indicate distances that were estimated since the coordinates of the receiver were unknown.

	47.9R	44.7R	43.7R	25.0L	16.9R	15.0R	12.7L	12.5L	10.7L
69.9 R	[12.0]	[15.2]	[16.2]	[24.9]	[53.0]	[54.9]	[57.2]	[57.4]	[59.2]
69.1L	27.9	[14.4]	[15.4]	[24.1]	66.6	69.0	72.0	72.4	74.7
47.9R	-	[2.2]	[4.2]	[22.9]	38.7	41.1	44.1	44.5	46.8
44.7R	-	-	[1.0]	[19.7]	[27.8]	[29.7]	[32.0]	[32.2]	[34.0]
43.7R	-	-	-	[18.7]	[26.8]	[28.7]	[31.0]	[31.2]	[33.0]
25.0L	-	-	-	-	[8.1]	[10]	[12.3]	[12.5]	[14.3]
16.9R	-	-	-	-	-	2.4	5.4	5.8	8.1
15.0R	-	-	-	-	-	-	3	3.4	5.7
12.7L	-	-	-	-	-	-	-	0.4	2.7
12.5L	-	-	-	-	-	-	-	-	2.3
10.7L	-	-	-	-	-	-	-	-	-

Types. Comparisons were based on data collected, as well as descriptive statistics performed on the habitat features measured.

5.3.2 Photo Documentation

Comparison photos from 2008 and 2011 are presented to document habitat changes over time. Panoramic photos taken for the first time in 2011 are provided for establishing baseline photos of sites in their entirety; these types of photos will augment individual photos taken from multiple perspectives.

6.0 RESULTS

6.1 Fisheries Monitoring (Fish Objectives 1 and 2)

Two sampling events were conducted in 2011: March 2-5 and April 18-22. The data obtained from electrofishing efforts at each sampling site included environmental conditions (Appendix C), physical parameters (Appendix D) and species counts (Appendix E).

Fish species presence (Fish Objective 1) and fish community abundance and composition (Fish Objective 2) are presented in section 6.1.1. Models used to evaluate habitat preference of Chinook salmon and piscivorous predators and subsequent relevant interpretations (Fish Objective 2) are presented in section 6.1.2.

6.1.1 Species Captured (Fish Objective 1) and Fish Community Abundance and Composition (Fish Objective 2)

A total of 24 fish species were collected in the two sampling events (Table 5). In both sampling events combined, focus fish species recorded (captured and missed) included 529 Chinook salmon (404 juveniles and one adult captured and 124 juveniles missed), seven juvenile steelhead, and six adult delta smelt. Predatory species included 136 black bass, eight striped bass, 79 Sacramento pikeminnow, and 63 inland silverside³. Species counts for each sampling site and event is provided in Tables E-1 and E-2 in Appendix E. Brief summaries for fish species categories (focus, predator, and other species) are presented below.

Focus Fish Species

In March, a total of 131 juvenile (67 fry and 64 parr/smolt) and one adult Chinook salmon (FL 675 mm) were captured, and 28 parr/smolt evaded capture. Assignment of measured Chinook salmon to race (using Fisher race tables) indicated that 27% (n= 36) were spring-run and 70% (n= 93) were fall-run fish, and 1% (n=2) were winter-run (Table E-3 in Appendix E). The race of the single adult Chinook salmon could not be determined. Approximately 51% (n=67) of measured Chinook salmon were fry, 44% (n=57) were classified as parr, and 5% as smolt (n=7). Mortality rates were low during this survey, with an overall rate of 3.8% (five individuals) for juvenile Chinook salmon (2.7% and 4.4% for spring- and fall-run, respectively). Detailed length-frequency distributions are depicted in Figures 6 and 7.

In March, six juvenile steelhead were captured with an average length of 184 mm (range: 112-230 mm FL). Five of the steelhead were of hatchery origin. No mortalities occurred. No delta smelt were captured during the March sampling event.

In April, a total of 273 juvenile (18 fry and 255 parr/smolt) Chinook salmon were captured and 136 parr/smolt Chinook salmon evaded capture. The majority of “missed” fish were observed at sites 24.0L (41%), 43.7R (13%), 47.0L (13%), 34.5 (9%) and 26.9L (8%). Based on the Fisher race tables, 90% of measured Chinook salmon were fall-run fish with the remainder designated as spring-run (Table E-3 in Appendix E). Approximately 7% of the Chinook salmon measured in April were classified as fry (n=18), 59% (n=160) classified as smolts, and 35% (n=95) as parr. There were 25 Chinook salmon mortalities (all fall-run fish), with an overall mortality rate of 9.1%. Detailed length-frequency distributions are depicted in Figures 6 and 7.

In April, one steelhead and five delta smelt were also captured. The steelhead measured 42 mm FL, while the average length of delta smelt was 75 mm (range: 70 to 80 mm TL). No mortalities occurred.

³ Inland silversides are potential predators of delta smelt eggs and/or larvae. However, sampling was not conducted in delta smelt habitat downstream of RM 20, so these fish are unlikely to function as predators in the sampling reach. Nonetheless, data was provided regarding their presence in sampled sites.

Table 5. Summary of all species captured during 2008-2011 including individual counts (N) and mean total lengths in millimeters. Sampling methods and number of sampling events varied per year (grid-point sampling in 2009 and 2010 and continuous unit sampling in 2011; three events in 2009, three events in 2010, and two events in 2011). Note: counts of some species may differ from those analyzed due to exclusion of counts from the single Setback Levee Site (RM 57.2R).

Common (Scientific) Names	Native Status (N)/ I)	Counts			Mean Total Lengths		
		2009	2010	2011	2009	2010	2011
Focus Species							
Chinook salmon (<i>Oncorhynchus tshawytscha</i>)	N	158	464	405	64	61	71
Steelhead/rainbow (<i>Oncorhynchus mykiss</i>)	N	6	9	7	200	227	175
Delta smelt (<i>Hypomesus transpacificus</i>)	N	0	2	6	-	76	80
Predatory Species							
Sacramento pikeminnow (<i>Ptychocheilus grandis</i>)	N	25	52	79	155	177	142
Striped bass (<i>Morone saxatilis</i>)	I	4	10	8	260	236	423
Black bass-Large and smallmouth (<i>Micropterus salmoides/ dolomieu</i>)	I	99	137	131	211.5	145.5	143
Inland silverside (<i>Menidia beryllina</i>)	I	96	160	63	74	61	63
Other species							
Bigscale logperch (<i>Percina macrolepida</i>)	I	1	0	0	85	-	-
Black bullhead (<i>Ameiurus melas</i>)	I	0	1	0	-	169	-
Black crappie (<i>Pomoxis nigromaculatus</i>)	I	1	1	0	174	166	-
Bluegill sunfish (<i>Lepomis macrochirus</i>)	I	14	75	14	92	70	86
Brown bullhead (<i>Ameiurus nebulosus</i>)	I	5	0	0	83	-	-
California roach (<i>Lavinia symmetricus</i>)	N	0	0	2	-	-	58
Common carp (<i>Cyprinus carpio</i>)	I	11	12	12	485	488	390
Channel catfish (<i>Ictalurus punctatus</i>)	I	2	9	0	200	306	-
Golden shiner (<i>Notemigonus crysoleucas</i>)	I	5	13	3	77	52	88
Goldfish (<i>Carassius auratus</i>)	I	9	5	0	233	176	-
Green sunfish (<i>Lepomis cyanellus</i>)	I	5	19	3	106	92	121
Hitch (<i>Lavinia exilicauda</i>)	N	3	5	3	274	250	100
Western mosquitofish (<i>Gambusia affinis</i>)	I	2	3	0	34	30	-
Pacific lamprey (<i>Lampetra tridentata</i>)	N	1336	811	1	n/a	n/a	-
Prickly sculpin (<i>Cottus asper</i>)	N	0	0	11	-	-	92
Pumpkinseed sunfish(<i>Lepomis gibbosus</i>)	I	16	0	0	121	-	-
Redear sunfish (<i>Lepomis microlophus</i>)	I	0	15	9	-	159	117
Red shiner (<i>Cyprinella lutrensis</i>)	I	0	0	1	-	-	54
Sacramento blackfish (<i>Orthodon microlepidotus</i>)	N	0	15	3	-	426	425
Sacramento splittail (<i>Pogonichthys macrolepidotus</i>)	N	1	22	5	290	341	369
Sacramento sucker (<i>Catostomus occidentalis</i>)	N	33	11	43	170	266	147
Spotted bass (<i>Micropterus punctulatus</i>)	I	0	1	0	-	245	-
Threadfin shad (<i>Dorosoma petenense</i>)	I	1	1	4	103	84	101
Tule perch (<i>Hysterocarpus traski</i>)	N	310	376	38	113	112	99
Warmouth (<i>Lepomis gulosus</i>)	I	1	14	3	118	63	77
White catfish (<i>Ameiurus catus</i>)	I	10	38	2	195	224	197
Yellow bullhead (<i>Ameiurus natalis</i>)		2	0	0	58	-	-

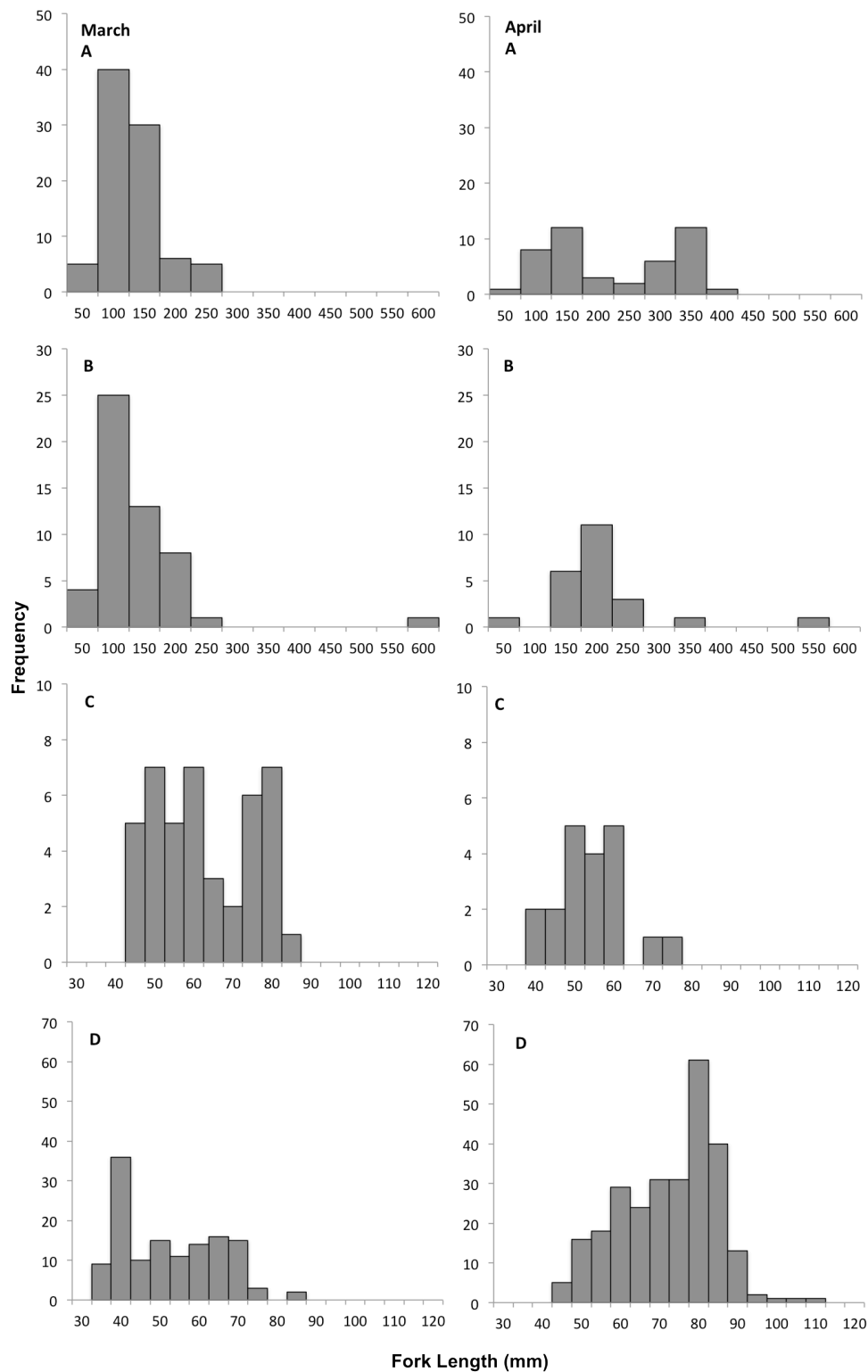


Figure 6. A comparison of length-frequency histograms between sampling events (March on left, April on right) for A) black bass, B) Sacramento pikeminnow, C) inland silversides and D) Chinook salmon. Catch data from five site types were combined (i.e., Historic Repair, Natural, 10:1 Slope/Bench, No Bench and Wetland Trench) within sampling events.

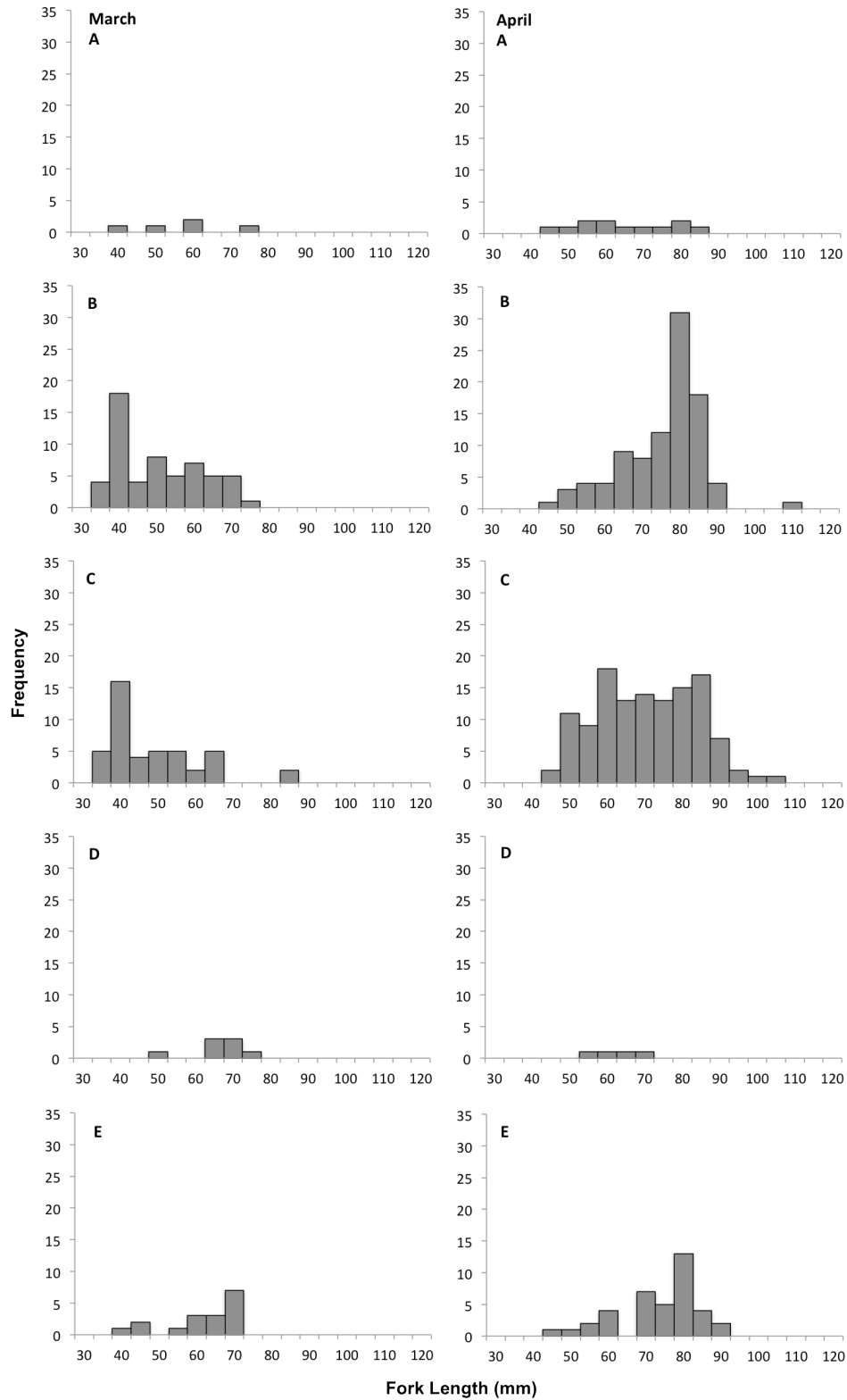


Figure 7. A comparison of length-frequency histograms for Chinook salmon between sampling events (March on left, April on right) and by site type: A) Historic Repair, B) Natural, C) 10:1 Slope/Bench, D) No Bench and E) Wetland Trench.

Predatory Fish Species. In March, a total of 185 predatory fish were captured: 56 Sacramento pikeminnow, 43 inland silverside, 83 black bass, and zero striped bass. In April, a total of 96 predatory fish were captured: 23 Sacramento pikeminnow, 20 inland silverside, 45 black bass, and 8 striped bass. Length frequency distributions by site type and sampling event for bass predators and combined piscivores are depicted in Figures 6-9.

Other Fish Species. In March, a total of 77 fish belonging to other species were captured in March. Two fish were unidentifiable. In April, a total of 81 fish belonging to other species were captured.

Morisita-Horn Indices

Morisita-Horn indices for pair-wise site type comparisons indicate that species composition was very similar in Natural, Wetland Trench, and 10:1 Slope Sites (Table 6)⁴. In general, index values were high, indicating a large degree of overlap or similarity between site types. The fish community at Historic Repair sites was markedly different from that at Natural sites, and most similar to No Bench sites.

Results indicated that Natural Sites were more similar to combined SAM Repair Sites compared to the Historic Sites (Table 7). These results reflect the higher proportion of black bass captured at the Historic Sites in comparison with Combined Repair and Natural Sites (Table 8). In addition, Chinook salmon constituted a much lower proportion of catch at the Historic Sites in comparison to 10:1 Slope and Natural Sites. The proportions of key species (i.e., black bass and Chinook salmon) were most similar between Natural and Combined Repair types, contributing greatly to their degree of community overlap and outweighing site specific differences in the presence of less abundant species.

A comparison of species composition between sampling events (i.e., March and April) indicates that species composition within the No Bench type was least consistent through time, while community composition of the Natural and Wetland Trench sites was characterized by temporal stability (Table 9).

⁴ NOTE: The community composition indices are parameters of similarity and indicate the degree of overlap between two samples; the parameters should not be interpreted as statistical comparisons (Horn 1966).

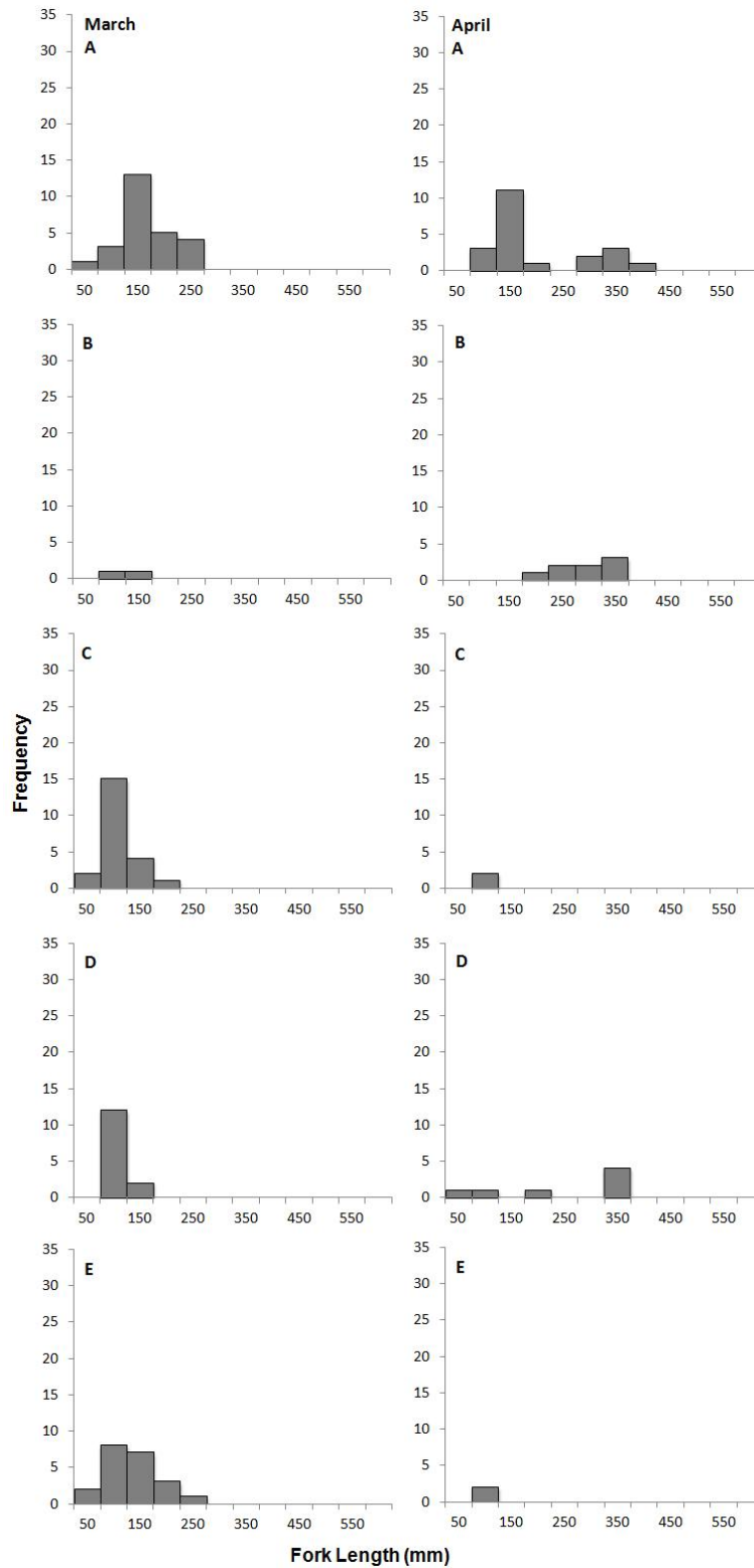


Figure 8. A comparison of length-frequency histograms for Black Bass between sampling events (March on left, April on right) and by site type: A) Historic Repair, B) Natural, C) 10:1 Slope/Bench, D) Wetland Trench and E) No Bench.

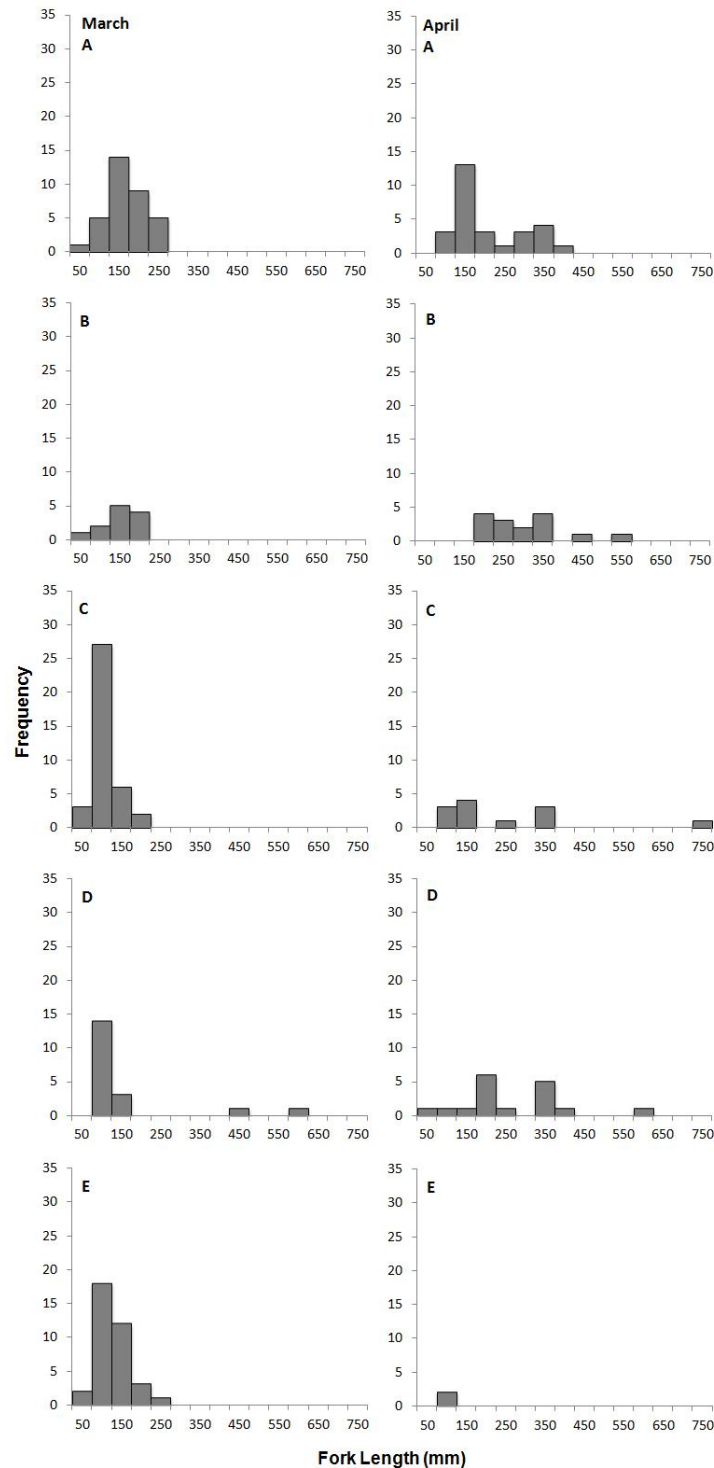


Figure 9. A comparison of length-frequency histograms for piscivorous species between sampling events (March on left, April on right) and by site type: A) Historic Repair, B) Natural, C) 10:1 Slope/Bench, D) Wetland Trench and E) No Bench.

Table 6. The Morisita-Horn index of similarity of species composition between pairs of site types. The index is 0 when the samples have no species in common and 1 when the samples have the same proportional species composition.

	10:1 Slope	Historic Repair	Natural	No Bench	Bench
Historic Repair	0.42	-	-	-	-
Natural	0.98	0.35	-	-	-
No Bench	0.42	0.86	0.4	-	-
Bench	0.82	0.62	0.79	0.65	-
Wetland Trench	0.96	0.59	0.93	0.59	0.88

Table 7. The Morisita-Horn index of similarity of species composition between pairs of general site types. The All Repair category includes the combined catch data from four site types: No Bench, Bench, Wetland Trench, 10:1 Slope. The index is 0 when the samples have no species in common and 1 when the samples have the same proportional species composition.

	All SAM Repairs	Historic Repairs
Historic Repairs	0.58	
Natural	0.93	0.35

Table 8. Total (March and April) capture by species and site type.

Species	Historic	Natural	10:1 Slope	No Bench	Bench	Wetland Tr.	All SAM Repairs
Black bass	48	10	23	23	4	22	72
Bluegill sunfish		2	6		3	2	11
California roach				2			2
Chinook salmon	17	152	152	12	15	56	235
Common carp	2	3	1	4	2		7
Delta smelt			5			1	6
Golden shiner	2					1	1
Green sunfish	1			2			2
Hitch		3					
Inland silverside	3	39	12	4		5	21
Prickly sculpin	1		5	3		2	10
Readear sunfish	1		2	3		2	7
Red shiner		1					
Sacramento blackfish	1		1			1	2
Sacramento pikeminnow	14	16	11	15	9	12	47
Sacramento sucker	24	3	7	3	5	1	16
Splittail			5				5
Steelhead		3		2	1	1	4
Striped bass	1	1	4			2	6
Threadfin shad			2	2			4
Tule perch	14	4	2	17	1		20
Warmouth		1	2				2
White catfish	1	1					
Grand Total	130	241	240	92	40	108	480

Table 9. Morisita-Horn index of similarity of species composition between sampling events (i.e., March vs. April) at each site type. The index is 0 when the samples have no species in common and 1 when the samples have the same proportional species composition.

Historic Repair	Natural	10:1 Slope	No Bench	Bench	Wetland Trench
0.34	0.04	0.24	0.55	0.23	0.15

6.1.2 Salmonid Presence and Abundance (Use) at Sites and Microhabitats, and Relative Risk of Predation to Juvenile Salmonids at Sites (Fish Objective 2)

Count data used in these analyses contains 1165 sampling points, representing samples for 21 sampling sites (Setback Levee Sites excluded due to limited data) and six different site types. A summary of results relevant to the evaluation of habitat use by Chinook salmon and predator fish is presented below. An underlying assumption for interpretation of these results is that fish abundance is reflective of habitat preference.

While a number of biologically important measurements were evaluated in these analyses, a multitude of additional factors may influence fish abundance and habitat preference. In the subsequent interpretation, significant effects having negative coefficients are considered to indicate lower fish abundance than the Reference Category, while those having positive coefficients indicate higher fish abundance than the Reference Category and possibly indicate habitat preferences.

Site Types

In the first step, the model incorporated the effect of Site Type (SAM Repair, Historic Repair, and Natural) to determine whether different types of repairs (i.e., Historic versus SAM Repairs) influence Chinook salmon fry and juvenile abundance in comparison with each other and with Natural Sites.

The predictive equations for evaluating the effect of Site Type on Chinook salmon fry and juvenile abundance were in the general form

$$\log(c) = \beta_0 + \beta_1 RegionA + \beta_2 RegionB + \beta_5 Month_{March} + \dots \\ \beta_6 SiteA + \beta_7 SiteB + \beta_{8:11} Region \times Month$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to River Regions Garcia Bend (RM40-70) and Knights Landing (RM70-95). Ko-Ket (RM 21-39) was used as the reference region. *Month* represents the sampling time, Sites A and B correspond to the Site Types not used as the reference category in each respective model (see Table 10 for a summary results of all models), and *Region x*

Month refers to the interaction between region and sampling month (two combinations, excluding the reference region).

Coefficients for the different Sites Types (regardless of the reference category used in the model) indicate that abundance of both Chinook salmon fry and juveniles was significantly lower at Historic Sites (Table 10). Therefore, data from Historic Sites was excluded from further analysis to focus on the parameters of main interest in this study.

Table 10. Influence of combined Site Type (SAM Repair, Historic Repair, and Natural) on abundance of Chinook salmon (fry and juvenile) at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” ($p<0.05$), “” ($p<0.01$), and “***” ($p<0.001$).**

Reference Category	Site Type	Fry			Juvenile		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
Historic	SAM R.	1.17	0.27	***	1.66	0.35	***
	Natural	1.07	0.28	***	2.08	0.35	***
SAM Repair	Historic	-0.89	0.29	**	-1.66	0.35	***
	Natural	0.79	0.26	**	0.42	0.16	**
Natural	Historic	-1.07	0.28	***	-2.08	0.35	***
	SAM R.	0.10	0.14	NS	-0.42	0.16	**

SAM Repair Site Types

The abundance models for Chinook salmon fry and juveniles were then used to assess whether individual SAM Repair Site Types (10:1 Slope/Bench, No Bench, WLT) influence Chinook salmon fry and/or juvenile abundance. Natural Sites were used as the reference category in all model fits.

The predictive equations for evaluating the effect of SAM Repair Site Types on Chinook salmon abundance were in the general form

$$\log(c) = \beta_0 + \beta_1 RegionA + \beta_2 RegionB + \beta_3 Month h_{March} + \dots \\ \beta_4 TypeA + \beta_5 TypeB + \beta_6 TypeC + \beta_{7:8} Region \times Month h$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]), *Month* represents the sampling time, Types A through C correspond to repair types 10:1 Slope/Bench, No Bench and WLT, respectively, and *Region x Month* refers to the interaction between region and sampling month (two combinations, excluding the reference region). Catch of Chinook salmon fry and juveniles (March and April combined) is illustrated in Figures 10 and 11, respectively.

Fry

In this model, two repair types appeared to significantly influence Chinook salmon fry abundance compared to Natural Site Types; the 10:1 Slope/Bench Site Type was associated with increased Chinook salmon abundance, while the No Bench Site Type was associated with reduced abundance of fry abundance.

Juveniles

Similar to the preceding model, this model predicts higher Chinook salmon juvenile abundance at the 10:1 Slope/Bench Site Type. The WLT Site Type also had significant effects in this model, and was associated with reduced abundance for juvenile Chinook salmon (Table 11).

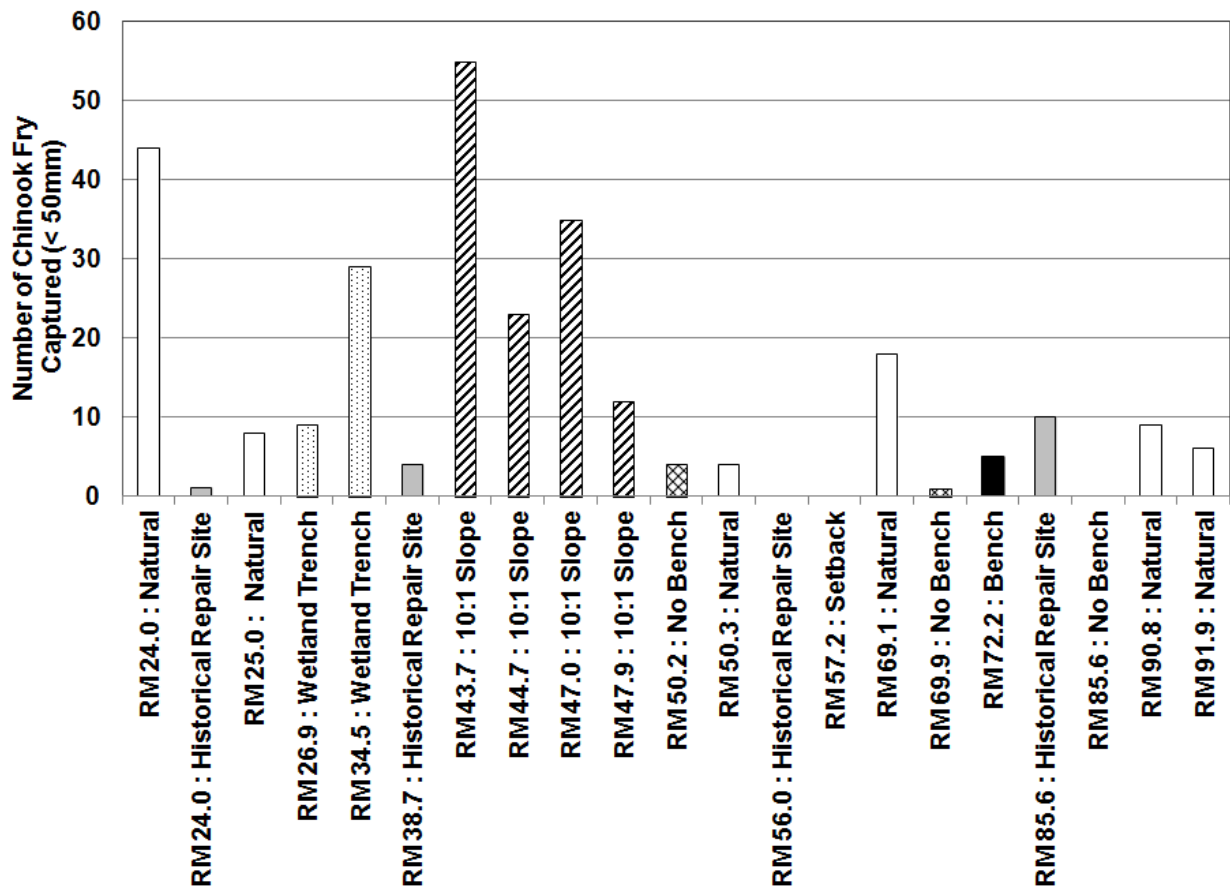


Figure 10. River Mile, Site Type, and Number of Chinook salmon fry captured during the March and April 2011 Electrofishing surveys.

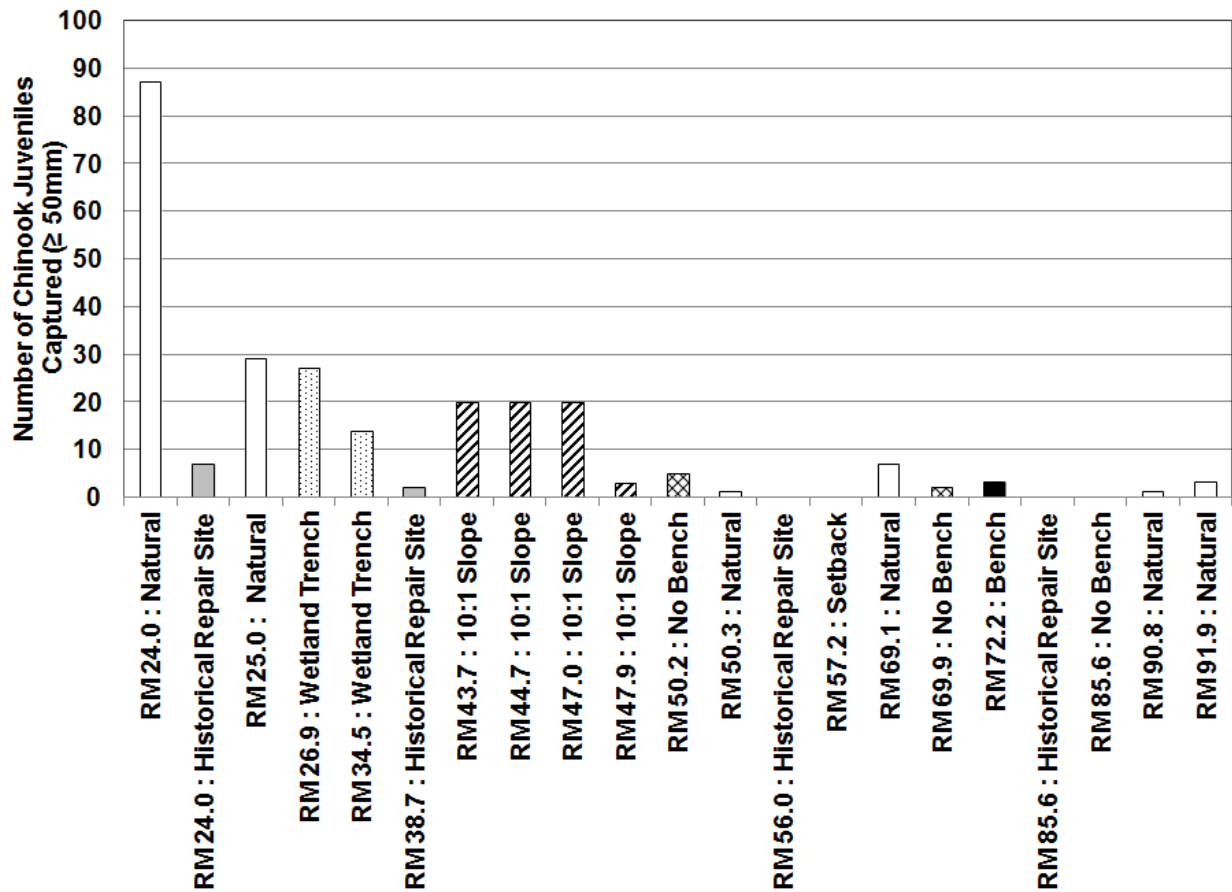


Figure 11. River Mile, Site Type, and Number of Chinook salmon juveniles captured during March and April 2011 Electrofishing surveys.

Table 11. Influence of SAM Repair Site Type (10:1 Slope/Bench, No Bench, WLT) on abundance of Chinook salmon fry and juveniles at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” ($p < 0.05$), “***” ($p < 0.01$) and “****” ($p < 0.001$).

Reference Category	Site Type	Fry			Juvenile		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
Natural	10:1 Slope/Bench	0.57	0.19	**	0.97	0.33	**
	No Bench	-1.59	0.48	***	-0.45	0.49	NS
	WLT	-0.10	0.21	NS	-0.98	0.18	***

Habitat Features

To investigate potential relationships between Chinook salmon fry and juvenile abundance and habitat features, the model used in this step included the following habitat variables (in addition to the spatial and temporal variables included in all models): shade (present/absent), density of emergent vegetation (absent/low/medium/high, absent was used as the reference category), density of emergent woody material (absent/low/medium/high, absent was used as the reference category), flow (ft/s), depth (feet), terrestrial vegetation (TV, percent bank coverage) and terrestrial woody material (percent bank coverage).

The predictive equations for evaluating the effect of habitat features on Chinook salmon abundance were in the general form

$$\log(c) = \beta_0 + \beta_1 RegionA + \beta_2 RegionB + \beta_3 RegionC + \beta_4 RegionD + \beta_5 Month_{March} + \dots$$

$$\beta_6 Shade_{present} + \beta_{7:9} EV_{dens.} + \beta_{10:12} EW_{dens.} + \beta_{13} Flow + \beta_{14} Depth + \beta_{15} TV_{\%} + \dots$$

$$\beta_{16} TW_{\%} + \beta_{17:19} Region \times Month$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]), $Month$ represents the sampling time, $Shade$ refers to shade presence, EV and EW correspond to the density of emergent vegetation and emergent woody material, respectively (low, medium and high, absence is used as the reference category), TV and TW refer to the percent coverage of terrestrial vegetation and terrestrial woody material, respectively, and $Region \times Month$ refers to the interaction between region and sampling month (four combinations, excluding the reference region).

Fry

Of the habitat variables analyzed in this model, average depth, percent bank coverage of terrestrial vegetation, and percent bank coverage of terrestrial woody material at the sampling site appeared to have a significant negative effect on the abundance of Chinook salmon fry (Table 12). Though these terrestrial habitat features appear to negatively impact Chinook salmon abundance, the mechanisms of this interaction are unclear. It is possible that these features affect variables not accounted for in the models presented here, or are reflective of associated habitat conditions below the water level. None of the other variables had a significant effect on fry abundance in this model.

Juveniles

Flow was significantly associated with the abundance of juvenile Chinook salmon, and the large coefficient (6.75) further indicates higher juvenile abundance in areas with faster current (Table 12). Depth showed a significant negative correlation, indicating higher abundance of juveniles in shallower water. While the density of emergent

vegetation had no effect on juvenile abundance in this model, abundance was significantly lower at all sites having emergent woody material, irrespective of density.

Table 12. Influence of microhabitat features on abundance of Chinook salmon fry and juvenile at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” (p<0.05), “*” (p<0.01) and “****” (p<0.001).**

Habitat Variable	Levels	Fry			Juvenile		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
Shade	-	-0.16	0.18	NS	-0.13	0.16	NS
Flow	-	-0.75	0.71	NS	6.75	0.91	***
Depth	-	-0.66	0.12	***	-1.09	0.14	***
Terr. Veg.	-	-1.83	0.34	***	0.09	0.30	NS
Terr. WM	-	-0.90	0.32	**	0.01	0.31	NS
Emergent Veg.	Low	-0.08	0.17	NS	-0.02	0.20	NS
	Medium	-0.09	0.19	NS	0.09	0.21	NS
	High	-0.43	0.24	NS	0.46	0.20	NS
Emergent WM	Low	-0.13	0.19	NS	-0.72	0.19	***
	Medium	0.11	0.22	NS	-0.69	0.26	**
	High	0.10	0.26	NS	-0.79	0.25	**

Combined SAM Repair Site Types/Habitat Features

To investigate how Chinook salmon fry and/or juvenile abundance is affected by both SAM Repair Site Types and habitat features, models that combined these variables were used in this step. Variables included the different SAM Repair Site Types (10:1 Slope/Bench, No Bench, WLT) with Natural Sites used as the reference category and those habitat features that were previously determined to have a significant effect on Chinook salmon abundance (Table 12), in addition to the spatial and temporal variables included in all models.

The predictive models for evaluating the effect of SAM Repair Site Types and habitat features on Chinook salmon abundance were in the general form

$$\begin{aligned} \log(c) &= \beta_0 + \beta_1 \text{Region } A + \beta_2 \text{Region } B + \beta_3 \text{Month} h_{\text{March}} + \beta_{4:6} \text{Type } A: C + \beta_{7:9} \text{EW}_{\text{dens}} \\ &\quad + \beta_{10} \text{Flow} + \dots \\ &\quad \beta_{11} \text{Depth} h + \beta_{12} \text{TV}_{\%} + \beta_{13} \text{TW}_{\%} + \beta_{14:15} \text{Region } x \text{Month } h \end{aligned}$$

where c is predicted Chinook salmon abundance (fry or juvenile), Regions A and B correspond to regions Garcia Bend (RM40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM21-39]); Month

represents the sampling time; Types *A* through *C* correspond to Repair Sites 10:1 Slope/Bench, No Bench and WLT, respectively; *EW* correspond to the density of emergent woody material (low, medium and high, absence was used as the reference category); *TV* and *TW* refer to the percent coverage of terrestrial vegetation and terrestrial woody material, respectively; and *Region x Month* refers to the interaction between region and sampling month (two combinations, excluding the reference region). Habitat variables without significant effects in the previous model cannot be significant in this combined model and were therefore not included.

Fry

All variables with significant effects on fry abundance showed a negative correlation in this analysis (Table 13). The No Bench Repair Site Types indicate a significant effect on fry abundance in both the SAM Repair Site model (Table 11) and this combined SAM Repair Site/Habitat Features model (Table 13), suggesting a strong effect. The same is true for the effects of Depth, Terrestrial Vegetation, and Terrestrial Woody Material. Interestingly, the 10:1 Slope/Bench Site Type had significant negative effects on fry abundance in this model, whereas, they had a positive correlation to fry abundance in the model excluding microhabitat variables. These conflicting results could be due to confounding effects caused as a result of combining the 10:1 Slope and Bench Repair Sites into one category.

Juveniles

The 10:1 Slope/Bench Repair Site Type shows a significant negative effect on juvenile abundance in both the SAM Repair Site Type model (Table 11) and this combined SAM Repair Site Types/Habitat Features model (Table 13), indicating a strong effect. The No Bench Site Type also appears to negatively influence juvenile Chinook salmon abundance, which stands in contrast to results from the SAM Repair Site Type model. This indicates that variation introduced by environmental factors outweigh the effects of this repair site type, yet if all environmental variables included in the model were controlled for, juvenile abundance at the No Bench Repair Site Type would be lower. The negative effect of Emergent Woody Material (across all densities) was consistently significant in both the Habitat Features and the combined SAM Repair Site Types/Habitat Features model. Likewise, prediction of higher juvenile abundance in faster current and shallower water was confirmed by the combined model. This suggests that both faster current and shallower water have a significant effect on juvenile Chinook salmon abundance, irrespective of Repair Site Type.

Table 13. Influence of Repair Site (10:1 Slope/Bench, No Bench, WLT) and microhabitat variables on abundance of Chinook salmon fry and juvenile at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” ($p<0.05$), “” ($p<0.01$) and “***” ($p<0.001$).**

Habitat Variable	Levels	Fry			Juvenile		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
SAM Repair Site	10:1 Slope/ Bench	-1.25	0.48	**	-1.26	0.47	**
	No Bench	-2.63	0.51	***	-1.46	0.54	**
	WLT	0.13	0.32	NS	-0.15	0.26	NS
Emergent WM	Low	0.04	0.19	NS	-0.56	0.19	**
	Medium	0.14	0.23	NS	-0.54	0.25	*
	High	-0.01	0.25	NS	-0.52	0.23	*
Flow	-	1.23	1.08	NS	7.40	1.01	***
Depth	-	-0.88	0.26	***	-1.27	0.23	***
Terr. Veg.	-	-2.41	0.32	***	0.35	0.28	NS
Terr. WM	-	-1.33	0.33	***	-0.33	0.30	NS

Predators

SAM Repair Site Type

A GLM was used to evaluate the effects of different SAM Repair Site Types on predator abundance. One model focused exclusively on the abundance of black bass, while another was used to evaluate the combined abundance of all potential salmonid piscivores, including black bass, striped bass and Sacramento Pikeminnow. SAM Repair Site Types evaluated in this model include 10:1 Slope/Bench, No Bench, and WLT. Natural Sites were used as the reference category in all model fits. Spatial and temporal variables (Region, Month) were included in initial model fits, but dropped from further analyses as they had no significant effect on predator abundance. A visual depiction of predator catch at various repair types can be found in Figure 12.

The final predictive model for evaluation predator abundance with respect to SAM Repair Site Types was in the form

$$\log(c) = \beta_0 + \beta_1 \text{TypeA} + \beta_2 \text{TypeB} + \beta_3 \text{TypeC}$$

where c is predicted predator abundance (black bass or combined black bass/striped bass/pikeminnow) and Types A through C correspond to SAM Repair Site Types 10:1 Slope/Bench, No Bench and WLT, respectively.

In the black bass only model, the No Bench and WLT Site Types had significant positive influences with bass abundance (Table 14). In the combined predator model, only the WLT repair type was significantly associated with increased predator abundance.

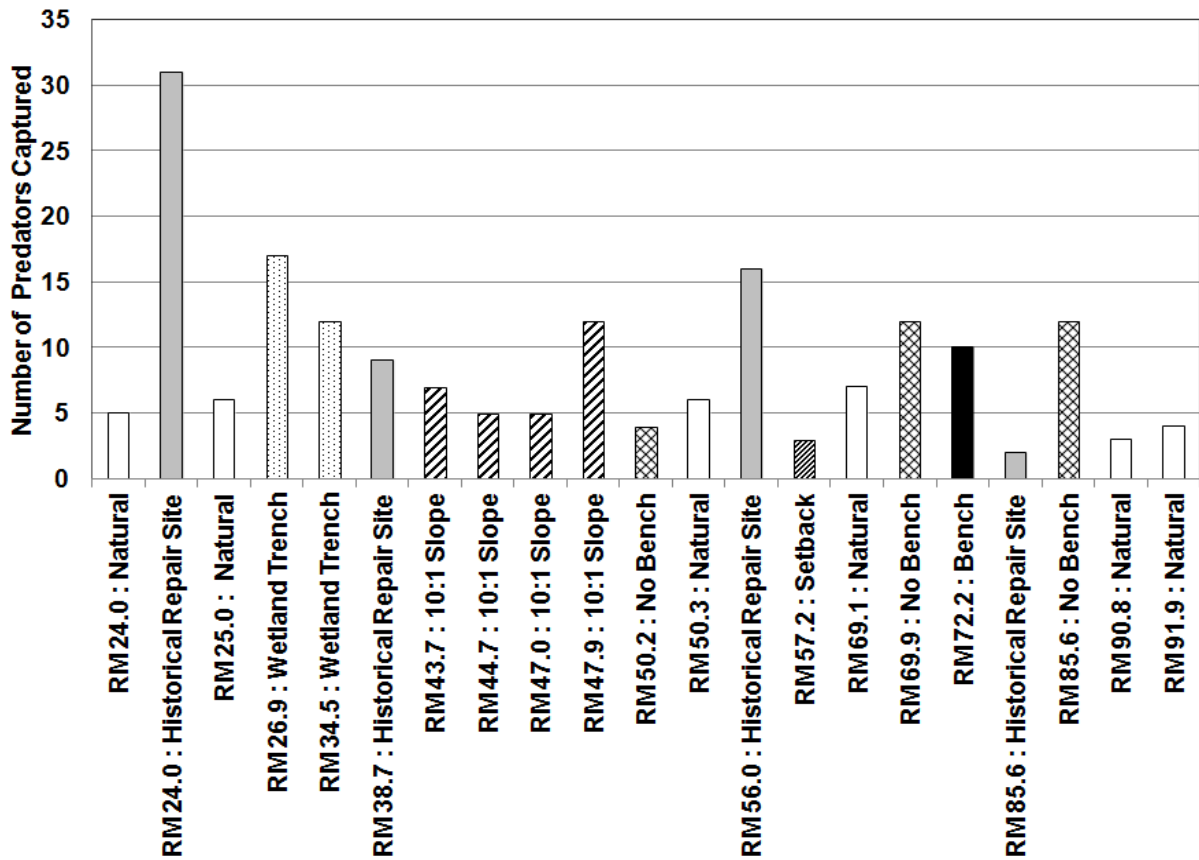


Figure 12. River Mile, Site Type, and Number of Predators captured during March and April 2011 Electrofishing surveys.

Table 14. Influence of SAM Repair Site Type (10:1 Slope/Bench, No Bench, WLT) on abundance of black bass and other piscivores at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” (p<0.05), “***” (p<0.01) and “****” (p<0.001).

SAM Repair Type	Black Bass Predators			All Predators		
	Coefficient	SE	Sig.	Coefficient	SE	Sig.
10:1 Slope/ Bench	0.54	0.36	NS	0.21	0.24	NS
No Bench	0.83	0.39	*	0.51	0.26	NS
WLT	1.26	0.37	***	0.74	0.26	**

Habitat Features

To investigate potential relationships between predator abundance and habitat variables, the model used in this step included the following habitat features (and spatial and temporal variables included in all models): shade (present/absent), density of emergent vegetation (absent/low/medium/high, absent was used as the reference category), density of emergent woody material (absent/low/medium/high, absent was used as the reference category), flow (ft/s), depth (ft), terrestrial vegetation (percent bank coverage) and terrestrial woody material (percent bank coverage).

The predictive equations for evaluating the effect of habitat features on predator abundance was in the general form

$$\log(c) = \beta_0 + \beta_1 \text{Shade}_{\text{present}} + \beta_{2:4} \text{EV}_{\text{dens.}} + \beta_{5:7} \text{EW}_{\text{dens.}} + \beta_8 \text{Flow} + \dots \\ \beta_9 \text{Depth} + \beta_{10} \text{TV}_{\%} + \beta_{11} \text{TW}_{\%}$$

where c is predicted predator abundance (black bass or combined black bass/striped bass/pikeminnow), Shade refers to shade presence, EV and EW correspond to the density of emergent vegetation and emergent woody material, respectively (low, medium and high, absence is used as the reference category), TV and TW refer to the percent coverage of terrestrial vegetation and terrestrial woody material, respectively.

In this analysis, none of the habitat variables significantly influenced predator abundance (either black bass only or combined black bass/striped bass/pikeminnow) (Table 15). This precludes inclusion of habitat features in a combined model.

Table 15. Influence of microhabitat features on abundance of piscivorous predators at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” (p<0.05), “” (p<0.01) and “***” (p<0.001).**

Habitat Variable	Levels	<u>All Predators</u>			<u>Black Bass</u>		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
Shade	-	-0.26	0.24	NS	-0.34	0.33	NS
Flow	-	-1.35	0.98	NS	-1.00	1.38	NS
Depth	-	0.13	0.11	NS	0.04	0.16	NS
Terrestrial Veg.	-	0.23	0.42	NS	0.70	0.57	NS
Terrestrial WM	-	-0.23	0.47	NS	0.16	0.67	NS
Emergent Veg.	Low	0.35	0.24	NS	0.33	0.33	NS
	Medium	-0.03	0.33	NS	-0.13	0.46	NS
	High	0.44	0.33	NS	0.34	0.45	NS
Emergent WM	Low	-0.27	0.22	NS	0.07	0.29	NS
	Medium	-0.37	0.30	NS	-0.23	0.42	NS
	High	-0.40	0.32	NS	-0.35	0.46	NS

Combined SAM Repair Site Types/Habitat Features

As none of the habitat variables evaluated in the preceding model showed a significant influence on predator abundance, a model including effects of both SAM Repair Site Type and Habitat Features would not provide additional insights on the habitat use of piscivores.

Fish Size

GLMs were used to evaluate the effects of different SAM Repair Site Types on juvenile (combined fry and juvenile) Chinook salmon size (mm FL). For this analysis, data collected in the two months (March and April) was analyzed separately, as fish size is expected to vary dramatically between sampling events. SAM Repair Site Types evaluated in this model include 10:1 Slope/Bench, No Bench, and WLT. Natural Sites were used as the reference category in all model fits. The spatial variable “Region” was used in this analysis to account for geographic variation in the fish size unrelated to SAM Repair Site Type (i.e., Chinook salmon encountered downstream may be of different sizes than those found farther upstream). Region Ko-Ket was used as the reference region in all model fits.

The predictive equations for evaluating the effect of repair type on Chinook salmon size were in the general form

$$\log(c) = \beta_0 + \beta_1 RegionA + \beta_2 RegionB + \beta_3 TypeA + \beta_4 TypeB + \beta_5 TypeC$$

where c is predicted Chinook salmon size (FL), Regions A and B correspond to regions Garcia Bend (RM 40-70) and Knights Landing (RM70-95), respectively (the reference region for this model was Ko-Ket [RM 21-39]), and Types A through C correspond to repair types 10:1 Slope/Bench, No Bench and WLT, respectively.

March

Based on data from the March sampling event, the GLM predicts significantly smaller fish size in the Garcia Bend and Knights Landing regions than at the downstream reference region (Table 16). Larger fish were associated with the No Bench Site Type, while the WLT and 10:1 Slope/Bench Site Types did not significantly affect fish size. Care should be taken when interpreting these model results: these results do not allow for valid conclusions regarding rearing conditions at these sites as the ages and origins (spawning sites) of captured fish are unknown (e.g., it cannot be said that the No Bench Site Type *does not produce* larger fish than other sites).

April

Similar to the model results for the March sample, the GLM predicts significantly smaller fish size in the Garcia Bend and Knights Landing regions. In analysis based exclusively on this sampling event, only the 10:1 Slope/Bench repair type was significant in predicting fish size, and larger juvenile Chinook salmon were generally found at those sites.

Table 16. Influence of river Region and SAM Repair Site Type on juvenile (combined fry and juvenile) Chinook salmon size at various sites sampled in March and April. Significance (Sig.) of categories in the GLM model fit is indicated as: “NS” (not significant), “*” (p<0.05), “*” (p<0.01) and “****” (p<0.001).**

Variable	Levels	March			April		
		Coefficient	SE	Sig.	Coefficient	SE	Sig.
Region	Garcia Bend	-20.72	3.28	***	-20.06	3.78	***
	Knights Landing	-17.84	3.18	***	-16.87	4.41	***
SAM Repair Type	10:1 Slope/Bench	4.20	2.26	NS	12.59	3.64	***
	No Bench	20.97	3.99	***	4.15	6.75	NS
	WLT	3.70	3.53	NS	-3.86	2.22	NS

6.2 Telemetry (Fish Objective 3)

Chinook salmon

Of the 154 detected Chinook salmon tags, 31.8 % (n=49) of the individuals did not exhibit residency at any site, while the remaining 68.2% (n=105) resulted in 398 observations of residency. Chinook salmon were more likely to reside at repair sites than at natural sites, with 25.8% of the individuals detected at repair sites taking up residency for at least one hour, in contrast to 17.4% of Chinook salmon detected at natural sites (Table 17). However, the duration of residency for Chinook salmon was generally longer at natural sites than at repair sites. These differences in duration of residency may be biologically significant, as fish (at the 75th percentile) remained for less than half a day at repair sites, compared to nearly two days at natural sites (Table 17).

During 2011, Chinook salmon were also found to be more likely to reside at sites in the Rio Vista region than in upstream areas (Table 17). This finding is similar to results of the 2009 and 2010 telemetry studies, and likely reflects a response to increasing tidal influence downstream of RM 20. However, while Chinook salmon were more likely to reside in the lower Sacramento River and Delta, the duration of residency was generally shorter than in upstream areas, with most Chinook salmon spending less than half of a day at these sites (Table 17). Although durations of residency at sites in the lower river were shorter, Chinook salmon tended to move between the three lowermost sites over extended periods of time (perhaps related to tidal influence), and the total amount of time spent at individual site was generally greater than for sites upstream where fish typically did not return after leaving.

Table 17. Summary of the proportion of acoustically tagged Chinook salmon exhibiting residency at various detection sites, site types and regions during 2011 (residency duration in hours).

Site	Site Type	% Individuals Detected	Number of Observations	<i>Quartile</i>				
				Minimum	25%	50%	75%	Maximum
<i>Residency By Site</i>								
69.9R	No Bench	0.0%	0	-	-	-	-	-
69.1L	Natural	3.3%	1	652.9	-	-	-	652.9
47.9R	10:1 slope	10.5%	6	7.5	36.3	63.6	237.0	370.9
44.7R	10:1 slope	0.9%	4	7.6	8.4	9.2	10.2	11.5
43.7R	10:1 slope	1.1%	5	9.8	15.2	35.8	45.1	235.8
25.0L	Natural	15.9%	10	1.0	7.7	31.9	99.6	1257.5
16.9R	Natural	6.8%	4	2.0	3.1	3.9	4.9	6.7
15.0R	Natural	34.8%	24	1.1	1.3	5.3	36.0	2098.2
12.7L	Bench	9.3%	15	1.0	1.5	3.3	6.0	24.2
12.5L	Bench	64.3%	224	1.0	1.8	3.5	9.4	989.8
10.7L	Wetland Trench	67.5%	105	1.0	3.5	6.9	13.7	1115.4
<i>Residency By Site Type</i>								
Repair		25.8%	359	1.0	2.0	4.8	10.9	1115.4
Natural		17.4%	39	1.0	2.1	6.9	42.9	2098.2
<i>Residency By Region</i>								
Garcia Bend		3.1%	16	7.5	9.8	34.8	121.1	652.9
Ko-ket		15.9%	10	1.0	7.7	31.9	99.6	1257.5
Rio Vista		40.2%	372	1.0	1.9	4.5	10.5	2098.2

Steelhead

Of the 61 steelhead tags detected, 44.3 % (n=27) of the individuals did not exhibit residency at any site, while the remaining, 55.7% (n=34) provided 73 observations of residency. Similar to Chinook salmon, steelhead were more likely to reside at repair sites than at natural sites, with 25.8% of the individuals detected at repair sites taking up residency for at least one hour, in contrast to only 9.6% of steelhead detected at natural sites. Duration of residency was similar at repair- and natural sites, with most individuals spending 12 hours or less at a site.

Like the Chinook salmon discussed above, steelhead were more likely to reside at sites in the Rio Vista region than in upstream areas, yet the duration of residency in the lower

river was shorter than that observed in upstream areas, with most steelhead spending less than half of a day at these sites (Table 18).

Table 18. Summary of the proportion of acoustically tagged steelhead exhibiting residency and the duration of residency by site, by site type, and by region during 2011.

Site	Site Type	% Individuals Detected	Number of Observations	<i>Quartile</i>				
				Minimum	25%	50%	75%	Maximum
<i>Residency By Site</i>								
69.9R	No Bench	100.0%	1	2.8	-	-	-	2.8
69.1L	Natural	0.0%	0	-	-	-	-	-
47.9R	10:1 slope	3.7%	1	3.6	-	-	-	3.6
44.7R	10:1 slope	2.2%	2	9.1	11.8	14.6	17.4	20.1
43.7R	10:1 slope	5.1%	3	2.0	46.7	91.4	137.2	183.0
25.0L	Natural	5.9%	2	1.0	2.8	4.7	6.5	8.4
16.9R	Natural	9.1%	3	1.3	1.3	1.3	256.4	511.5
15.0R	Natural	18.8%	6	1.0	1.6	2.5	5.0	69.3
12.7L	Bench	10.7%	6	1.3	2.6	20.3	145.8	331.6
12.5L	Bench	35.1%	22	1.0	1.9	2.9	6.1	22.6
10.7L	Wetland Trench	58.6%	27	1.3	2.5	5.0	15.7	120.7
<i>Residency By Site Type</i>								
Repair		18.4%	62	1.0	2.2	3.8	12.9	331.6
Natural		9.6%	11	1.0	1.25	2.5	7.07	511.5
<i>Residency By Region</i>								
Garcia Bend		3.9%	7	2.0	3.21	9.1	55.7	183.0
Ko-ket		5.9%	2	1.0	2.84	4.7	6.53	8.4
Rio Vista		26.4%	64	1.0	2.08	3.5	11.5	511.5

Green Sturgeon

Of the five green sturgeon tags detected, 60% (n=3) exhibited residency (8 observations, Table 19). The few observations of residency were not sufficient for meaningful comparisons between site type or region; therefore, no proportions of acoustically tagged steelhead exhibiting residency were generated.

Table 19. Summary of residency duration (in hours) of acoustically tagged green sturgeon duration by site during 2011.

Site	Site Type	Number of Observations	<i>Quartile</i>				Maximum
			Minimum	25%	50%	75%	
69.9R	No Bench	0	-	-	-	-	-
69.1L	Natural	0	-	-	-	-	-
47.9R	10:1 slope	0	-	-	-	-	-
44.7R	10:1 slope	0	-	-	-	-	-
43.7R	10:1 slope	0	-	-	-	-	-
25.0L	Natural	1	8.1	-	-	-	8.1
16.9R	Natural	0	-	-	-	-	-
15.0R	Natural	0	-	-	-	-	-
12.7L	Bench	1	1.4	-	-	-	1.4
12.5L	Bench	2	2.5	3.0	3.5	4.0	4.5
10.7L	Wetland Trench	4	2.0	2.0	5.2	8.7	9.9

Striped Bass

Striped bass were detected at all sites monitored with acoustic receivers during the 2011 study period; however, note that subsequent interpretations of telemetry data are solely based on detections of small numbers of individuals, and do not allow for strong conclusions. Of the nine striped bass tags detected, 78% (n=7) exhibited residency and provided 77 observations (Table 20). The duration of residency was less than one day for most observations, and durations were similar between repair and natural sites. While approximately the same number of individuals was detected in the Garcia Bend- and Rio Vista regions, duration of residency was slightly longer in the Garcia Bend region. However, the difference of only 5-10 hours is unlikely to be of biological significance. More importantly, striped bass appeared to be more likely to reside at repair sites than at natural sites (a similar trend to that exhibited by Chinook salmon and steelhead), possible indicating increased predator/prey interactions at repair sites. Also, the possibility exists that at least some of the detections believed to be salmon or steelhead may in reality have been striped bass, which had preyed upon those individuals.

Table 20. Summary of the proportion of acoustically tagged striped bass exhibiting residency at various detection sites, site types and regions during 2011 (residency duration in hours).

Site		% Individuals Detected	Number of Observations	<u>Quartile</u>				
				Minimum	25%	50%	75%	Maximum
Residency By Site								
69.9R	No Bench	100.0%	10	1.6	11.8	22.6	42.1	276.9
69.1L	Natural	50.0%	7	3.6	5.9	9.2	21.5	213.6
47.9R	10:1 slope	50.0%	1	4.8	-	-	-	4.8
44.7R	10:1 slope	0.0%	0	-	-	-	-	-
43.7R	10:1 slope	50.0%	2	201.7	318.8	435.9	553.0	670.0
25.0L	Natural	0.0%	0	-	-	-	-	-
16.9R	Natural	0.0%	0	-	-	-	-	-
15.0R	Natural	50.0%	2	1.0	1.0	1.1	1.1	1.2
12.7L	Bench	33.3%	2	1.6	1.6	1.6	1.6	1.6
12.5L	Bench	33.3%	16	1.0	1.3	2.0	4.0	12.3
10.7L	Wetland Trench	60.0%	37	1.6	4.8	8.2	24.8	295.8
Residency By Site Type								
Repair		45.0%	68	1.0	3.0	5.8	21.9	670.0
Natural		33.3%	9	1.0	3.6	7.8	17.2	213.6
Residency By Region								
Garcia Bend		45.5%	20	1.6	7.0	15.8	15.1	670.0
Ko-ket		0.0%	0	-	-	-	-	-
Rio Vista		42.9%	57	1.0	2.1	5.1	10.7	295.8

6.3 Habitat Monitoring (Habitat Objective 1)

In 2011, the SAM habitat survey was conducted September 12-16 and 27 to characterize habitat features at seasonal water surface elevations. A number of comparisons are presented below for many of the habitat features, between Natural and Repair Sites, and between the four levee repair design types (Wetland Trench, 10:1 Slope, Bench, No Bench).

6.3.1 Bank Substrate Size

The results of the substrate particle size bank-line coverage for the Repair versus Natural Sites and by Repair Site Type (Wetland Trench, 10:1 Slope, Bench, No Bench) for the

September 2011 surveys are presented graphically in Figures 13 through 16. The available results for the 2008 through 2011 surveys are presented in Table 21. Note that detailed repair site substrate values for the 2008 and 2009 surveys could not be located in the H.T. Harvey & Associates and PRBO (2010) report.

In 2011, both bank-line transects at all Natural Sites were dominated by silt/clay substrate category with lesser amounts of sand (Figure 13). At the Repair Sites, there was a distinct difference between substrates at the low water elevation and high water elevation bank-lines (Figure 13). Substrates at the high water elevation bank-lines were very similar to those at the Natural Sites and were almost exclusively composed of fine silt/clay particles with lesser amounts of sand. Substrate particle size for the low water elevation transects at Repair Sites were also dominated by the fine silt/clay substrate particle category, but with significant amounts of boulder and cobble substrates present as well.

Average substrate size varied between Repair Sites in 2011 (Figures 14 through 16; Table 21). In 2011, the Wetland Trench and 10:1 Slope Repair Sites exhibited similar substrate characteristics (Figure 14). At both these Repair Sites, while there were some minor amounts of boulder and cobble particle sizes present along the low water transects, both bank-line transects were still dominated by fine sediment elements in the sand and silt/clay particle size categories. At the remaining Bench, No Bench, and Historic Repair Sites, the fine sediments continued to dominate the high water elevation transects, but the low water transects had less fine sediment and higher levels of boulder and cobble elements (Figure 15 and 16; Table 21).

Examination of the bank-line substrate data over time suggests that there has been virtually no change to the Natural Sites since 2008 (Table 21). In all four years, fine sediments have dominated both the low water and high water transects. Over this same four year period, the Repair Site bank-line transects have been changing and becoming dominated by finer sediment particle sizes (i.e., silt/clay and sand categories). This change toward finer substrate categories along the bank-line transects is especially evident at all the high water elevation repair transects and at the low water transects at the Wetland Trench and the 10:1 Slope Repair Sites. Coarse substrate elements (boulder and cobble particles) dominate the substrates at only the low water transects at the Bench, No Bench, and Historic Repair Sites. This data seems to indicate that during the recent winter and spring high flow periods, depositional events have led to the levee Repair Sites becoming more similar to the Natural Sites in terms of substrate particle sizes.

Evidence of fine sediment deposition at the Repair Sites can be clearly observed in several of the photographs taken during the September 2011 habitat surveys. At both Wetland Trench Repair Sites, portions of the ditch originally excavated to provide shallow-water off channel fish habitat, had become completely filled with mud and silt (Figures 17 and 18). Fine sediment deposition was also noted at several of the 10:1 Slope Repair Sites where a thick layer of recently deposited sand and mud covered the original rock substrate laid down as part of the original repair (Figures 19 and 20).

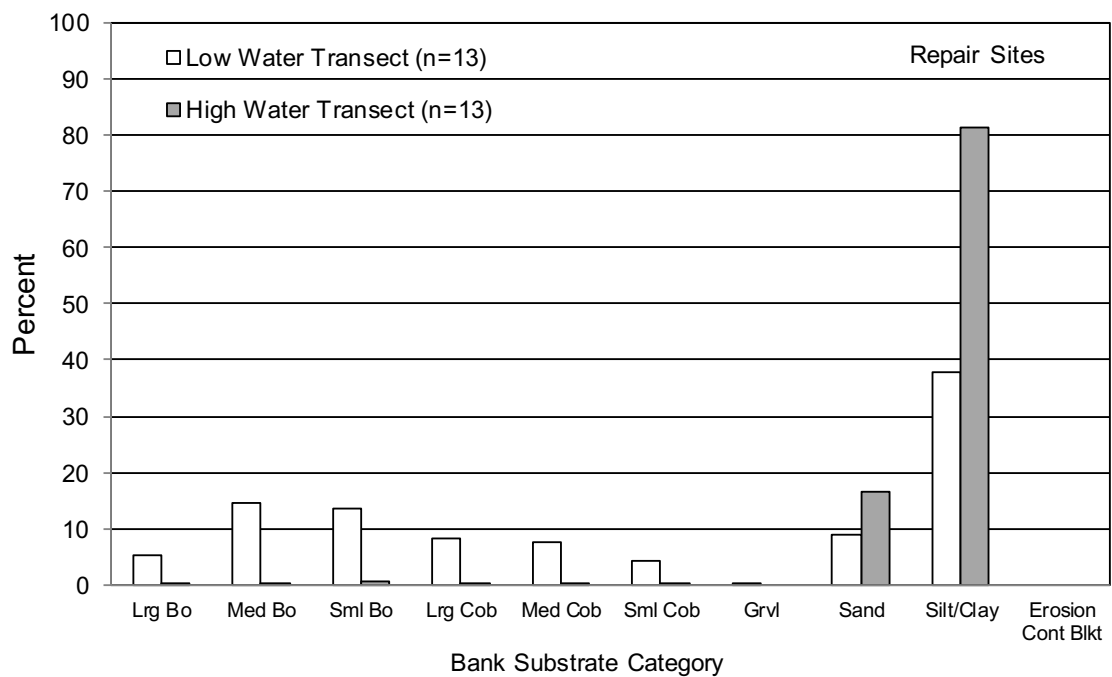
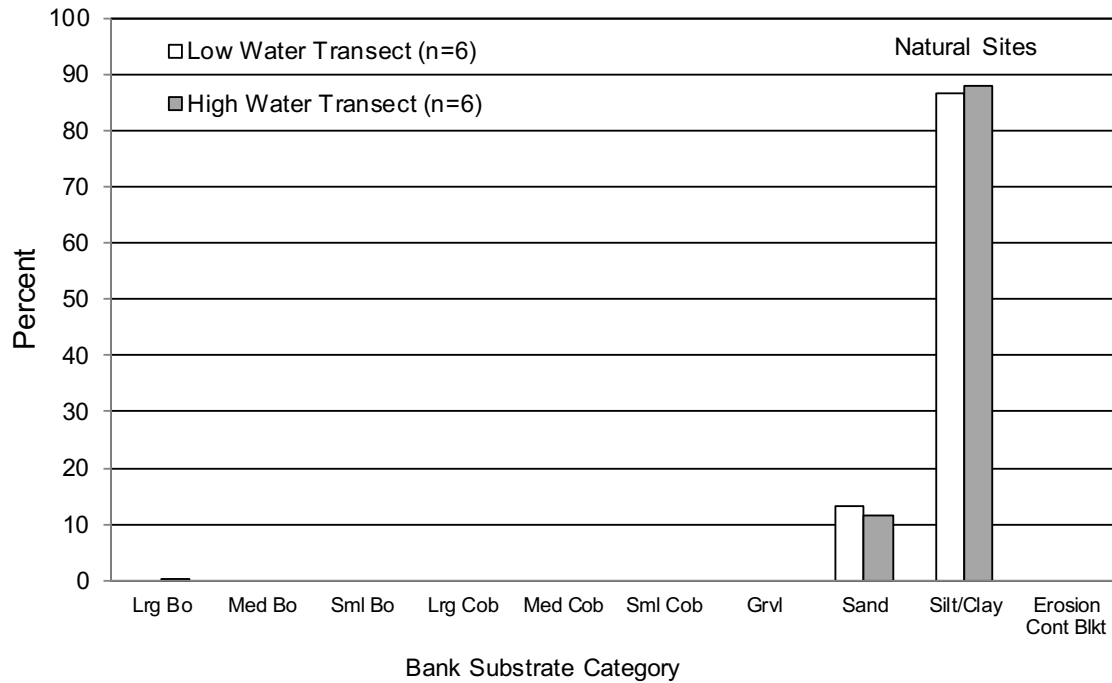


Figure 13. Average percent substrate particle categories at the low water and high water elevation transects for the six Natural (top) and 13 Repair (bottom) Sacramento River levee habitat monitoring sites surveyed in September 2011.

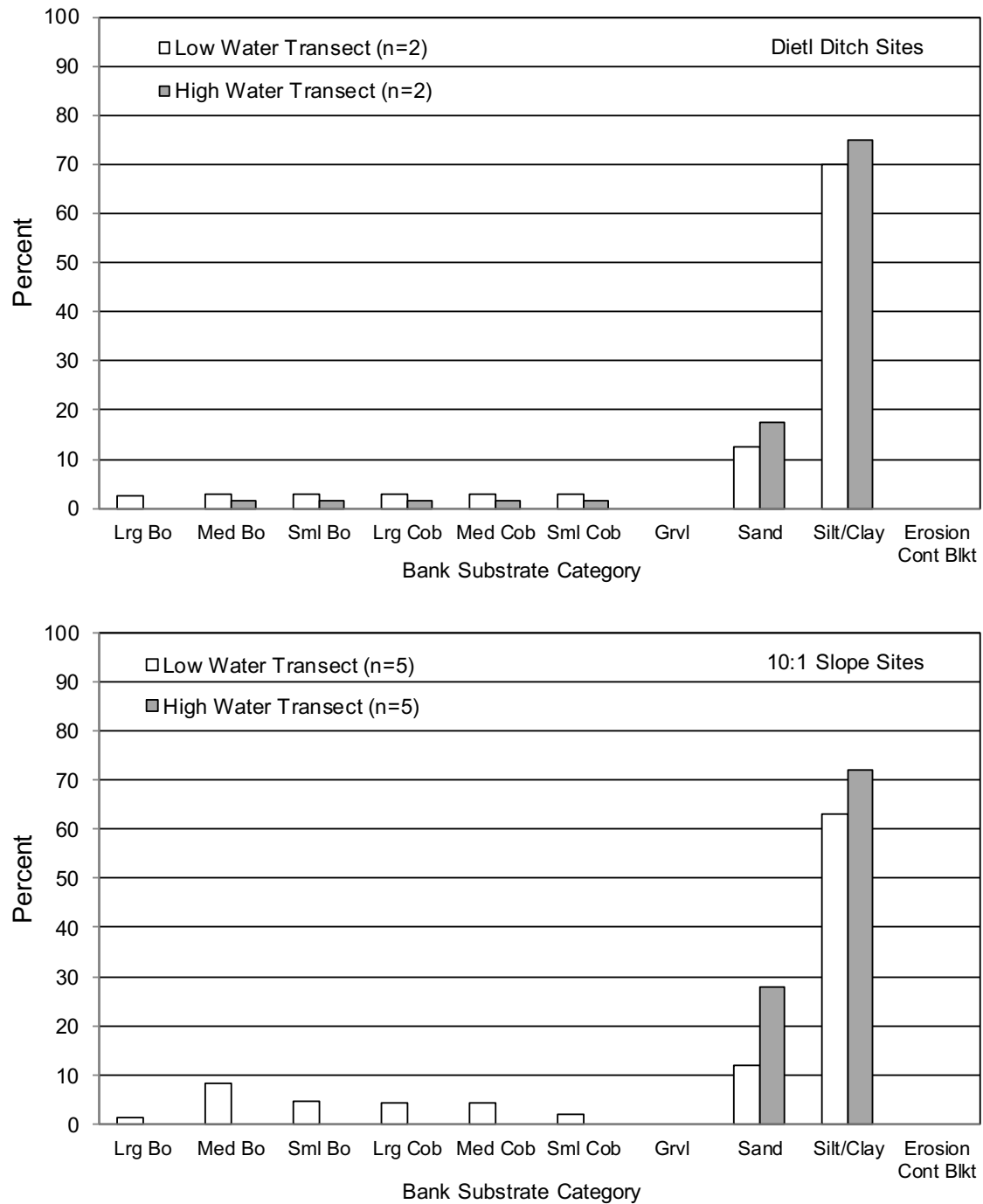


Figure 14. Average percent substrate particle categories at the low water and high water elevation transects for the Wetland Trench (a.k.k., Dietl ditch)(top) and 10:1 slope (bottom) repair Sacramento River levee habitat monitoring sites surveyed in September 2011.

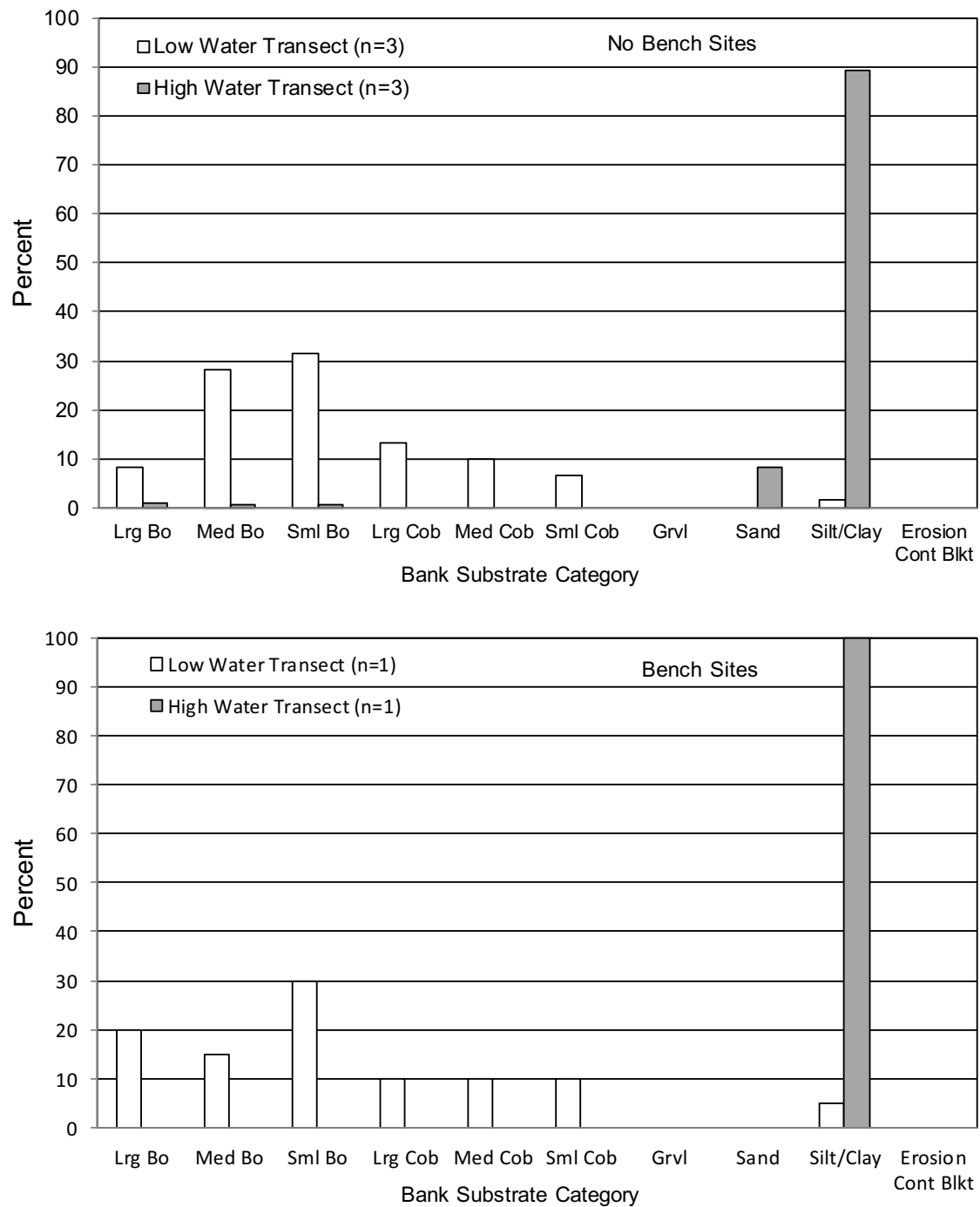


Figure 15. Average percent substrate particle categories at the low water and high water elevation transects for the no bench (top) and bench (bottom) repair Sacramento River levee habitat monitoring sites surveyed in September 2011.

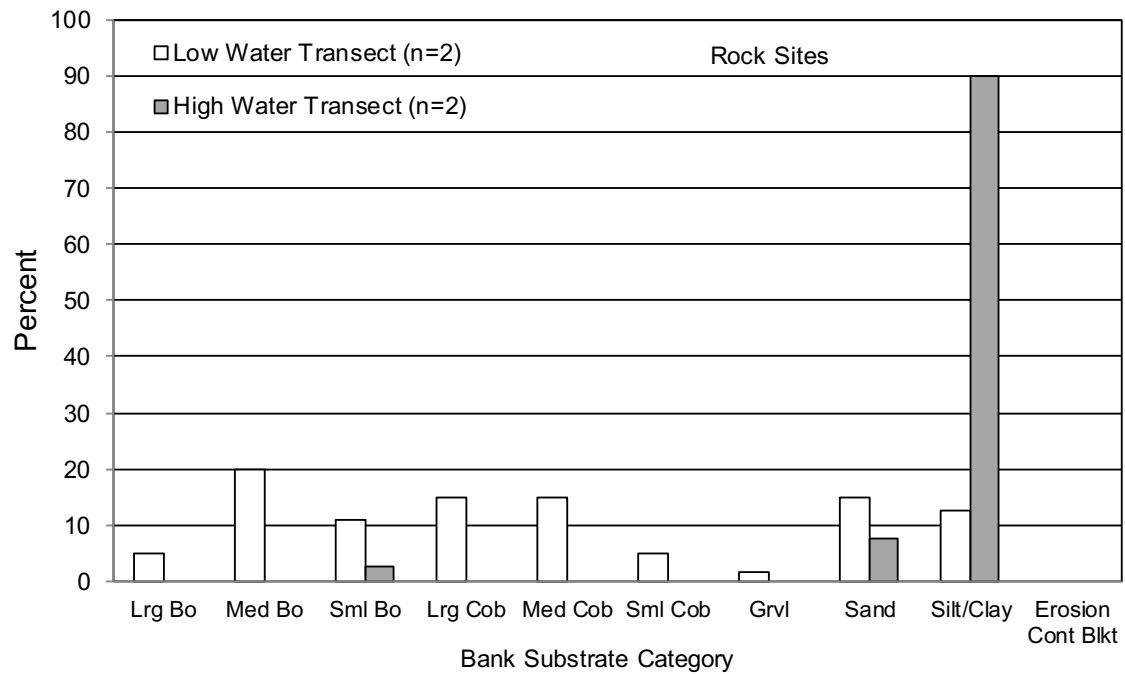


Figure 16. Average percent substrate particle categories at the low water and high water elevation transects for the Historic Repair (a.k.a., Rock) Sacramento River levee habitat monitoring sites surveyed in September 2011.

Table 21. Average percent bank-line coverage of substrate particle categories at low water and high water elevation transects by site type, 2008-2011. The 2008 and 2009 Natural versus repair site data from Table D-2 in H.T. Harvey & Associates and PRBO (2010); no individual repair type substrate values were included in this report. The 2010 data was derived from H.T. Harvey & Associates data sheets.

Site Type	2008												2009											
	Low water elevation transect						High water elevation transect						Low water elevation transect						High water elevation transect					
	EB	Silt/ Clay	Snd	Grvl	Cob	Bo	EB	Silt/ Clay	Snd	Grvl	Cob	Bo	EB	Silt/ Clay	Snd	Grvl	Cob	Bo	EB	Silt/ Clay	Snd	Grvl	Cob	Bo
	Natural	0.8	89.6	5.6	0.0	2.5	1.5	0.5	77.9	16.3	0.0	2.3	3.2	0.4	87.8	6.3	0.0	1.5	4.1	0.3	81.4	11.6	0.0	2.2
	n = 8						n = 8						n = 8						n = 8					
Repair	0.2	9.5	10.8	0.8	18.9	60.5	1.6	38.3	6.8	0.4	15.5	38.6	0.2	8.5	12.9	0.5	21.6	59.8	0.2	30.6	11.8	0.6	16.7	46.2
	n = 11						n = 11						n = 11						n = 11					

Site Type	2010										2011									
	Low water elevation transect					High water elevation transect					Low water elevation transect					High water elevation transect				
	Silt/ Clay	Snd	Grvl	Cob	Bo	Silt/ Clay	Snd	Grvl	Cob	Bo	Silt/ Clay	Snd	Grvl	Cob	Bo	Silt/ Clay	Snd	Grvl	Cob	Bo
Natural	87.5	5.5	0.0	3.5	3.5	82.5	10.8	0.0	4.1	3.3	86.7	13.3	0.0	0.0	0.0	88.0	11.7	0.0	0.0	0.3
	n = 8					n = 8					n = 6					n = 6				
Repair	26.1	6.0	5.1	40.9	21.9	49.9	0.9	3.6	24.5	21.1	37.7	8.8	0.2	20.1	33.2	81.4	16.5	0.0	0.7	1.4
	n = 11					n = 11					n = 13					n = 13				
Wetland Tr.	50.0	0.0	0.0	30.3	19.7	25.0	0.7	1.3	24.3	50.3	70.0	12.5	0.0	9.0	8.5	75.0	17.5	0.0	4.5	3.0
	n = 3					n = 3					n = 2					n = 2				
10:1 Slope	23.6	9.4	5.0	34.4	27.6	52.8	1.0	4.4	24.4	17.4	63.0	12.0	0.0	10.6	14.4	72.0	28.0	0.0	0.0	0.0
	n = 5					n = 5					n = 5					n = 5				
Bench	---	---	---	---	---	---	---	---	---	---	5.0	0.0	0.0	30.0	65.0	100	0.0	0.0	0.0	0.0
	not sampled					not sampled					n = 1					n = 1				
No Bench	0.3	0.3	5.3	69.3	24.7	48.3	0.0	3.0	29.7	19.0	1.7	0.0	0.0	30.0	68.3	89.3	8.3	0.0	0.0	2.3
	n = 3					n = 3					n = 3					n = 3				
Historic	---	---	---	---	---	---	---	---	---	---	12.5	15.0	1.5	35.0	36.0	90.0	7.5	0.0	0.0	2.5
	not sampled					not sampled					n = 2					n = 2				



Figure 17. Example of Wetland Trench habitat at Site 26.9L filled in with fine sediment and terrestrial vegetation so that it is now completely out of water during the summer low flow period. 27 September 2011.



Figure 18. Example of Wetland Trench habitat at Site 34.5R filled in with fine sediment and terrestrial vegetation so that it is now completely out of water during the summer low flow period. 16 September 2011.



Figure 19. Example of fine sediment layer deposited over original repair boulder substrates at Site 43.7R. 16 September 2011.



Figure 20. Example of fine sediment layer deposited over original repair boulder substrates at Site 44.7R. 16 September 2011.

6.3.2 Aquatic Vegetation

Aquatic vegetation bank-line coverage for the various site types during the 2011 SAM Habitat Surveys are presented graphically in Figures 21 through 24. Values for 2008-2011 are presented in Table 22.

A comparison of the 2011 data for the Natural versus Repair Sites shows that the mean bank-line coverage for the low water transects were quite similar for both the Natural and Repair Sites, as were the high water transect averages for the Natural and Repair Sites (Figure 21). At both the Natural and Repair Sites, the high water transects had slightly higher average bank line coverage of aquatic vegetation than the respective low water transects, but the differences were likely not significant and were within one standard error of the means.

When examined by Repair Type, the low water transects tended to have higher average bank-line coverage of aquatic vegetation than the seasonally adjusted values at the high water transects for the Wetland Trench, 10:1 Slope, and Bench Repair Sites, though only at the Wetland Trench sites were these differences in the low and high water bank-line averages outside one standard error of the mean (Figures 22 and 23). Significant differences between the low and high water average bank-line coverage of aquatic vegetation were noted in 2011 for at the No Bench and the Historic Repair Sites (Figures 23 and 24). The higher mean aquatic vegetation bank-line coverage at high water transects for the No Bench and Historic Repair Sites were well beyond one standard error of the mean.

Since 2009, the average bank-line coverage of aquatic vegetation at the both the low water and high water transects at both the Natural and Repair Sites have remained relatively stable (Table 22). The one noteworthy change in the aquatic vegetation bank coverage observed in 2011 was for the low water transects at the Wetland Trench Sites. From 2008 through 2010, the average low water bank-line coverage has remained stable at around 65 percent; while in 2011, the average appeared to increase to 85 percent. However, this apparent increase is associated with exclusion in 2011 of one of the lower river Wetland Trench Sites (10.7L) that had been included in the prior surveys. When this sample site effect is taken into account, the average aquatic vegetation bank-line coverage at all the repair site types appears to be very stable over the past four years.

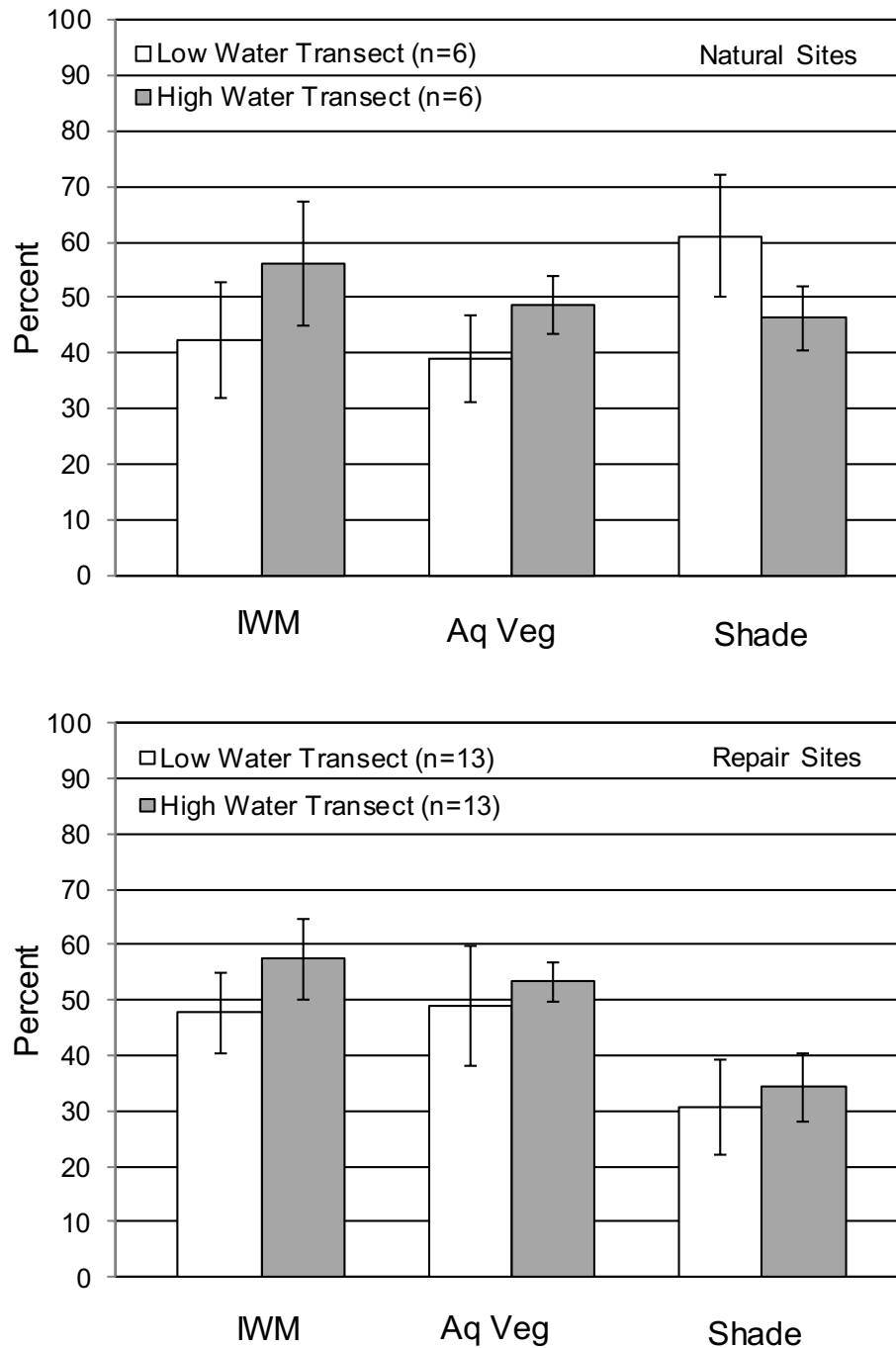


Figure 21. Average percentage bank-line coverage with standard error at the low water and high water elevation transects for instream woody material (IWM), aquatic vegetation (Aq Veg) and overhanging shade for six Natural (top) and 13 Repair (bottom) Sites surveyed in September 2011. High water transect aquatic vegetation and shade data is seasonally adjusted.

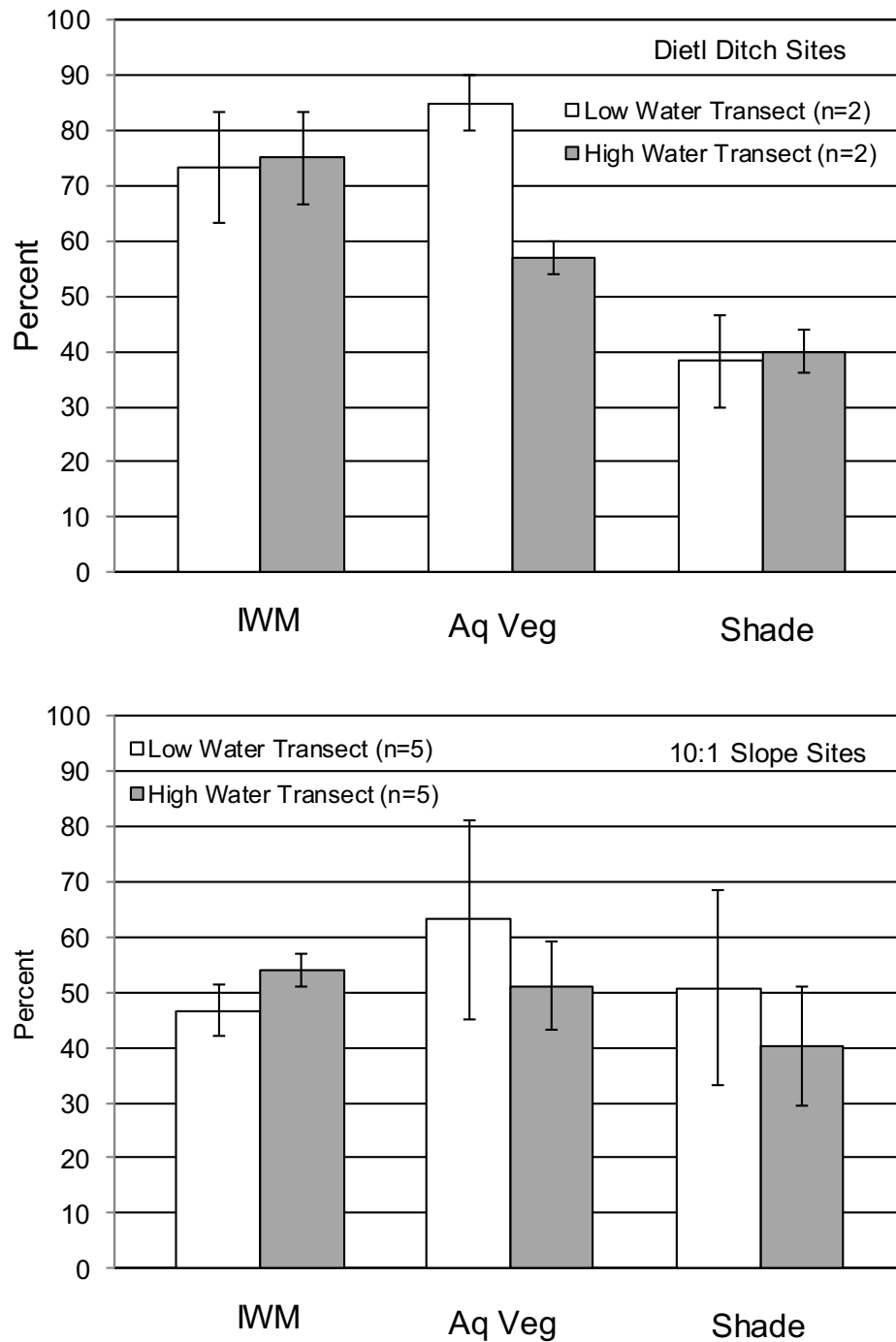


Figure 22. Average percentage bank-line coverage with standard error at the low water and high water elevation transects for instream woody material (IWM), aquatic vegetation (Aq Veg) and overhanging shade for the Wetland Trench (a.k.a., Dietl ditch) (top) and 10:1 Slope (bottom) Sites surveyed in September 2011. High water transect aquatic vegetation and shade data is seasonally adjusted.

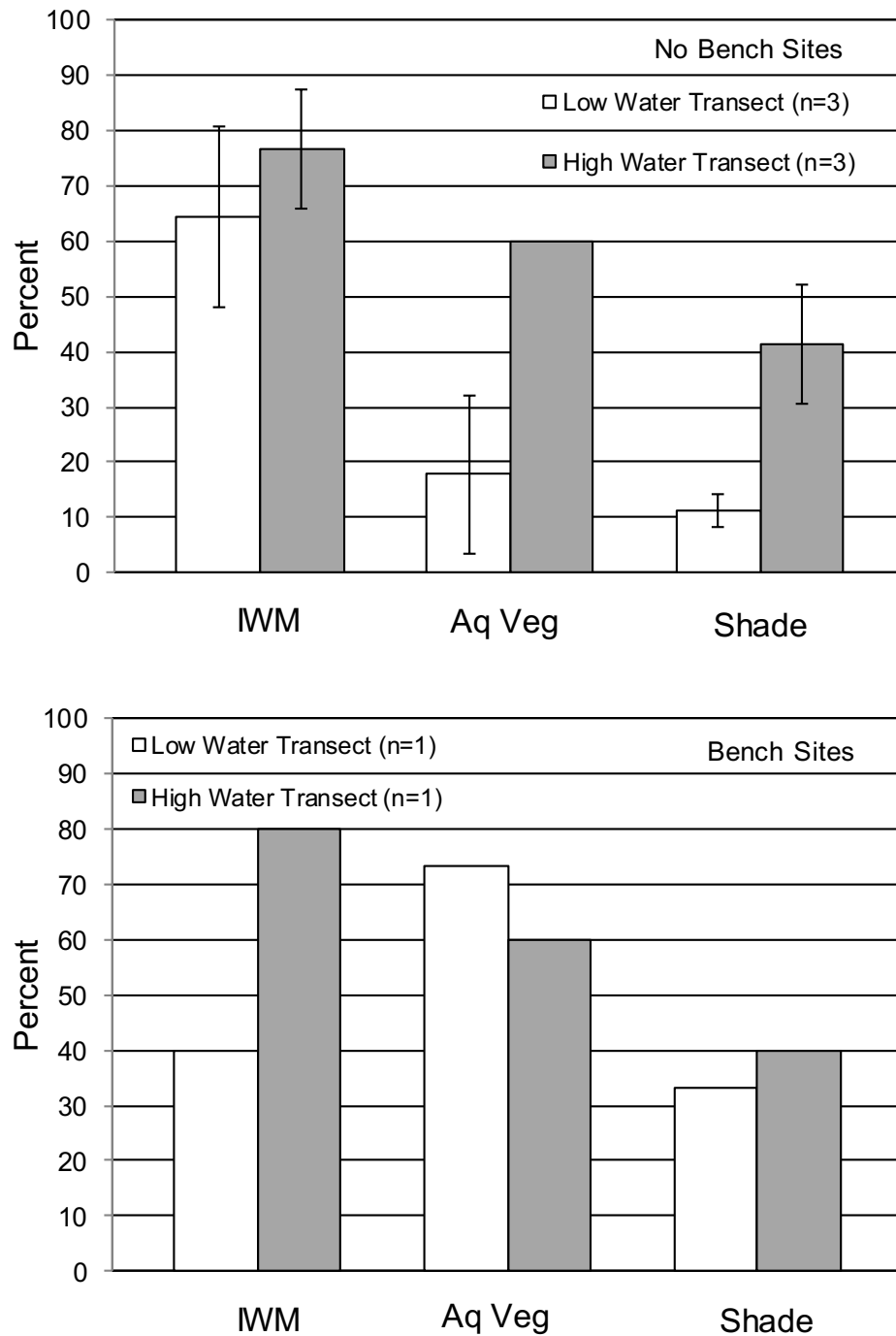


Figure 23. Average percentage bank-line coverage with standard error at the low water and high water elevation transects for instream woody material (IWM), aquatic vegetation (Aq Veg) and overhanging shade for the No Bench (top) and Bench (bottom) Sites surveyed in September 2011. High water transect aquatic vegetation and shade data is seasonally adjusted.

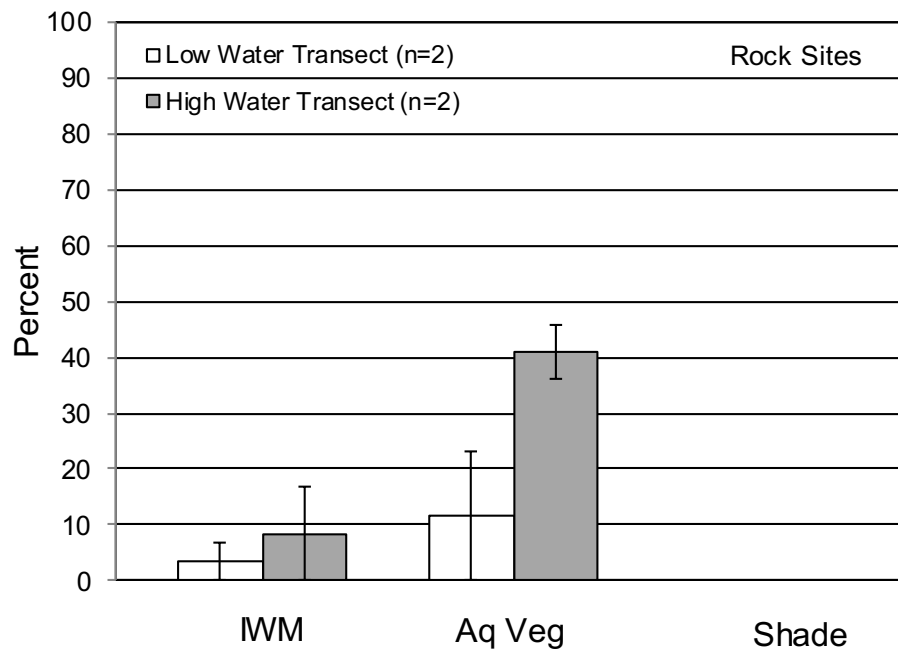


Figure 24. Average percentage bank-line coverage with standard error at the low water and high water elevation transects for instream woody material (IWM), aquatic vegetation (Aq Veg) and overhanging shade at the Historic Repair (a.k.a., Rock) Sites surveyed in September 2011. High water transect aquatic vegetation and shade data is seasonally adjusted.

Table 22. Average percent bank-line coverage (with associated standard error, SE) for aquatic vegetation at low water and high water elevation transects by site type, 2008-2011. High water transect data is seasonally adjusted. Source: 2008 and 2009 data from H.T. Harvey & Associates and PRBO (2010) and 2010 data from H.T. Harvey & Associates data sheets.

Site Type	Water Elevation Transects							
	2008		2009		2010		2011	
	Low	High	Low	High	Low	High	Low	High
Natural	41.7 (7.8) n = 8	50.3 (2.6)	57.1 (12.5) n = 8	53.2 (2.5)	45.0 (10.2) n = 8	46.5 (2.6)	38.9 (7.8) n = 6	48.7 (5.1)
Repairs	34.4 (10.3) n = 13	40.6 (5.2)	44.8 (11.1) n = 13	53.7 (2.4)	55.8 (11.0) n = 11	56.0 (1.8)	48.9 (10.8) n = 13	53.2 (3.4)
Wetland Tr.	62.2 (31.1) n = 3	59.3 (0.6)	64.4 (32.3) n = 3	60.0 (0.0)	67.8 (27.2) n = 3	60.0 (0.0)	85.0 (5.0) n = 2	57.0 (3.0)
10:1 Slope	47.3 (11.9) n = 5	38.0 (6.5)	70.7 (7.4) n = 5	53.2 (1.9)	70.0 (10.9) n = 5	56.0 (2.6)	63.1 (18.1) n = 5	51.2 (7.8)
Bench	0.0 (0.0) n = 2	47.0 (9.0)	0.0 (0.0) n = 2	60.0 (0.0)	- not sampled	-	73.3 (0.0) n = 1	60.0 (0.0)
No Bench	7.8 (4.0) n = 3	22.0 (13.1)	12.2 (4.4) n = 3	44.0 (8.0)	20.0 (13.3) n = 3	52.0 (4.6)	17.8 (14.4) n = 3	60.0 (0.0)
Historic Repair	- not sampled	-	- not sampled	-	- not sampled	-	11.7 (11.7) n = 2	41.0 (5.0)

6.3.3 Shade

The results of the overhanging shade bank-line coverage for the repair versus Natural Sites and by repair site type for the September 2011 surveys are presented graphically in Figures 21 through 24. The results for the 2011 survey, as well as those from the prior three years, are presented in Table 23.

In 2011, overhanging shade coverage was greater at the Natural Sites than at the Repair Sites for both transects (Figure 21 and Table 23). The majority of the overhanging shade at Repair Sites was a result of plantings throughout the riparian restoration sites (86.8% of the low water elevation shade and 74.1% of the high water elevation shade).

Conversely, all the overhanging shade identified at the Natural Sites during 2011 was provided from large, pre-existing trees that appear to have been growing for decades.

Examination of the 2011 overhanging shade data by repair site suggests similar average bank-line coverage at the low and high water elevation transects within most repair types (Figure 22 through 24). The exception is for the No Bench repair design, where the mean shade coverage at the high water elevation transects (43.1%) is four times higher than that at the low water elevation transects (11.1%). This difference in transect mean shade coverage exceeds the standard error of the averages. No shade was identified along any of the Historic Repair Sites during the 2011 survey.

Examination of the overhanging shade data over time suggest that shade has remained very stable over time at the Natural Sites (Table 23). The annual variation noted at both the low water elevation and high water elevation transects since 2008 are within the standard error of the means. At the Repair Sites, there has been a consistent increase in overhanging shade at the low water transects since 2008 (Table 23). This increase in low water transect shade is evident at all the different SAM Repair Sites but not the Historic Repair Sites. The amount of shade noted at the various repair high water transects has remained relatively stable in recent years and has changed little since 2009.

Table 23. Average percent bank-line coverage (with associated standard error, SE) for overhanging shade at low water and high water elevation transects by site type, 2008-2011. High water transect data is seasonally adjusted. 2008 and 2009 data from H.T. Harvey & Associates and PRBO (2010); 2010 data from H.T. Harvey & Associates data sheets.

Site Type	Water Elevation Transects							
	2008		2009		2010		2011	
	Low	High	Low	High	Low	High	Low	High
Natural	63.8 (9.0) n = 8	44.0 (6.6)	70.8 (5.7) n = 8	47.0 (3.5)	56.3 (11.9) n = 8	39.3 (5.0)	61.1 (10.9) n = 6	46.3 (5.8)
Repair	1.8 (1.0) n = 13	13.1 (4.1)	9.5 (7.3) n = 13	32.8 (5.3)	13.0 (4.9) n = 11	33.3 (5.2)	30.5 (8.6) n = 13	34.2 (6.1)
Wetland Tr.	5.6 (4.0) n = 3	19.3 (9.7)	33.3 (31.7) n = 3	34.7 (17.5)	17.8 (17.8) n = 3	23.3 (11.7)	38.3 (8.3) n = 2	40.0 (4.0)
10:1 Slope	0.7 (0.7) n = 5	3.2 (1.8)	2.7 (1.2) n = 5	26.4 (4.4)	14.0 (5.0) n = 5	36.4 (5.1)	50.8 (17.5) n = 5	40.1 (10.8)
Bench	0.0 (0.0) n = 2	12.0 (12.0)	0.0 (0.0) n = 2	33.0 (27.0)	--- not sampled	---	33.3 (0.0) n = 1	40.0 (0.0)
No Bench	1.1 (1.1) n = 3	24.0 (12.2)	3.3 (1.9) n = 3	41.3 (7.5)	6.7 (1.9) n = 3	38.0 (14.0)	11.1 (2.9) n = 3	41.3 (18.2)
Historic	--- not sampled	---	--- not sampled	---	--- not sampled	---	0.0 (3.3) n = 2	0.0 (0.0)

6.3.4 Instream Woody Material

The results of the instream woody material (IWM) bank-line coverage for the repair versus Natural Sites and by repair site type for the September 2011 surveys are presented graphically in Figures 21 through 24. The results for the 2011 survey, as well as those from the prior three years, are presented in Table 24.

In September 2011, both the Natural and the Repair Sites tended to have higher percentages of IWM present along the high water transect compared to the low water transects (Figure 21; Table 24). The mean percentage bank-line coverage of IWM at the low water transects was nearly identical at the Natural and the Repair Sites (42.2% versus 47.7%, respectively). The average percentage bank-line coverage for IWM at the high water elevation transects were also nearly identical at the Natural and Repair Sites (56.1% versus 57.4%). Despite the similarity between the bank-line coverage of IWM at the Natural and Repair Sites, most of the IWM surveyed at the Natural Sites was identified as recruit material, while most of the IWM at the Repair Sites was material that had been placed (anchored or buried) as part of the repair design. At the Natural Sites, over 95% of the IWM was recruited material, compared to only 16% of the IWM at the Repair Sites.

An examination of the various Repair Sites shows that relatively high levels of IWM were present at both the low and high water bank-line transects at the Wetland Trench and the No Bench design Repair Sites. The high water bank-line transect at the one Bench repair site that was surveyed in 2011 also showed relatively high levels of IWM (Figures 22 and 23). Moderate levels of IWM were noted at the 10:1 Slope Repair Sites (both low and high water bank-lines) and at the low water bank-line at the Bench repair site (Figures 22 and 23). Low levels of IWM were present along either of the bank-line transects at the Historic Repair Sites in 2011 (Figure 24). The Historic Repair Sites did not have any placed wood installed as part of their design, so all the IWM noted at this repair type was recruited.

Since the 2008 survey, the average bank-line coverage of IWM has declined at the Natural Sites, though the 2011 percentage were a little higher than the lowest levels noted at the Natural Sites in the 2010 survey (Table 24). Since virtually all the IWM noted at the Natural Sites is recruited material, this decline over time suggests that there may be a woody debris deficit (i.e., IWM recruitment < IWM loss) operating within the lower Sacramento River system, and may indicate a lack of recruitment input sources.

Examination of the mean IWM bank-line coverage by repair site since 2008 shows a general decline in IWM at the 10:1 Slope Repair Sites at both the low and high water transects (Table 24). Since only a small fraction of the IWM at this site is recruited, the loss of IWM over time suggests a loss of the design-placed wood that was installed as part of the original repair. One unexplained inconsistency in the data is the large increase in the average percentage of IWM coverage for the Wetland Trench low water transects noted in 2011 compared to previous years. The 2011 increase is larger than the standard error of the data and so appears to be a significant increase. Since only 11% of the IWM

Table 24. Average percent bank-line coverage (with associated standard error, SE) for instream woody material at low water and high water elevation transects by site type, 2008-2011. 2008 and 2009 data from H.T. Harvey & Associates and PRBO (2010); 2010 data from H.T. Harvey & Associates data sheets.

Site Type	Water Elevation Transects							
	2008		2009		2010		2011	
	Low	High	Low	High	Low	High	Low	High
Natural	75.4 (6.6) n = 8	76.7 (6.6)	55.8 (8.1) n = 8	56.7 (7.6)	37.1 (SE 5.6) n = 8	42.5 (5.8)	42.2 (10.5) n = 6	56.1 (11.2)
Repair	44.4 (9.8) n = 13	73.6 (4.1)	42.8 (8.6) n = 13	81.5 (3.3)	31.2 (6.6) n = 11	66.7 (4.7)	47.7 (7.3) n = 13	57.4 (7.3)
Dietl Ditch	15.6 (8.7) n = 3	82.2 (2.9)	20.0 (10.7) n = 3	86.7 (5.1)	10.0 (5.1) n = 3	86.7 (1.9)	73.3 (10.0) n = 2	75.0 (8.3)
10:1 Slope	62.7 (13.1) n = 5	72.0 (7.0)	58.0 (7.6) n = 5	70.0 (2.3)	32.7 (8.3) n = 5	54.0 (4.0)	46.7 (4.6) n = 5	54.0 (2.9)
Bench	0.0 (0.0) n = 2	53.3 (6.7)	0.0 (0.0) n = 2	93.3 (3.3)	--- not sampled	---	40.0 (0.0) n = 1	80.0 (0.0)
No Bench	72.2 (9.7) n = 3	81.1 (5.9)	68.9 (14.7) n = 3	87.8 (7.3)	50.0 (11.5) n = 3	67.8 (2.2)	64.4 (16.4) n = 3	76.7 (10.7)
Historic	--- not sampled	---	--- not sampled	---	--- not sampled	---	3.3 (3.3) n = 2	8.3 (8.3)

was identified as recruited, it is unlikely recruited wood accounted for the recent increase. Perhaps some difference in identifying the low water transect at these Wetland Trench Sites during the 2011 surveys may have been responsible for this inconsistency.

A similar increase over time in the amount of IWM at the low water transect of the Bench repair site was also evident. This, however was probably due to the fact that a different Bench site was surveyed in 2011 (newly introduced Site 72.2R) than in the previous surveys (Sites 12.5L and 12.7L).

6.3.5 Individual Large Woody Debris Pieces

The results of the individual large woody debris pieces survey are presented in Table 25. Approximately the same number of pieces of wood per 100 feet of bank-line were documented at Natural and Repair Sites in 2011. The average diameter of the woody debris continues to be greater at the Natural Sites compared to the Repair Sites. Data suggest there was little change in the overall number or diameter of the woody material at the Natural Sites between 2010 and 2011. This recent stability in LWD pieces at the Natural Sites is in contrast with the earlier decline in woody material at the Natural Sites noted during the 2009 surveys. Conversely, the recent survey data has suggested there has been a large decline in the amount of LWD at the Repair Sites. The prolonged periods of high flows in the lower Sacramento River during the winter and spring of 2010-2011 may have resulted in the loss of large woody debris at the Repair Sites. Despite the loss of wood at the Repair Sites since 2010, the average diameter of the pieces of wood has remained unchanged.

Table 25. Average number, diameter, area in square feet, and volume in cubic feet of large woody debris pieces (with associated standard error, SE) below the high water elevation transects by site, 2008-2011. 2008 and 2009 data from H.T. Harvey & Associates and PRBO (2010); 2010 data from H.T. Harvey & Associates data sheet.

Site Type	2008				2009				2010				2011			
	Number of pieces per 100 feet of bank-line	Mean diam. (inches)	Total area per 100 feet of bank-line	Total volume per 100 feet of bank-line	Number of pieces per 100 feet of bank-line	Mean diam. (inches)	Total area per 100 feet of bank-line	Total volume per 100 feet of bank-line	Number of pieces per 100 feet of bank-line	Mean diam. (inches)	Total area per 100 feet of bank-line	Total volume per 100 feet of bank-line	Number of pieces per 100 feet of bank-line	Mean diam. (inches)	Total area per 100 feet of bank-line	Total volume per 100 feet of bank-line
Natural	36 (10.9)	7.7 (1.4)	953 (416.5)	---	18 (6.3)	9.1 (0.7)	36 (10.9)	---	7.8 (1.7)	7.8 (0.7)	221.9 (54.1)	---	8.2 (2.7)	8.1 (0.7)	87.3 (26.5)	77.1 (17.7)
	n = 8				n = 8				n = 8				n = 6			
Repair	12 (2.3)	6.2 (0.6)	432 (116.7)	---	18 (3.5)	7.4 (0.4)	36 (10.9)	---	19.3 (7.3)	5.7 (0.3)	364.2 (48.5)	---	8.0 (1.0)	5.5 (0.5)	63.2 (8.5)	29.2 (4.5)
	n = 13				n = 13				n = 11				n = 13			
Wetland Tr.	15 (10.5)	7.2 (2.1)	365 (102.0)	---	12 (8.6)	8.5 (0.4)	36 (10.9)	---	36.1 (27.8)	6.3 (0.7)	488.3 (152.0)	---	7.3 (1.0)	7.1 (0.5)	88.2 (10.5)	45.0 (4.7)
	n = 3				n = 3				n = 3				n = 2			
10:1 Slope	11 (2.2)	5.3 (0.9)	594 (288.4)	---	13 (2.5)	7.2 (0.9)	36 (10.9)	---	12.6 (1.9)	5.1 (0.4)	326.7 (51.3)	---	8.7 (0.5)	5.5 (0.5)	70.0 (6.0)	30.8 (5.6)
	n = 5				n = 5				n = 5				n = 5			
Bench	11 (1.3)	6.2 (0.5)	99 (13.3)	---	33 (7.1)	6.6 (0.6)	36 (10.9)	---	---	---	---	---	12.6 (0.0)	5.7 (0.0)	75.9 (0.0)	31.1 (0.0)
	n = 2				n = 2				not sampled				n = 1			
No Bench	12 (1.2)	6.8 (1.2)	450 (72.2)	---	22 (8.7)	7.0 (0.8)	36 (10.9)	---	13.7 (0.4)	6.0 (0.5)	36 (21.6)	---	10.6 (1.1)	6.4 (0.6)	71.9 (11.7)	34.3 (5.9)
	n = 3				n = 3				n = 3				n = 3			
Historic	---	---	---	---	---	---	---	---	---	---	---	---	0.6 (0.6)	5.0 (2.5)	1.7 (1.7)	0.5 (0.5)
	not sampled				not sampled				not sampled				n = 2			

6.3.6 Photo Documentation

Representative photographs showing comparisons of sites between August/September 2008 and September 2011, as well photographs of newly added sites in 2011 are presented in Appendix B.

At four of the five most upstream project sites, the opposing bank provided an unobstructed view of the entire site, providing the ability to establish traditional land based photo points. These sites are located above the Feather River confluence where channel widths are typically narrower than those encountered at areas further downstream. At each of these sites, two permanent photo points were established: one near the upstream and another near the downstream boundaries of the target site, along the opposing bank of the river.

Panoramic images lose their resolution and are difficult to view in a document (Figure 25); therefore, panoramic images are available on cd where they can be viewed in high resolution within a photo program. At one site (26.0L), poor lighting conditions from all angles necessitated taking several overlapping close-up images while floating through the site. Due to the moving perspective, these images do not lend themselves to being stitched into a single panoramic photo.



Figure 25. An example of a project site (91.9R) where a panoramic photo was produced from several consecutive images taken from a stationary photo point.

7.0 CONCLUSIONS

Fish communities were most similar between Natural Sites and SAM Repair Sites.

Based on electrofishing catch, fish community abundance and composition were more similar between Natural Sites and SAM Repair Sites (10:1 Slope, Bench, No Bench, Wetland Trench) than to Historic Sites. This result was mainly driven by the proportions of predator and focus species (e.g., black bass and Chinook salmon) at Natural and SAM Repair Sites, which outweighed site-specific differences in the presence of less abundant species.

Chinook salmon appear to prefer SAM Repair Sites containing mitigated habitat features compared to Historic Repair Sites consisting primarily of rip-rap. Chinook salmon presence and abundance was documented at various sites (SAM Repair, Historic Repair, and Natural Sites) by electrofishing, and this is the first Program study conducted to examine differences in salmon abundance between Historic Repair and SAM Repair Sites. Based on GLM analysis, it appears that the abundance of both Chinook salmon fry and juveniles was significantly higher at

the SAM Repair Sites than at the Historic Repair Sites, suggesting that Chinook salmon may prefer SAM Repair Sites containing mitigated habitat features compared to Historic Repair Sites consisting primarily of rip-rap.

Based on electrofishing, Chinook salmon appear to prefer the 10:1 Slope/Bench Site Type over both Natural Sites and other SAM Repair Site Types. Among SAM Repair Sites electrofished in 2011, Chinook salmon fry and juveniles were strongly associated with 10:1 Slope/Bench Site Types, which is consistent with results from previous years (H.T. Harvey & Associates and PRBO 2011). Unlike previous years where there was higher abundance of Chinook salmon fry and juveniles associated with Wetland Trench Site Types compared to other site types, there was no significant association of fry, and a lower abundance of juveniles associated with this site type in 2011. Also, we found that fry were significantly less likely to be found at No Bench sites in 2011, but this was not found in previous years. The mechanism(s) influencing differences observed between years is unclear, but possibilities include different inundation levels sampled or differences in sampling strategies between years.

Regardless of SAM Repair Site Type, Chinook salmon preferentially inhabit shallow water areas, and juveniles are more commonly found in areas with higher velocity. Both Chinook salmon fry and juveniles are more frequently found in shallow water, which is consistent with results from previous years (H.T. Harvey & Associates and PRBO 2011). Unlike previous years, we did not find a strong association with submerged vegetation but did find that juveniles showed significant preference for areas with higher water velocity.

Chinook salmon fry abundance decreased as bank coverage of terrestrial vegetation and terrestrial woody material increased. Based on electrofishing, fry abundance decreased as bank coverage of terrestrial vegetation and terrestrial woody material increased. Though these terrestrial habitat features appear to negatively impact Chinook abundance, the mechanisms of this interaction are unclear. It is possible that these features affect variables not accounted for in the models presented here, or are reflective of associated habitat conditions below the water level.

Predation risk to Chinook salmon fry and juveniles appears to be highest at Wetland Trench Sites, with some potential at No Bench Sites. In 2011, we examined black bass capture data from electrofishing, as well as the combined capture data of all potential salmonid predators (black bass, striped bass, and Sacramento pikeminnow) and found that, higher black bass and combined predator abundance was associated with Wetland Trench sites and higher black bass abundance was also associated with the No Bench design type. Previous predator analysis also found black bass to be strongly associated with Wetland Trench Sites, and given the similarity of results between years, it appears that predation risk for Chinook salmon is highest at Wetland Trench Sites. None of the microhabitat features examined in this study appeared to significantly affect predator abundance.

Effect of river region and SAM Repair Site Type on Chinook salmon size. The GLM analysis suggests that during both sampling events, Chinook salmon captured in downstream regions (Ko-ket and Garcia Bend) were larger than those sampled farther upstream larger (Knights Landing). This observed pattern may be due to age differences in observed Chinook

salmon, as fish are expected to be older (and therefore larger) as they migrate downstream. In March, larger fish were associated with the No Bench Site Type compared to other site types while in April, the 10:1 Slope/Bench repair type showed a strong positive correlation with fish size. However, it is important to note that these results do not allow for valid conclusions regarding rearing conditions at these sites (e.g., the No Bench design type does not *produce* larger fish than other sites), as the ages and origins (spawning sites) of captured fish are unknown.

Based on telemetry, SAM Repair Sites appear to be preferred by Chinook salmon, steelhead, and striped bass over Natural Sites. Acoustic tagged Chinook salmon (140 mm to 180 mm), steelhead (181 mm to 263 mm), and striped bass (245 mm to 625 mm) were each more likely to reside at SAM Repair Sites than at Natural Sites based on the proportions of detected fish that remained at a site for a minimum of one hour. This result is consistent with findings of acoustic telemetry studies in 2009 and 2010, but contrasts somewhat with electrofishing results, which suggest that Chinook salmon fry had no significant preference while juvenile Chinook salmon had a slight preference for Natural Sites over combined SAM Repair Sites. This difference may be a result of preferences changing with fish size, since acoustic tagged Chinook salmon were much larger (>140 mm) than the individuals observed during electrofishing (<110 mm).

Chinook salmon and steelhead residency is more likely in sites with tidal influence.

Acoustic tagged yearling Chinook salmon and steelhead were each more likely to reside at sites in the Rio Vista region, which is tidally influenced, than at upstream sites in the Sacramento River. These observations were based on the proportions of detected fish that remained at a site for a minimum of one hour, and are consistent with findings of acoustic telemetry studies conducted during 2009 and 2010.

Residency at sites generally short, but fish in tidal areas frequently visited individual sites more than once. Overall, regardless of site type or location, few Chinook salmon (140 mm-180 mm), steelhead (181 mm to 263 mm), and striped bass (245 mm to 625 mm) stayed at a site for more than a few hours to a day. However, fish detected at tidally influenced sites in the Rio Vista reach were frequently detected visiting different individual sites more than once, perhaps moving between sites with tidal excursions, which likely results in underestimation of the cumulative residency duration of individuals at given sites.

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Appendix A. Latitude and longitudes for the SAM habitat monitoring sites surveyed in September 2011

Site	Downstream high water elevation reference pin		Upstream high water elevation reference pin		Downstream low water elevation reference pin		Upstream low water elevation reference pin		Source
	Latitude	Longitude	Latitude	Longitude	Latitude	Longitude	Latitude	Longitude	
24.0L	38.232498	-121.557536	38.234177	-121.558216	---	---	---	---	USACE
25.0L	38.240662	-121.546634	38.240355	-121.544392	---	---	---	---	USACE/FISHBIO
26.9L	38.24373	-121.512745	38.244782	-121.511898	38.243738	-121.512761	38.244797	-121.511919	USACE/FISHBIO
34.5R	38.335106	-121.569155	38.336183	-121.568149	38.335092	-121.569142	38.336181	-121.568135	USACE/FISHBIO
43.7R	38.43585	-121.525841	38.435704	-121.52345	38.43581	-121.525852	38.435678	-121.523453	USACE/FISHBIO
44.7R	38.4371	-121.509437	38.438085	-121.507405	38.437077	-121.50942	38.438049	-121.507375	USACE/FISHBIO
47.0L	38.468641	-121.503853	38.470345	-121.504227	38.468632	-121.50392	38.470324	-121.504354	USACE/FISHBIO
47.9R	38.473319	-121.516998	38.47348	-121.519147	38.473366	-121.516985	38.473505	-121.519143	USACE/FISHBIO
50.2L	38.487987	-121.551057	38.489433	-121.552117	38.487954	-121.551155	38.489396	-121.552216	USACE
50.3R	38.48854	-121.554257	38.489556	-121.554981	---	---	---	---	USACE/FISHBIO
69.1L	38.657653	-121.604547	38.659381	-121.606208	---	---	---	---	USACE
69.9R	38.666891	-121.619445	38.668257	-121.621251	38.666912	-121.619416	38.66828	-121.621224	USACE/FISHBIO
72.2R	38.695743	-121.633292	38.697404	-121.632265	38.695748	-121.633273	38.697397	-121.632244	USACE
85.6L	38.765448	-121.689432	38.766591	-121.691009	38.765413	-121.689461	38.76656	-121.691052	FISHBIO
85.6R	38.76495	-121.690188	38.766102	-121.691701	38.764985	-121.690146	38.766136	-121.691653	USACE/FISHBIO
87.0L	38.780619	-121.687243	38.782112	-121.688143	38.780611	-121.687274	38.78211	-121.688165	USACE/FISHBIO
90.8R	38.812216	-121.724019	38.813722	-121.724267	---	---	---	---	FISHBIO
90.8L	38.812217	-121.723037	38.813953	-121.723282	38.81222	-121.723103	38.813933	-121.723307	FISHBIO
91.9R	38.825302	-121.72422	38.826963	-121.724783	---	---	---	---	FISHBIO

Note: Site numbers indicate river mile and right (R) or left (L) bank (looking downstream).

Appendix B. Photo Documentation of Sampling Sites 2008 and 2011

Site 24.0L (Natural Site)



Figure 26. Site 24.0L (Natural Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 27. Site 24.0L (Natural Site), September 2011. Photo Source: FISHBIO database.

Site 24.0R (Historic Repair Site)

Site 24.0R was a newly added site in 2011; therefore, no previous photos exist of the site.



Figure 28. Site 24.0R (Historic Repair), September 2011. Photo Source: FISHBIO database.

Site 25.0L (Natural Site)



Figure 29. Site 25.0L (Natural Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 30. Site 25.0L (Natural Site), September 2011. Photo Source: FISHBIO database.

Site 26.9L (Wetland Trench Site)



Figure 31. Site 26.9L (Wetland Trench Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 32. Site 26.9L (Wetland Trench Site), September 2011. Photo Source: FISHBIO database.

Site 34.5R (Wetland Trench Site)



Figure 33. Site 34.5R (Wetland Trench Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 34. Site 34.5R (Wetland Trench Site), September 2011. Photo Source: FISHBIO database.

Site 38.7 (Historic Repair)

Site 38.7R was a newly added site in 2011; therefore, no previous photos exist of the site.



Figure 35. Site 38.7R (Historic Repair), March 2011. Photo Source: FISHBIO database.

Site 43.7R (10:1 Slope Site)



Figure 36. Site 43.7R (10:1 Slope Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 37. Site 43.7R (10:1 Slope Site), September 2011. Photo Source: FISHBIO database.

Site 44.7R (10:1 Slope Site)



Figure 38. Site 44.7R (10:1 Slope Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 39. Site 44.7R (10:1 Slope Site), September 2011. Photo Source: FISHBIO database.

Site 47.0L (10:1 Slope Site)



Figure 40. Site 47.0L (10:1 Slope Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 41. Site 47.0L (10:1 Slope Site), September 2011. Photo Source: FISHBIO database.

Site 47.9R (10:1 Slope Site)



Figure 42. Site 47.9R (10:1 Slope Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 43. Site 47.9R (10:1 Slope Site), September 2011. Photo Source: FISHBIO database.

Site 50.2L (No Bench Site)



Figure 44. Site 50.2L (No Bench Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 45. Site 50.2L (No Bench Site), September 2011. Photo Source: FISHBIO database.

Site 50.3R (Natural Site)



Figure 46. Site 50.3R (Natural Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 47. Site 50.3R (Natural Site), September 2011. Photo Source: FISHBIO database.

Site 56.0R (Historic Repair Site)

Site 56.0R was a newly added site in 2011; therefore, no previous photos exist of the site.



Figure 48. Site 56.0R (Historic Repair Site), April 2011. Photo Source: FISHBIO database.

Site 57.2R (Setback Levee Site)

Site 57.2R was a newly added site in 2011; therefore, no previous photos exist of the site. This site was surveyed in March but not in April, as directed by USACE.



Figure 49. Site 57.2R (Setback Levee Site), March 2011.

Site 69.1L (Natural Site)



Figure 50. Site 69.1L (Natural Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 51. Site 69.1L (Natural Site), September 2011.

Site 69.9R (No Bench Site)



Figure 52. Site 69.9R (No Bench Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 53. Site 69.9R (No Bench Site), September 2011. Photo Source: FISHBIO database.

Site 72.2R (Bench Site)

Site 72.2R was a newly added site in 2011; therefore, no previous photos exist of the site.



Figure 54. Site 72.2R (Bench Site), March 2011. Photo Source: FISHBIO database.

Site 85.6L (Historic Repair Site)

Site 85.6L was a newly added site in 2011; therefore, no previous photos exist of the site.



Figure 55. Site 85.6L (Historic Repair Site), April 2011. Photo Source: FISHBIO database.

Site 85.6R (No Bench Site)



Figure 56. Site 85.6R (No Bench Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 57. Site 85.6R (No Bench Site), September 2011. Photo Source: FISHBIO database.

Site 90.8L (Historic Repair Site)

Site 90.8L was a newly added site in 2011; therefore, no previous photos exist of the site.



Figure 58. Site 90.8L (Historic Repair Site), September 2011. Photo Source: FISHBIO database.

Site 90.8R (Natural Site)



Figure 59. Site 90.8R (Natural Repair Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 60. Site 90.8R (Natural Repair Site), September 2011. Photo Source: FISHBIO database.

Site 91.9R (Natural Site)



Figure 61. Site 91.9R (Natural Site), August/September 2008. Photo Source: H.T. Harvey & Associates and PRBO (2008).



Figure 62. Site 91.9R (Natural Site), September 2011. Photo Source: FISHBIO database.

Appendix C. Environmental Conditions at Each Sampling Site during March 2-5, 2011, and April 18-22, 2011, Surveys

Site ID	Date	Turbidity (ntu)	Temp (°C)	Dissolved Oxygen (mg/L)	Conductivity (µS/cm)	Salinity (ppmS)	Secchi (ft.)
24.0L	22-Apr	19.44	14.3	10.2	110.8	55.2	2
24.0R	5-Mar	26.5	11.1	10.61	142.8	70.6	1.7
	22-Apr	19.44	14.3	10.2	110.8	55.2	2
25.0L	5-Mar	19.56	11.4	10.51	146.2	74.5	1.8
	22-Apr	20.7	14.2	10.14	138.5	67.8	2
26.9L	5-Mar	21.1	11.1	10.62	140.6	70.8	1.5
	22-Apr	13.85	14	10.08	153.3	108	2.3
34.5R	5-Mar	23.8	10.9	10.63	137.9	72.7	1.5
	21-Apr	15.51	15	10.13	155.9	70.9	2.2
38.7R	5-Mar	16.96	10.5	10.82	140.3	69.8	1.8
	21-Apr	18.57	14.4	10.26	171	91.8	2
43.7R	4-Mar	14.68	10.6	10.61	142.6	72	1.9
	21-Apr	15.81	14.2	10.33	108.9	54.1	2.3
44.7R	4-Mar	19.63	10.7	10.72	137.6	68.3	2
	21-Apr	18.13	14	10.35	146.6	71.8	2.3
47.0L	4-Mar	18	10.9	10.75	143.2	71.4	2
	20-Apr	18.95	14.3	10.19	117.2	63.6	2.2
47.9R	4-Mar	18.67	2	10.71	147.3	63	1.7
	20-Apr	16.78	14.1	10.21	125.2	61.2	2.3
50.2L	4-Mar	19.87	10.4	10.77	156	65.2	1.5
	20-Apr	15.58	14	10.24	117.8	63.6	2.2
50.3R	4-Mar	19.68	10.2	10.85	161.66	70.6	1.4
	20-Apr	15.58	14	10.24	117.8	63.6	2.3
56.0R	3-Mar	26	10.6	10.76	158.83	67.9	0
	20-Apr	15.15	13.8	10.25	181.6	89.9	2.5
57.2R	3-Mar	26.8	10.8	10.61	150.16	65.2	1.25
69.1L	3-Mar	30.2	10.6	10.46	171.5	73.4	1.25
	19-Apr	23.9	14.4	9.91	143.5	88.1	2
69.9R	3-Mar	25.4	10.4	10.47	172.16	84.2	1
	19-Apr	20.1	14.9	9.77	180.9	82.6	1.5
72.2R	3-Mar	25.6	10.3	10.48	173.3	86.5	1.5
	19-Apr	15.84	14.5	9.99	131.5	74.3	1.75
85.6L	2-Mar	21.2	9.8	10.76	177.83	75.8	2.25
	19-Apr	23.7	15.4	9.64	147.5	73.7	1.75
85.6R	2-Mar	21.2	9.8	10.76	177.8	75.8	2.25
	19-Apr	18.84	15.9	9.66	195.1	98.8	1.25
90.8L	19-Apr	18.94	15.6	9.65	97.8	9.4	1.3
90.8R	2-Mar	16.83	9.7	10.78	176.5	76.5	2
	18-Apr	18.94	15.6	9.65	97.8	9.45	1.75
91.9R	2-Mar	34.1	9.6	10.81	156	74.8	1.5
	18-Apr	24.8	15.5	9.66	267	154	1.8

Appendix D. Physical Parameters at Each Sampling Site during March 2-5, 2011, and April 18-22, 2011, Surveys

Site ID	Date	Average Depth (ft.)	Average Velocity (ft/s)	Substrate Percent Composition			Percent Vegetation
				Rip Rap	Sand/Silt	Other	
24.0L	22-Apr	2.2	0.30	0%	100%	0%	0%
24.0R	5-Mar	5.0	-	100%	0%	0%	-
	22-Apr	3.1	0.56	93%	7%	0%	0%
25.0L	5-Mar	3.2	-	0%	100%	0%	-
	22-Apr	2.6	0.13	0%	100%	0%	60%
26.9L	5-Mar	4.1	-	25%	75%	0%	-
	22-Apr	3.1	0.31	39%	61%	0%	44%
34.5R	5-Mar	2.9	-	100%	0%	0%	-
	21-Apr	2.3	0.03	40%	60%	0%	40%
38.7R	5-Mar	6.6	0.59	100%	0%	0%	-
	21-Apr	3.9	0.49	100%	0%	0%	0%
43.7R	4-Mar	3.1	0.13	0%	100%	0%	-
	21-Apr	1.0	0.02	0%	100%	0%	67%
44.7R	4-Mar	3.8	0.25	67%	33%	0%	-
	21-Apr	1.8	0.40	0%	100%	0%	60%
47.0L	4-Mar	3.5	0.45	22%	78%	0%	-
	20-Apr	1.9	0.25	20%	73%	7%	67%
47.9R	4-Mar	3.8	0.15	100%	0%	0%	-
	20-Apr	0.8	0.09	0%	100%	0%	100%
50.2L	4-Mar	4.9	0.31	33%	67%	0%	-
	20-Apr	3.2	0.34	40%	60%	0%	27%
50.3R	4-Mar	4.0	0.13	22%	78%	0%	-
	20-Apr	4.6	0.12	0%	100%	0%	67%
56.0R	3-Mar	5.6	0.62	100%	0%	0%	-
	20-Apr	4.0	0.58	100%	0%	0%	0%
57.2R	3-Mar	4.1	0.20	67%	33%	0%	-
69.1L	3-Mar	3.7	-	0%	100%	0%	-
	19-Apr	3.5	0.23	20%	80%	0%	40%
69.9R	3-Mar	4.1	-	100%	0%	0%	-
	19-Apr	3.3	0.17	33%	67%	0%	53%
72.2R	3-Mar	4.2	-	89%	11%	0%	-
	19-Apr	1.8	0.08	7%	93%	0%	100%
85.6L	2-Mar	5.0	-	100%	0%	0%	-
	19-Apr	4.7	0.33	73%	27%	0%	33%
85.6R	2-Mar	4.6	-	67%	0%	33%	-
	19-Apr	3.9	0.19	80%	20%	0%	33%
90.8L	19-Apr	3.3	-0.07	0%	100%	0%	60%
90.8R	2-Mar	8.0	0.60	0%	100%	0%	-
	18-Apr	5.5	0.25	0%	100%	0%	27%
91.9R	2-Mar	5.9	0.34	0%	100%	0%	-
	18-Apr	5.7	0.15	0%	100%	0%	53%

Appendix E. Number of Species Captured at Each Sampling Site during March 2-5, 2011, and April 18-22, 2011, Surveys.

Table E-1. Number of focus and predator species captured at each sampling site during March 2-5, 2011, and April 18-22, 2011, surveys.

Site ID	Date	Focus Species				Predator Species					
		Chinook Juvenile	Chinook Adult	<i>O. mykiss</i>	Delta Smelt	Sacramento Pikeminnow	Striped Bass	Black Bass Unidentified	Largemouth Bass	Smallmouth Bass	Inland Silverside
24.0L	22-Apr	64	0	0	0	1	0	2	0	0	14
24.0R	5-Mar	1	0	0	0	0	0	11	0	0	0
	22-Apr	1	0	0	0	0	0	15	0	0	0
25.0L	5-Mar	14	0	0	0	0	0	0	0	0	18
	22-Apr	19	0	0	0	2	1	2	0	0	3
26.9L	5-Mar	12	0	0	0	3	0	11	0	0	5
	22-Apr	14	0	1	1	1	1	6	0	0	0
34.5R	5-Mar	5	0	0	0	2	0	3	0	0	0
	21-Apr	25	0	0	0	6	1	2	0	0	0
38.7R	5-Mar	1	0	0	0	1	0	5	0	0	0
	21-Apr	4	0	0	0	0	0	6	0	0	0
43.7R	4-Mar	2	0	0	0	2	0	2	0	0	6
	21-Apr	51	0	0	1	1	1	1	0	0	2
44.7R	4-Mar	26	0	0	0	2	0	3	0	0	3
	21-Apr	18	0	0	0	1	3	0	0	0	0
47.0L	4-Mar	2	0	0	0	4	0	1	0	0	0
	20-Apr	38	0	0	3	0	0	3	0	0	0
47.9R	4-Mar	5	0	0	0	0	0	12	0	0	1
	20-Apr	10	0	0	0	1	0	1	0	0	0
50.2L	4-Mar	4	0	0	0	1	0	4	0	0	1
	20-Apr	3	0	0	0	0	0	0	0	0	0
50.3R	4-Mar	6	0	0	0	1	0	1	0	0	0
	20-Apr	3	0	0	0	0	0	3	0	1	0
56.0R	3-Mar	0	0	0	0	7	0	10	0	0	0

Site ID	Date	Focus Species				Predator Species					
		Chinook Juvenile	Chinook Adult	<i>O. mykiss</i>	Delta Smelt	Sacramento Pikeminnow	Striped Bass	Black Bass Unidentified	Largemouth Bass	Smallmouth Bass	Inland Silverside
	20-Apr	0	0	0	0	1	0	0	0	0	0
57.2R	3-Mar	0	1	0	0	2	0	1	0	0	0
69.1L	3-Mar	13	0	0	0	3	0	1	0	0	3
	19-Apr	5	0	0	0	0	0	0	0	0	0
69.9R	3-Mar	4	0	1	0	5	0	14	0	0	3
	19-Apr	1	0	0	0	0	0	0	0	0	0
72.2R	3-Mar	9	0	1	0	8	0	4	0	0	0
	19-Apr	6	0	0	0	1	0	0	0	0	0
85.6L	2-Mar	3	0	0	0	0	0	0	0	0	2
	19-Apr	5	0	0	0	0	1	1	0	0	1
85.6R	2-Mar	0	0	1	0	9	0	0	3	0	0
	19-Apr	0	0	2	0	0	0	0	0	0	0
90.8L	19-Apr	2	0	0	0	5	0	0	0	0	0
90.8R	2-Mar	12	0	0	0	4	0	0	0	0	0
	18-Apr	1	0	0	0	1	0	0	0	0	0
91.9R	2-Mar	12	0	3	0	2	0	0	0	0	1
	18-Apr	3	0	0	0	2	0	0	0	0	0

Table E-2. Number of other species captured at each sampling site during March 2-5, 2011, and April 18-22, 2011, surveys.

Site	Bluegill	Carp	Ca. roach	Green sunfish	Hitch	Golden shiner	Prickly sculpin	Redear sunfish	Red shiner	Sac. sucker	Sac. blackfish	Sac. splittail	Threadfin shad	Tule perch	Warmouth	White catfish	Unidentified
March 2-5, 2011, Sampling Event																	
91.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1
90.8R	1	3	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1
90.8L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
85.6R	0	0	0	1	0	0	0	2	0	1	0	0	1	12	0	0	0
85.6L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
72.2R	2	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0
69.9R	0	2	2	0	0	0	0	1	0	0	0	0	1	0	0	0	0
69.1L	1	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
57.2R	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
56.0R	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
50.3R	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
50.2L	0	0	0	0	0	0	3	0	0	0	0	0	0	5	0	0	0
47.9R	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0
47.0L	1	0	0	0	0	0	2	0	0	1	0	0	0	1	0	0	0
44.7R	3	0	0	0	0	0	3	0	0	0	1	0	0	0	0	0	0
43.7R	1	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0
38.7R	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	0
34.5R	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
26.9L	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
25.0L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24.0R	0	0	0	0	0	1	0	1	0	0	1	0	0	0	0	0	0
24.0L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total	10	5	2	1	0	3	9	6	0	8	2	0	3	23	2	1	2

Table E-2. cont.

Site	Bluegill	Carp	Ca. roach	Green sunfish	Hitch	Golden shiner	Prickly sculpin	Red ear sunfish	Red shiner	Sac. sucker	Sac. blackfish	Sac. splittail	Threadfin shad	Tule perch	Warmouth	White catfish	Unidentified
April 18-22, 2011, Sampling Event																	
91.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
90.8R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
90.8L	0	1	0	0	0	0	0	0	0	22	0	0	0	12	0	0	0
85.6R	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
85.6L	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
72.2R	1	2	0	0	0	0	0	0	0	2	0	0	0	1	0	0	0
69.9R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
69.1L	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
57.2R	n/a	n/a	0	n/a	n/a	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0
56.0R	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
50.3R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
50.2L	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
47.9R	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
47.0L	1	0	0	0	0	0	0	1	0	3	0	0	0	0	1	0	0
44.7R	0	1	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0
43.7R	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0
38.7R	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0
34.5R	2	0	0	0	0	0	0	2	0	1	1	0	0	0	0	0	0
26.9L	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
25.0L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24.0R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
24.0L	0	0	0	0	3	0	0	0	0	0	0	0	0	1	0	1	0
Total	4	7	0	2	3	0	2	3	1	35	1	5	1	15	1	1	0

Table E-3. Chinook salmon race assignments and mortalities at each sampling site during March 2-5, 2011, and April 18-22, 2011, surveys.

Site	<u>Fall-Run</u>		<u>Spring-Run</u>		<u>Adult</u>	
	Live	Mortality	Live	Mortality	Live	Mortality
March 2-5, 2011						
91.9R	12	0	0	0	0	0
90.8R	12	0	0	0	0	0
90.8L	n/a	n/a	n/a	n/a	0	0
85.6L	1	0	2	0	0	0
72.2R	6	0	2	1	0	0
69.9R	1	0	3	0	0	0
69.1L	12	0	1	0	0	0
57.2R	0	0	0	0	1	0
50.3R	5	1	0	0	0	0
50.2L	0	0	4	0	0	0
47.9R	5	0	0	0	0	0
47.0L	2	0	0	0	0	0
44.7R	21	2	3	0	0	0
43.7R	1	0	1	0	0	0
38.7R	1	0	0	0	0	0
34.5R	2	0	3	0	0	0
26.9L	5	0	7	0	0	0
25.0L	3	1	10	0	0	0
24.0R	0	0	1	0	0	0
24.0L	n/a	n/a	n/a	n/a	0	0
Total	89	4	37	1	1	0

Table E-3. cont.

Site	<u>Fall-Run</u>		<u>Spring-Run</u>		<u>Adult</u>	
	Live	Mortality	Live	Mortality	Live	Mortality
April 18-22, 2011						
91.9R	2	1	0	0	0	0
90.8R	1	0	0	0	0	0
90.8L	2	0	0	0	0	0
85.6L	5	0	0	0	0	0
72.2R	5	0	1	0	0	0
69.9R	0	1	0	0	0	0
69.1L	5	0	0	0	0	0
57.2R	n/a	n/a	n/a	n/a	0	0
50.3R	3	0	0	0	0	0
50.2L	2	1	0	0	0	0
47.9R	8	2	0	0	0	0
47.0L	23	5	10	0	0	0
44.7R	12	2	4	0	0	0
43.7R	40	7	4	0	0	0
38.7R	4	0	0	0	0	0
34.5R	22	0	3	0	0	0
26.9L	12	1	1	0	0	0
25.0L	13	2	4	0	0	0
24.0R	1	0	0	0	0	0
24.0L	51	3	10	0	0	0
Total	211	25	37	0	0	0